

RIEMANNIAN GEOMETRY EXERCISES

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1 HW1

Problem 1.1

Given a metric

$$g = \sum_{i,j=1}^n g_{ij} dx^i \otimes dx^j, \quad g_{ij} = \delta_{ij} + \frac{x^i x^j}{\kappa^2 - \sum_{i=1}^n (x^i)^2}$$

where $\kappa^2 - \sum_{i=1}^n (x^i)^2 > 0$. Compute Γ_{ij}^k , R_{ijkl} , R_{ij} and scalar curvature S . Note that $R_{ij} = g^{kl} R_{kijl}$ and $S = g^{ij} R_{ij}$.

Proof. We have the Sherman-Morrison formula

$$(A + \alpha \beta^t)^{-1} = A^{-1} - \frac{A^{-1} \alpha \beta^t A^{-1}}{1 + \beta^t A^{-1} \alpha}.$$

We set $A = I$ and

$$\alpha = \beta = \frac{x}{\sqrt{\kappa^2 - |x|^2}}.$$

Then $(g_{ij}) = A + \alpha \beta^t$, and Sherman-Morrison formula indicates that

$$(g^{ij}) = I - \frac{\alpha \alpha^t}{1 + |\alpha|^2} \quad \text{i.e.} \quad g^{ij} = \delta^{ij} - \frac{x^i x^j}{\kappa^2}.$$

Note that

$$\frac{\partial g_{ij}}{\partial x^k} = \frac{\delta_{ik} x^j + \delta_{jk} x^i}{\kappa^2 - |x|^2} + \frac{2x^i x^j x^k}{(\kappa^2 - |x|^2)^2}.$$

Therefore, we know

$$\begin{aligned} \Gamma_{ij}^k &= \frac{1}{2} g^{kl} \left(\frac{\partial g_{jl}}{\partial x^i} + \frac{\partial g_{il}}{\partial x^j} - \frac{\partial g_{ij}}{\partial x^l} \right) \\ &= \frac{1}{2} \sum_{l=1}^n \left(\delta_{kl} - \frac{x^k x^l}{\kappa^2} \right) \left(\frac{\delta_{jl} x^i + \delta_{li} x^j + \delta_{ij} x^l + \delta_{lj} x^i - \delta_{il} x^j - \delta_{jl} x^i}{\kappa^2 - |x|^2} + \frac{2x^i x^j x^l}{(\kappa^2 - |x|^2)^2} \right) \\ &= \sum_{l=1}^n \left(\delta_{kl} - \frac{x^k x^l}{\kappa^2} \right) \left(\frac{\delta_{ij} x^l}{\kappa^2 - |x|^2} + \frac{x^i x^j x^l}{(\kappa^2 - |x|^2)^2} \right) \\ &= \frac{x^k}{\kappa^2} \left(\delta_{ij} + \frac{x^i x^j}{\kappa^2 - \sum_{i=1}^n (x^i)^2} \right) = \frac{x^k}{\kappa^2} g_{ij}. \end{aligned}$$

Also,

$$g_{ij} x^j = \sum_{j=1}^n x^j \left(\delta_{ij} + \frac{x^i x^j}{\kappa^2 - |x|^2} \right) = x^i + \frac{x^i |x|^2}{\kappa^2 - |x|^2} = \frac{x^i \kappa^2}{\kappa^2 - |x|^2},$$

and hence

$$\begin{aligned} R_{ijkl} &= g_{lp} R_{ijk}^p = g_{lp} \left(\frac{\partial \Gamma_{jk}^p}{\partial x^i} + \Gamma_{jk}^s \Gamma_{is}^p - \frac{\partial \Gamma_{ik}^p}{\partial x^j} - \Gamma_{ik}^s \Gamma_{js}^p \right) \\ &= g_{lp} \left(\frac{1}{\kappa^2} \frac{\partial (x^p g_{jk})}{\partial x^i} + \frac{x^s x^p g_{jk} g_{is}}{\kappa^4} - \frac{1}{\kappa^2} \frac{\partial (x^p g_{ik})}{\partial x^j} - \frac{x^s x^p g_{ik} g_{js}}{\kappa^4} \right) \\ &= \frac{g_{lp} x^s x^p (g_{jk} g_{is} - g_{ik} g_{js})}{\kappa^4} + \frac{g_{lp}}{\kappa^2} \left(\delta_i^p g_{jk} + x^p \frac{\partial g_{jk}}{\partial x^i} - \delta_j^p g_{ik} - x^p \frac{\partial g_{ik}}{\partial x^j} \right) \\ &= \frac{g_{il} g_{jk} - g_{jl} g_{ik}}{\kappa^2} + \frac{x^i x^l (g_{jk} - \delta_{jk}) - x^j x^l (g_{ik} - \delta_{ik})}{(\kappa^2 - |x|^2)^2} \\ &= \frac{g_{il} g_{jk} - g_{jl} g_{ik}}{\kappa^2}. \end{aligned}$$

That is,

$$R_{ijkl} = \frac{1}{\kappa^2} (g_{il}g_{jk} - g_{jl}g_{ik}).$$

Thus

$$R_{ij} = g^{kl} R_{kijl} = \frac{g^{kl}g_{kl}g_{ij} - g^{kl}g_{il}g_{kj}}{\kappa^2} = \frac{ng_{ij} - g_{ij}}{\kappa^2} = \frac{n-1}{\kappa^2} g_{ij}$$

and

$$S = g^{ij} R_{ij} = \frac{n(n-1)}{\kappa^2}.$$

□

Problem 1.2

Let $\mathbb{B}^n = \{x = (x^1, \dots, x^n) \in \mathbb{R}^n : |x|^2 < 1\}$. We put the hyperbolic metric g on $\mathbb{B}^n$:

$$g = \sum g_{ij} dx^i \otimes dx^j, \quad g_{ij} = \frac{\delta_{ij}}{(1 - |x|^2)^2}.$$

Show that the curvature tensor R of g satisfies:

$$R_{ijkl} = -4(g_{il}g_{jk} - g_{ik}g_{jl}).$$

Proof. Since $g^{ij} = (1 - |x|^2)^2 \delta^{ij}$, we have

$$\begin{aligned} \Gamma_{ij}^k &= \frac{1}{2} g^{kl} \left(\frac{\partial g_{jl}}{\partial x^i} + \frac{\partial g_{il}}{\partial x^j} - \frac{\partial g_{ij}}{\partial x^l} \right) \\ &= \sum_{l=1}^n \delta_{kl} \frac{2\delta_{jl}x^i + 2\delta_{il}x^j - 2\delta_{ij}x^l}{1 - |x|^2} \\ &= \frac{2\delta_{jk}x^i + 2\delta_{ik}x^j - 2\delta_{ij}x^k}{1 - |x|^2}. \end{aligned}$$

It follows that

$$\begin{aligned} R_{ijkl} &= g_{lp} R_{ijk}^p = g_{lp} \left(\frac{\partial \Gamma_{jk}^p}{\partial x^i} + \Gamma_{jk}^s \Gamma_{is}^p - \frac{\partial \Gamma_{ik}^p}{\partial x^j} - \Gamma_{ik}^s \Gamma_{js}^p \right) \\ &= (1 - |x|^2)^{-2} \delta_{lp} \left(\frac{4\delta_{jp}\delta_{ik}}{1 - |x|^2} + \frac{4(\delta_{kp}x^j + \delta_{jp}x^k - \delta_{jk}x^p)x^i}{(1 - |x|^2)^2} \right) \\ &\quad - (1 - |x|^2)^{-2} \delta_{lp} \left(\frac{4\delta_{ip}\delta_{jk}}{1 - |x|^2} + \frac{4(\delta_{kp}x^i + \delta_{ip}x^k - \delta_{ik}x^p)x^j}{(1 - |x|^2)^2} \right) \\ &\quad + (1 - |x|^2)^{-2} \delta_{lp} \frac{(2\delta_{ks}x^j + 2\delta_{js}x^k - 2\delta_{jk}x^s)(2\delta_{sp}x^i + 2\delta_{ip}x^s - 2\delta_{is}x^p)}{(1 - |x|^2)^2} \\ &\quad - (1 - |x|^2)^{-2} \delta_{lp} \frac{(2\delta_{ks}x^i + 2\delta_{is}x^k - 2\delta_{ik}x^s)(2\delta_{sp}x^j + 2\delta_{jp}x^s - 2\delta_{js}x^p)}{(1 - |x|^2)^2} \\ &= 4(1 - |x|^2)^{-3} (\delta_{jl}\delta_{ik} - \delta_{il}\delta_{jk}) + 4(1 - |x|^2)^{-4} (\delta_{kl}x^j + \delta_{jl}x^k - \delta_{jk}x^l)x^i \\ &\quad - 4(1 - |x|^2)^{-4} (\delta_{kl}x^i + \delta_{il}x^k - \delta_{ik}x^l)x^j \\ &\quad + (1 - |x|^2)^{-4} (2\delta_{ks}x^j + 2\delta_{js}x^k - 2\delta_{jk}x^s)(2\delta_{sl}x^i + 2\delta_{il}x^s - 2\delta_{is}x^l) \\ &\quad - (1 - |x|^2)^{-4} (2\delta_{ks}x^i + 2\delta_{is}x^k - 2\delta_{ik}x^s)(2\delta_{sl}x^j + 2\delta_{jl}x^s - 2\delta_{js}x^l) \\ &= 4(1 - |x|^2)^{-3} (\delta_{jl}\delta_{ik} - \delta_{il}\delta_{jk}) - 4(1 - |x|^2)^{-4} \delta_{jk}\delta_{il}|x|^2 + 4(1 - |x|^2)^{-4} \delta_{ik}\delta_{jl}|x|^2 \\ &= 4(1 - |x|^2)^{-4} (\delta_{jl}\delta_{ik} - \delta_{il}\delta_{jk}) = -4(g_{il}g_{jk} - g_{ik}g_{jl}). \end{aligned}$$

□

Problem 1.3

Let (M, g) be an Einstein manifold, i.e. $R_{ij} = \lambda g_{ij}$ for some $\lambda \in \mathbb{R}$. If $\dim M = 3$, show that (M, g) is of constant sectional curvature.

Proof. For any two dimensional subspace $\sigma \subset T_p M$, let $\{e_1, e_2, e_3\}$ be an orthonormal basis of $T_p M$ such that σ is spanned by e_1 and e_2 .

Since $R_{ij} = \lambda g_{ij}$, we have

$$\begin{aligned} R_{1221} + R_{1331} &= Ric(e_1, e_1) = \lambda, \\ R_{1221} + R_{2332} &= Ric(e_2, e_2) = \lambda, \\ R_{1331} + R_{2332} &= Ric(e_3, e_3) = \lambda. \end{aligned}$$

From these we get $\sec(\sigma) = R_{1221} = \lambda/2$. Thus M is of constant sectional curvature. $\square$

Problem 1.4

Let (M^n, g) be a Riemannian manifold. Prove that the scalar curvature $s(p)$ is given by

$$s(p) = \frac{n}{\omega_{n-1}} \int_{S^{n-1}} Ric_p(x, x) dS.$$

Proof. Let $\{e_i\}_{i=1}^n$ be an orthonormal basis of $T_p M$. Then

$$\begin{aligned} s(p) &= Ric_p(e_i, e_j) \delta^{ij} = \frac{n}{\omega_{n-1}} Ric_p(e_i, e_j) \int_{S^{n-1}} x^i x^j dS \\ &= \frac{n}{\omega_{n-1}} \int_{S^{n-1}} Ric(x^i e_i, x^j e_j) dS = \frac{n}{\omega_{n-1}} \int_{S^{n-1}} Ric_p(x, x) dS \end{aligned}$$

where we use the fact that

$$\int_{S^{n-1}} x^i x^j dS = \frac{\delta^{ij}}{n} \omega_{n-1}.$$

$\square$

Problem 1.5

Let (M^n, g) be a Riemannian manifold and $p \in M$. For each unit vector $v \in T_p M$ prove that

$$Ric_p(v, v) = \frac{n-1}{\text{vol}(S^{n-2})} \int_{w \in S_v^\perp} \sec(v, w) dV_{\hat{g}}$$

where $S_v^\perp$ denotes the set of unit vectors in $T_p M$ that are orthogonal to v and $\hat{g}$ denotes the Riemannian metric on $S_v^\perp$ induced from the flat metric g_p on $T_p M$.

Proof. Let $\{e_i\}_{i=0}^{n-1}$ be an orthonormal basis of $T_p M$ such that $e_0 = v$. Then

$$w \in S_v^\perp \iff w = \sum_{i=1}^{n-1} w^i e_i \quad \text{where} \quad \sum_{i=1}^{n-1} (w^i)^2 = 1.$$

Therefore, we know

$$\begin{aligned}
\text{Ric}_p(v, v) &= \sum_{i,j=1}^{n-1} \delta^{ij} R(v, e_i, e_j, v) \\
&= \frac{n-1}{\text{vol}(S^{n-2})} \sum_{i,j=1}^{n-1} R(v, e_i, e_j, v) \int_{S^{n-2}} x^i x^j dS \\
&= \frac{n-1}{\text{vol}(S^{n-2})} \int_{S^{n-2}} \sum_{i,j=1}^{n-1} R(v, x^i e_i, x^j e_j, v) dS \\
&= \frac{n-1}{\text{vol}(S^{n-2})} \int_{w \in S_v^\perp} R(v, w, w, v) dV_{\hat{g}} \\
&= \frac{n-1}{\text{vol}(S^{n-2})} \int_{w \in S_v^\perp} \sec(v, w) dV_{\hat{g}}
\end{aligned}$$

where we use the fact that

$$\int_{S^{n-2}} x^i x^j dS = \frac{\delta^{ij}}{n-1} \text{vol}(S^{n-2}).$$

□

Problem 1.6

Consider an Einstein metric (N^{n-1}, g_N) with $\text{Ric} = \frac{n-2}{n-1} \lambda g_N, \lambda < 0$. Find a $\rho : \mathbb{R} \rightarrow (0, \infty)$ such that $(M^n, g) = (\mathbb{R} \times N, dr^2 + \rho^2(r) g_N)$ becomes an Einstein metric with $\text{Ric} = \lambda g$.

Proof. Step 1. Compute the Ricci curvature of M .

Fix $(r_0, p) \in M$. Let $(x^0, x^1, \dots, x^{n-1})$ be local coordinates near (r_0, p) on M , where x^0 is the canonical coordinate on $\mathbb{R}$ and $(x^1, \dots, x^{n-1})$ are normal coordinates at p on (N, g_N) . Let $\Gamma_{ij}^k(0 \leq i, j, k \leq n-1)$ denote the Christoffel symbols on M , and let $\tilde{\Gamma}_{ij}^k(1 \leq i, j, k \leq n-1)$ denote the Christoffel symbols on N . Then for $1 \leq i, j, l \leq n-1$ we have

$$\begin{aligned}
\Gamma_{ij}^0(r, q) &= -\rho(r) \rho'(r) (g_N)_{ij}(q), \\
\Gamma_{i0}^j(r, q) &= \frac{\rho'(r)}{\rho(r)} \delta_i^j, \\
\Gamma_{ij}^l(r, q) &= \tilde{\Gamma}_{ij}^l(q)
\end{aligned}$$

and $\Gamma_{00}^i(r, q) = \Gamma_{00}^0(r, q) = \Gamma_{0i}^0(r, q) = 0$. Also note $\tilde{\Gamma}_{ij}^l(p) = 0$.

Hence for $1 \leq i, j \leq n-1$ at (r_0, p) we have

$$\begin{aligned}
R^M \left(\frac{\partial}{\partial x^0}, \frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^0} \right) &= g \left(\nabla_0 \nabla_i \frac{\partial}{\partial x^j} - \nabla_i \nabla_0 \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^0} \right) \\
&= g \left(\nabla_0 \left(-\rho \rho' (g_N)_{ij} \frac{\partial}{\partial x^0} + \sum_{l=1}^{n-1} \tilde{\Gamma}_{ij}^l \frac{\partial}{\partial x^l} \right) - \nabla_i \left(\frac{\rho'}{\rho} \frac{\partial}{\partial x^j} \right), \frac{\partial}{\partial x^0} \right) \\
&= -((\rho')^2 + \rho \rho'') \delta_{ij} + (\rho')^2 \delta_{ij} \\
&= -\rho \rho'' \delta_{ij},
\end{aligned}$$

and at (r_0, p) we have

$$\begin{aligned}
& \frac{1}{\rho^2} \sum_{k=1}^{n-1} R^M \left(\frac{\partial}{\partial x^k}, \frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k} \right) = \frac{1}{\rho^2} \sum_{k=1}^{n-1} g \left(\nabla_k \nabla_i \frac{\partial}{\partial x^j} - \nabla_i \nabla_k \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k} \right) \\
& = \frac{1}{\rho^2} \sum_{k=1}^{n-1} g \left(\nabla_k \left(-\rho \rho' (g_N)_{ij} \frac{\partial}{\partial x^0} + \sum_{l=1}^{n-1} \tilde{\Gamma}_{ij}^l \frac{\partial}{\partial x^l} \right) - \nabla_i \left(-\rho \rho' (g_N)_{kj} \frac{\partial}{\partial x^0} + \sum_{l=1}^{n-1} \tilde{\Gamma}_{kj}^l \frac{\partial}{\partial x^l} \right), \frac{\partial}{\partial x^k} \right) \\
& = \frac{1}{\rho^2} \sum_{k=1}^{n-1} g \left(R^N \left(\frac{\partial}{\partial x^k}, \frac{\partial}{\partial x^i} \right) \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k} \right) + \sum_{k=1}^{n-1} (-(\rho')^2 \delta_{ij} + (\rho')^2 \delta_{ik} \delta_{kj}) \\
& = Ric^N \left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j} \right) - (n-2)(\rho')^2 \delta_{ij} = \frac{n-2}{n-1} \lambda \delta_{ij} - (n-2)(\rho')^2 \delta_{ij}.
\end{aligned}$$

It follows that

$$\begin{aligned}
Ric^M \left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j} \right) \Big|_{(r_0, p)} &= R^M \left(\frac{\partial}{\partial x^0}, \frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^0} \right) + \frac{1}{\rho^2} \sum_{k=1}^{n-1} R^M \left(\frac{\partial}{\partial x^k}, \frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k} \right) \\
&= \left(-\rho \rho'' - (n-2)(\rho')^2 + \frac{n-2}{n-1} \lambda \right) \delta_{ij}.
\end{aligned}$$

Similarly one has

$$Ric^M \left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^0} \right) \Big|_{(r_0, p)} = 0,$$

and

$$Ric^M \left(\frac{\partial}{\partial x^0}, \frac{\partial}{\partial x^0} \right) \Big|_{(r_0, p)} = -(n-1) \frac{\rho''}{\rho}.$$

Step 2. Find the differential equation for $\rho(r)$ and solve it.

We want $Ric_{ij}^M = \lambda \rho^2 \delta_{ij}$, $Ric_{i0}^M = 0$, $Ric_{00}^M = \lambda$ for $1 \leq i, j \leq n-1$. This is equivalent to

$$-(n-1) \frac{\rho''}{\rho} = \lambda, \tag{1.1}$$

$$-\rho \rho'' - (n-2)(\rho')^2 + \frac{n-2}{n-1} \lambda = \lambda \rho^2. \tag{1.2}$$

We can solve (1.1) and obtain

$$\rho(r) = C_1 \exp \left(\sqrt{\frac{-\lambda}{n-1}} r \right) + C_2 \exp \left(-\sqrt{\frac{-\lambda}{n-1}} r \right).$$

Plugging in (1.2), we have

$$\frac{n-2}{n-1} \lambda \left(1 - \rho^2 + \frac{n-1}{-\lambda} (\rho')^2 \right) = 0,$$

that is $4C_1 C_2 = 1$. We may choose $C_1 = C_2 = 1/2$, hence $\rho(r) = \cosh \left(\sqrt{\frac{-\lambda}{n-1}} r \right)$. □

Problem 1.7

Let $T_{\mathbb{C}}M = TM \otimes \mathbb{C}$ be the complexified tangent bundle to a manifold. A vector $v \in T_{\mathbb{C}}M$ looks like $v = v_1 + \sqrt{-1}v_2$, where $v_1, v_2 \in TM$, and can be conjugated $\bar{v} = v_1 - iv_2$. Any tensorial object on TM can be complexified. For example, if S is a $(1, 1)$ -tensor, then its complexification is given by

$$S_{\mathbb{C}}(v) = S_{\mathbb{C}}(v_1 + \sqrt{-1}v_2) = S(v_1) + \sqrt{-1}S(v_2).$$

A Riemannian structure g on TM gives a natural Hermitian structure on $T_{\mathbb{C}}M$ by

$$\begin{aligned} g(v, w) &= g_{\mathbb{C}}(v, \bar{w}) = g_{\mathbb{C}}(v_1 + \sqrt{-1}v_2, w_1 - \sqrt{-1}w_2) \\ &= g(v_1, w_1) + g(v_2, w_2) + \sqrt{-1}(g(v_2, w_1) - g(v_1, w_2)). \end{aligned}$$

A vector is called isotropic if it is Hermitian orthogonal to its conjugate

$$\begin{aligned} 0 &= g(v, \bar{v}) = g_{\mathbb{C}}(v, v) = g_{\mathbb{C}}(v_1 + \sqrt{-1}v_2, v_1 + \sqrt{-1}v_2) \\ &= g(v_1, v_1) - g(v_2, v_2) + \sqrt{-1}(g(v_2, v_1) + g(v_1, v_2)). \end{aligned}$$

More generally, isotropic subspaces are defined as subspaces on which $g_{\mathbb{C}}$ vanishes. The complex sectional curvature spanned by Hermitian orthonormal vectors v, w is given by the expression

$$R_{\mathbb{C}}(v, w, \bar{w}, \bar{v}).$$

It is called isotropic sectional curvature when v, w span an isotropic plane.

- (1) Show that a vector $v = v_1 + iv_2$ is isotropic if v_1, v_2 are orthogonal and have the same length.
- (2) An isotropic plane can be spanned by two Hermitian orthonormal vectors v, w that are isotropic. Show that if $v = v_1 + iv_2$ and $w = w_1 + iw_2$, then $\sqrt{2}v_1, \sqrt{2}v_2, \sqrt{2}w_1, \sqrt{2}w_2$ are orthonormal.
- (3) Show that $R_{\mathbb{C}}(v, w, \bar{w}, \bar{v})$ is always a real number.
- (4) Show that if the original metric is strictly quarter pinched, i.e., all sectional curvatures lie in an open interval of the form $(k/4, k)$ with $k > 0$, then the complex sectional curvatures are positive.
- (5) Show that the complex sectional curvatures are nonnegative (resp. positive) if the curvature operator is nonnegative (resp. positive).

Proof. 1. By the definition we know

$$\begin{aligned} v = v_1 + iv_2 \text{ is isotropic} \\ \iff g(v_1, v_1) - g(v_2, v_2) + 2\sqrt{-1}g(v_1, v_2) &= 0 \\ \iff g(v_1, v_1) = g(v_2, v_2) \text{ and } g(v_1, v_2) &= 0. \end{aligned}$$

2. Since $g_{\mathbb{C}}$ vanishes on $\text{span}\{v, w\}$, we know

$$g_{\mathbb{C}}(av + bw, av + bw) = 0 \quad \forall a, b \in \mathbb{C}$$

and hence $g_{\mathbb{C}}(v, v) = g_{\mathbb{C}}(w, w) = g_{\mathbb{C}}(v, w) = 0$. On the other hand, since v, w are Hermitian orthonormal, we know

$$g(v, w) = 0, \quad g(v, v) = g(w, w) = 1.$$

It follows that

$$\begin{aligned} g(v_1, v_1) - g(v_2, v_2) + \sqrt{-1}(g(v_2, v_1) + g(v_1, v_2)) &= 0, \\ g(w_1, w_1) - g(w_2, w_2) + \sqrt{-1}(g(w_2, w_1) + g(w_1, w_2)) &= 0, \\ g(v_1, w_1) + g(v_2, w_2) + \sqrt{-1}(g(v_2, w_1) - g(v_1, w_2)) &= 0, \\ g(v_1, w_1) - g(v_2, w_2) + \sqrt{-1}(g(v_2, w_1) + g(v_1, w_2)) &= 0, \\ g(v_1, v_1) + g(v_2, v_2) + \sqrt{-1}(g(v_2, v_1) - g(v_1, v_2)) &= 1, \\ g(w_1, w_1) + g(w_2, w_2) + \sqrt{-1}(g(w_2, w_1) - g(w_1, w_2)) &= 1. \end{aligned}$$

Hence $\sqrt{2}v_1, \sqrt{2}v_2, \sqrt{2}w_1, \sqrt{2}w_2$ are orthonormal.

3. Since $\overline{R_{\mathbb{C}}(v, w, \bar{w}, \bar{v})} = R_{\mathbb{C}}(\bar{v}, \bar{w}, w, v) = R_{\mathbb{C}}(v, w, \bar{w}, \bar{v})$ we know $R_{\mathbb{C}}(v, w, \bar{w}, \bar{v}) \in \mathbb{R}$.

4. It follows from Proposition 1.10.

5. Note that

$$\begin{aligned} f(R(X \wedge Y), Z \wedge W) &= R(X, Y, W, Z) \\ g(\mathcal{R}(x \wedge u - y \wedge v), x \wedge u - y \wedge v) &+ g(\mathcal{R}(x \wedge v + y \wedge u), x \wedge v + y \wedge u) \\ &= R(x, u, u, x) - R(x, u, v, y) - R(y, v, u, x) + R(y, v, v, y) \\ &\quad + R(x, v, v, x) + R(x, v, u, y) + R(y, u, v, x) + R(y, u, u, y) \\ &= R(z, w, \bar{w}, \bar{z}) \end{aligned}$$

Then the conclusion follows. □

Lemma 1.8

If there exist $Z, W \in T_p M \otimes \mathbb{C}$ with $\sec(Z, W) < 0$ then there exist $z, w \in T_p M \otimes \mathbb{C}$ such that

$$g(x, x) = 1 \geq g(y, y), \quad g(u, u) = 1 \geq g(v, v), \quad \sec(z, w) < 0$$

and that x, y, u, v are pairwise orthogonal, where $z = x + iy$ and $w = u + iv$.

Proof. 1. Define $f : (\mathbb{C}^m)^* \times (\mathbb{C}^m)^* \rightarrow \mathbb{R}$ by

$$f(Z, W) = \frac{R_{\mathbb{C}}(Z, W, \bar{W}, \bar{Z})}{g_{\mathbb{C}}(Z, \bar{Z})g_{\mathbb{C}}(W, \bar{W})},$$

where $\mathbb{C}^m$ is identified with $T_p M \otimes \mathbb{C}$ via an orthonormal basis $(e_j)_{j=1}^m$ of $T_p M$. Clearly, f is also defined on $\mathbb{CP}^{n-1} \times \mathbb{CP}^{n-1}$. Thus by compactness, the following conditional extremum problem admits a solution (z, w) :

$$g_{\mathbb{C}}(z, w) = 0, \quad f(z, w) = \min_{g_{\mathbb{C}}(Z, W)=0} f(Z, W).$$

2. By hypothesis, there exist $Z, W \in T_p M \otimes \mathbb{C}$ such that $R_{\mathbb{C}}(Z, W, \bar{W}, \bar{Z}) < 0$. If $g_{\mathbb{C}}(Z, Z) \neq 0$, we replace (Z, W) by $(Z, W - \frac{g_{\mathbb{C}}(Z, W)}{g_{\mathbb{C}}(Z, Z)} Z)$ while if both $g_{\mathbb{C}}(Z, Z) = g_{\mathbb{C}}(W, W) = 0$ we replace (Z, W) by $(Z + W, Z - W)$. Then by the formula

$$R_{\mathbb{C}}(aV + bW, cV + dW, \overline{cV + dW}, \overline{aV + bW}) = |ad - bc|^2 R_{\mathbb{C}}(V, W, \bar{W}, \bar{V})$$

we know $f(z, w) = \min_{g_{\mathbb{C}}(Z, W)=0} f(Z, W) < 0$.

3. Moreover, note that (z, w) is also the critical point of Lagrange multiplier: $F(Z, W) = f(Z, W) + \lambda |g_{\mathbb{C}}(Z, W)|^2$, $(Z, W) \in (\mathbb{C}^m)^* \times (\mathbb{C}^m)^*$. Therefore,

$$0 = \frac{\partial F}{\partial \bar{W}^j}(z, w) = \frac{\partial f}{\partial \bar{W}^j}(z, w) + \lambda \frac{\partial}{\partial \bar{W}^j} \Big|_{(z, w)} (|g_{\mathbb{C}}(Z, W)|^2).$$

A direct computation shows that

$$R_{\mathbb{C}}(z, w) \bar{z} + f(z, w) g(z, z) w = 0.$$

It follows that

$$0 = g_{\mathbb{C}}(R_{\mathbb{C}}(z, w) \bar{z} + f(z, w) g(z, z) w, \bar{z}) = f(z, w) g(z, z) g_{\mathbb{C}}(w, \bar{z}).$$

Therefore $g_{\mathbb{C}}(w, \bar{z}) = 0$ and hence $g(z, w) = 0$. To sum up, we found z, w such that

$$g(z, w) = g(z, \bar{w}) = 0 \quad \text{and} \quad \sec(z, w) < 0. \tag{1.3}$$

4. Now we replace (z, w) by $(\lambda z, \mu w)$ for some $\lambda, \mu \in \mathbb{C}^*$ such that (1.3) still holds and

$$g(x, y) = g(u, v) = 0, \quad g(x, x) = 1 \geq g(y, y), \quad g(u, u) = 1 \geq g(v, v),$$

where $z = x + iy$ and $w = u + iv$. Now, clearly (z, w) is as required. □

Lemma 1.9 (Berger)

Suppose that for any $X, Y \in T_p M$ and for some $k, \delta > 0$ we have

$$\frac{1+\delta}{4}k \leq \frac{R(X, Y, Y, X)}{|X \wedge Y|^2} \leq (1-\delta)k.$$

If $X, Y, U, V \in T_p M$ are linearly independent and

$$\Delta = \langle X, U \rangle \langle Y, V \rangle - \langle X, V \rangle \langle Y, U \rangle = 0,$$

then

$$\begin{aligned} 6|R(X, Y, U, V)| &\leq \frac{3-5\delta}{5-3\delta} (2R(X, Y) + 2R(U, V) \\ &\quad + R(X, V) + R(V, Y) + R(X, U) + R(U, Y)) \end{aligned}$$

where $R(X, Y) = R(X, Y, Y, X)$.

Proof. This is linear algebra. One can refer to [Yau91, Lemma 1]. □

Proposition 1.10

If the sectional curvature is $1/4$ pinched at p , in the sense that

$$\frac{k}{4} \leq \frac{R(X, Y, Y, X)}{|X \wedge Y|^2} \leq k \quad \forall X, Y \in T_p M$$

for some constant $k > 0$, then

$$R_{\mathbb{C}}(Z, W, \bar{W}, \bar{Z}) \geq 0 \quad \forall Z, W \in T_p M \otimes \mathbb{C}.$$

Moreover if

$$\frac{1+\delta}{4}k \leq \frac{R(X, Y, Y, X)}{|X \wedge Y|^2} \leq (1-\delta)k \quad \forall X, Y \in T_p M$$

for some $\delta > 0$ then there exists $\epsilon > 0$ such that

$$(R - \epsilon R_0)_{\mathbb{C}}(Z, W, \bar{W}, \bar{Z}) \geq 0 \quad \forall Z, W \in T_p M \otimes \mathbb{C},$$

where

$$\begin{aligned} (R_0)_{\mathbb{C}}(Z, W, \bar{W}, \bar{Z}) &= g_{\mathbb{C}}(Z, \bar{Z})g_{\mathbb{C}}(W, \bar{W}) - g_{\mathbb{C}}(Z, \bar{W})g_{\mathbb{C}}(W, \bar{Z}) \\ &= g(Z, Z)g(W, W) - g(Z, W)g(W, Z) = g(Z \wedge W, Z \wedge W). \end{aligned}$$

Proof. For the first assertion, suppose for contradiction that there exist $Z, W \in T_p M \otimes \mathbb{C}$ such that

$$R_{\mathbb{C}}(Z, W, \bar{W}, \bar{Z}) < 0.$$

Then by Lemma 1.8 we know there exist $z, w \in T_p M \otimes \mathbb{C}$ such that

$$g(x, x) = 1 \geq g(y, y), \quad g(u, u) = 1 \geq g(v, v), \quad R_{\mathbb{C}}(z, w, \bar{w}, \bar{z}) < 0,$$

and that x, y, u, v are pairwise orthogonal, where $z = x + iy$ and $w = u + iv$. Then via Bianchi identity and Lemma 1.9 we have

$$\begin{aligned}
R(z, w, \bar{w}, \bar{z}) &= R(x, u) + R(x, v) + R(y, v) + R(y, u) + 2R(x, v, u, y) - 2R(x, u, v, y) \\
&= R(x, u) + R(x, v) + R(y, v) + R(y, u) + 2R(x, y, u, v) \\
&\geq R(x, u) + R(x, v) + R(y, v) + R(y, u) \\
&\quad - \frac{1}{5}(2R(x, y) + 2R(u, v) + R(x, v) + R(v, y) + R(x, u) + R(u, y)) \\
&= \frac{4}{5}(R(x, u) + R(x, v) + R(y, u) + R(v, y)) - \frac{2}{5}(R(x, y) + R(u, v)) \\
&\geq \frac{k}{5}(1 + |y|^2 + |u|^2 + |v|^2|y|^2) - \frac{2k}{5}(|y|^2 + |v|^2) \\
&= \frac{k}{5}(1 - |v|^2)(1 - |y|^2) \geq 0.
\end{aligned}$$

A contradiction. Hence we get the first assertion.

For the second assertion, picking $0 < \epsilon < \delta k/4$ then $\tilde{R} = R - \epsilon R_0$ satisfies the weaker $1/4$ pinching condition. Applying the preceding proof to $\tilde{R}$, then the result follows. $\square$

2 HW2

Problem 2.1

Compute the Ricci curvature and scalar of $(\mathbb{R}^2, g_a = e^{a(x^2+y^2)}(dx^2 + dy^2))$. Find a such that $(\mathbb{R}^2, g_a)$ has negative Ricci curvature.

Solution. 1. Since $g_{11} = g_{22} = e^{a(x^2+y^2)}$ and $g_{12} = g_{21} = 0$, we know $g^{11} = g^{22} = e^{-a(x^2+y^2)}$ and $g^{12} = g^{21} = 0$. Therefore,

$$\begin{aligned}
\Gamma_{11}^1 &= \frac{1}{2}g^{11}\frac{\partial g_{11}}{\partial x} = ax, & \Gamma_{22}^2 &= ay, \\
\Gamma_{11}^2 &= \frac{1}{2}g^{22}\left(-\frac{\partial g_{11}}{\partial y}\right) = -ay, & \Gamma_{22}^1 &= -ax, \\
\Gamma_{12}^1 &= \Gamma_{21}^1 = \frac{1}{2}g^{11}\frac{\partial g_{11}}{\partial y} = ay, & \Gamma_{12}^2 &= \Gamma_{21}^2 = ax.
\end{aligned}$$

It follows that

$$R_{1221} = R_{122}g_{11} = \left(\frac{\partial \Gamma_{22}^1}{\partial x} - \frac{\partial \Gamma_{21}^1}{\partial y} - \Gamma_{p2}^1\Gamma_{21}^p + \Gamma_{p1}^1\Gamma_{22}^p\right)g_{11} = -2a \cdot e^{a(x^2+y^2)}.$$

Therefore,

$$R_{11} = g^{22}R_{1221} = -2a, \quad R_{22} = g^{11}R_{1221} = -2a, \quad R_{12} = R_{21} = g^{pq}R_{p12q} = 0$$

and hence

$$\text{Ric} = -2a(dx \otimes dx + dy \otimes dy). \quad (2.1)$$

It follows that

$$\text{scal} = g^{pq}R_{pq} = g^{11}R_{11} + g^{22}R_{22} = -4a \cdot e^{-a(x^2+y^2)}. \quad (2.2)$$

2. By formula (2.1) we know Ric is negative iff $a > 0$. $\square$

Problem 2.2

Let $f : (M, g) \rightarrow (N, h)$ be a smooth map and $\tilde{\nabla}$ be the affine connection on the vector bundle $T^*M \otimes f^*TN$ induced by Levi-Civita connections ∇^M and ∇^N . Then

$$B = \tilde{\nabla} df$$

where df is regarded as a smooth section in $\Gamma(M, T^*M \otimes f^*TN)$.

Proof. Note that $df = \frac{\partial f^\alpha}{\partial x^i} dx^i \otimes \hat{e}_\alpha \in \Gamma(M, T^*M \otimes f^*TN)$ where $\hat{e}_\alpha = f^* \frac{\partial}{\partial y^\alpha}$. Let $\hat{\nabla}$ be the pullback connection.

It follows that

$$\begin{aligned} (\tilde{\nabla} df) \left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j} \right) &= \left(\tilde{\nabla}_{\frac{\partial}{\partial x^i}} \left(\frac{\partial f^\alpha}{\partial x^k} dx^k \otimes \hat{e}_\alpha \right) \right) \left(\frac{\partial}{\partial x^j} \right) \\ &= \left(\frac{\partial^2 f^\alpha}{\partial x^i \partial x^k} dx^k \otimes \hat{e}_\alpha + \frac{\partial f^\alpha}{\partial x^k} \left(\nabla_{\frac{\partial}{\partial x^i}} dx^k \right) \otimes \hat{e}_\alpha + \frac{\partial f^\alpha}{\partial x^k} dx^k \otimes \hat{\nabla}_{\frac{\partial}{\partial x^i}} \hat{e}_\alpha \right) \left(\frac{\partial}{\partial x^j} \right) \\ &= \frac{\partial^2 f^\alpha}{\partial x^i \partial x^j} \hat{e}_\alpha - \frac{\partial f^\alpha}{\partial x^k} \Gamma_{ij}^k \hat{e}_\alpha + \frac{\partial f^\alpha}{\partial x^j} \frac{\partial f^\beta}{\partial x^i} \Gamma_{\beta\alpha}^\gamma \hat{e}_\gamma \\ &= \left(\frac{\partial^2 f^\alpha}{\partial x^i \partial x^j} + \frac{\partial f^\beta}{\partial x^i} \frac{\partial f^\gamma}{\partial x^j} \Gamma_{\beta\gamma}^\alpha - \Gamma_{ij}^k \frac{\partial f^\alpha}{\partial x^k} \right) \hat{e}_\alpha \\ &= B_{ij}^\alpha \hat{e}_\alpha \end{aligned}$$

where we use the fact that $\nabla_X(\theta \otimes \eta) = (\nabla_X \theta) \otimes \eta + \theta \otimes \nabla_X \eta$.

It follows that $B = \tilde{\nabla} df$ since these two tensors have the same components. $\square$

3 HW3

Problem 3.1

Let (M, g) be a Riemannian manifold and (U, φ, x^i) be any normal coordinate chart centered at $p \in M$. Show that around $p \in M$

$$\begin{aligned} g_{ij} &= \delta_{ij} - \frac{1}{3} \sum_{k,l} R_{iklj}(p) x^k x^l + O(|x|^3), \\ g^{ij} &= \delta_{ij} + \frac{1}{3} \sum_{k,l} R_{iklj}(p) x^k x^l + O(|x|^3). \end{aligned} \tag{3.1}$$

In particular, one has

$$\begin{aligned} \det(g_{ij}) &= 1 - \frac{1}{3} R_{ij}(p) x^i x^j + O(|x|^3), \\ \frac{\partial^2 g_{ij}}{\partial x^k \partial x^l}(p) &= \frac{1}{3} (R_{ikjl} + R_{iljk})(p). \end{aligned} \tag{3.2}$$

Proof. 1. Using Gauss Lemma 3.2, by taking derivatives of (3.12), we obtain

$$\begin{aligned} \frac{\partial g_{is}}{\partial x^k} x^s + g_{ik} &= \delta_k^i \Rightarrow \frac{\partial^2 g_{is}}{\partial x^j \partial x^k} x^s + \frac{\partial g_{ij}}{\partial x^k} + \frac{\partial g_{ik}}{\partial x^j} = 0 \\ &\Rightarrow \frac{\partial^3 g_{is}}{\partial x^l \partial x^j \partial x^k} x^s + \frac{\partial^2 g_{ij}}{\partial x^l \partial x^k} + \frac{\partial^2 g_{il}}{\partial x^j \partial x^k} + \frac{\partial^2 g_{ik}}{\partial x^l \partial x^j} = 0. \end{aligned}$$

Hence at p we have

$$\frac{\partial^2 g_{ij}}{\partial x^l \partial x^k}(p) + \frac{\partial^2 g_{il}}{\partial x^j \partial x^k}(p) + \frac{\partial^2 g_{ik}}{\partial x^l \partial x^j}(p) = 0. \tag{3.3}$$

Changing indices in (3.3), we obtain

$$\frac{\partial^2 g_{ji}}{\partial x^l \partial x^k}(p) + \frac{\partial^2 g_{jl}}{\partial x^i \partial x^k}(p) + \frac{\partial^2 g_{jk}}{\partial x^i \partial x^l}(p) = 0, \quad (3.4)$$

$$\frac{\partial^2 g_{kj}}{\partial x^i \partial x^l}(p) + \frac{\partial^2 g_{kl}}{\partial x^j \partial x^i}(p) + \frac{\partial^2 g_{ki}}{\partial x^j \partial x^l}(p) = 0, \quad (3.5)$$

$$\frac{\partial^2 g_{lj}}{\partial x^i \partial x^k}(p) + \frac{\partial^2 g_{li}}{\partial x^j \partial x^k}(p) + \frac{\partial^2 g_{lk}}{\partial x^j \partial x^i}(p) = 0. \quad (3.6)$$

Considering (3.3) + (3.4) - (3.5) - (3.6), we obtain

$$\frac{\partial^2 g_{ij}}{\partial x^l \partial x^k}(p) = \frac{\partial^2 g_{kl}}{\partial x^j \partial x^i}(p). \quad (3.7)$$

Recall that

$$R_{ijkl} = \frac{1}{2} \left(\frac{\partial^2 g_{jl}}{\partial x^i \partial x^k} + \frac{\partial^2 g_{ik}}{\partial x^j \partial x^l} - \frac{\partial^2 g_{il}}{\partial x^j \partial x^k} - \frac{\partial^2 g_{jk}}{\partial x^i \partial x^l} \right) + g_{pq}(\Gamma_{ik}^p \Gamma_{jl}^q - \Gamma_{il}^q \Gamma_{kj}^p).$$

Then at p , from (3.7) we know

$$\begin{aligned} R_{iklj}(p) &= \frac{\partial^2 g_{jk}}{\partial x^i \partial x^l}(p) - \frac{\partial^2 g_{kl}}{\partial x^i \partial x^j}(p), \\ R_{ilkj}(p) &= \frac{\partial^2 g_{jl}}{\partial x^i \partial x^k}(p) - \frac{\partial^2 g_{kl}}{\partial x^i \partial x^j}(p), \end{aligned}$$

and then by (3.4) and (3.7) we obtain

$$R_{iklj}(p) + R_{ilkj}(p) = -\frac{\partial^2 g_{jl}}{\partial x^i \partial x^k}(p) - 2\frac{\partial^2 g_{kl}}{\partial x^i \partial x^j}(p) = -3\frac{\partial^2 g_{ij}}{\partial x^k \partial x^l}(p).$$

Thus we get

$$\frac{\partial^2 g_{ij}}{\partial x^k \partial x^l}(p) = -\frac{1}{3}(R_{iklj} + R_{ilkj})(p) = \frac{1}{3}(R_{ikjl} + R_{iljk})(p). \quad (3.8)$$

Hence by Taylor expansion we get

$$g_{ij} = \delta_{ij} - \frac{1}{3} \sum_{k,l} R_{iklj}(p) x^k x^l + O(|x|^3). \quad (3.9)$$

2. Then we can get the other assertions by linear algebra. Or, using the facts that

$$\frac{\partial g^{ij}}{\partial x^k} = -g^{ip} g^{qj} \frac{\partial g_{pq}}{\partial x^k} \quad \text{and} \quad \frac{\partial}{\partial x^k} \log \det g = g^{ji} \frac{\partial g_{ij}}{\partial x^k}$$

we can compute the Taylor expansion of g^{ij} and $\det g$ at p directly.

An alternative proof of Formula (3.8). Set $\sigma_v(t) = (tv^1, \dots, tv^n)$. Then by the geodesic equation we know $\Gamma_{ij}^k(\sigma_v(t))v^i v^j = 0$. Thus

$$0 = \frac{d}{dt} \Big|_{t=0} \Gamma_{ij}^k(\sigma_v(t))v^i v^j = \frac{\partial \Gamma_{ij}^k}{\partial x^l}(p) v^i v^j v^l.$$

Since $\frac{\partial \Gamma_{ij}^k}{\partial x^l} = \frac{\partial \Gamma_{ji}^k}{\partial x^l}$, we know

$$0 = \sum_{i \leq j \leq l} \left(\frac{\partial \Gamma_{ij}^k}{\partial x^l} + \frac{\partial \Gamma_{jl}^k}{\partial x^i} + \frac{\partial \Gamma_{li}^k}{\partial x^j} \right) (p) v^i v^j v^l.$$

Therefore, we get

$$\frac{\partial \Gamma_{ij}^k}{\partial x^l}(p) + \frac{\partial \Gamma_{jl}^k}{\partial x^i}(p) + \frac{\partial \Gamma_{li}^k}{\partial x^j}(p) = 0 \quad \forall i, j, k, l. \quad (3.10)$$

Recall that

$$R_{ijkl} = g_{im} \left(\frac{\partial \Gamma_{jl}^m}{\partial x^k} - \frac{\partial \Gamma_{ik}^m}{\partial x^j} + \Gamma_{jl}^p \Gamma_{kp}^m - \Gamma_{ik}^p \Gamma_{jp}^m \right).$$

Hence we know

$$\begin{aligned} R_{kl ij}(p) + R_{lk ij}(p) &= \left(\frac{\partial \Gamma_{jl}^k}{\partial x^i}(p) - \frac{\partial \Gamma_{il}^k}{\partial x^j}(p) \right) + \left(\frac{\partial \Gamma_{jk}^l}{\partial x^i}(p) - \frac{\partial \Gamma_{ik}^l}{\partial x^j}(p) \right) \\ &= 2 \frac{\partial \Gamma_{kl}^j}{\partial x^i}(p) - \frac{\partial \Gamma_{li}^j}{\partial x^k}(p) - \frac{\partial \Gamma_{ik}^j}{\partial x^l}(p) \stackrel{(3.10)}{=} 3 \frac{\partial \Gamma_{kl}^j}{\partial x^i}(p). \end{aligned}$$

Therefore,

$$\begin{aligned} \frac{\partial^2 g_{ij}}{\partial x^l \partial x^k}(p) &= \frac{\partial}{\partial x^l} \Big|_p (\Gamma_{ki}^m g_{mj} + \Gamma_{kj}^m g_{mi}) = \frac{\partial \Gamma_{ki}^j}{\partial x^l}(p) + \frac{\partial \Gamma_{kj}^i}{\partial x^l}(p) \\ &= \frac{1}{3} (R_{iklj} + R_{il kj} + R_{jkil} + R_{jl ki})(p) = \frac{1}{3} (R_{il kj} + R_{jl ki})(p). \end{aligned}$$

Hence we get (3.8).

Another alternative proof of Formula (3.8). Set $\sigma_v(t) = (v^1 t, \dots, v^n t)$ for each v . Note that

$$\begin{aligned} (3.8) &\iff v^k v^l \frac{\partial^2 g_{ij}}{\partial x^k \partial x^l}(p) = \frac{2}{3} R_{ikjl}(p) v^k v^l \quad \forall v \\ &\iff g_{ij} \circ \sigma_v(t) = \delta_{ij} - \frac{1}{3} R_{iklj}(p) v^k v^l t^2 + O(t^3) \quad \forall v. \end{aligned}$$

Thus to prove (3.8) it suffices to show that

$$g_{ij} \circ \sigma_v(t) = \delta_{ij} - \frac{1}{3} R_{iklj}(p) v^k v^l t^2 + O(t^3) \quad \forall v. \quad (3.11)$$

Fix some v . For each i we set $V_i(t) = t \frac{\partial}{\partial x^i} \Big|_{\sigma_v(t)}$. Then each V_i is a Jacobi field along γ , so

$$\hat{\nabla}_{\frac{d}{dt}} \hat{\nabla}_{\frac{d}{dt}} V_i + R(V_i, \gamma') \gamma' = 0.$$

Set $u(t) = \hat{g}(V_i(t), V_j(t))$. Then $u(t) = t^2 \cdot g_{ij} \circ \sigma_v(t)$ and we easily get

$$u(0) = 0, \quad u'(0) = 0, \quad u''(0) = 2\delta_{ij}, \quad u^{(3)}(0) = 0, \quad u^{(4)}(0) = 8R(v, e_i, v, e_j)(p).$$

Then we get (3.11), which indicates our conclusion. $\square$

Lemma 3.2

Let (M, g) be a Riemannian manifold and let (U, φ, x^i) be a normal coordinate chart centered at $p \in M$. Then in U we have

$$g_{ij} x^j = x^i. \quad (3.12)$$

Proof. Set $\sigma_v(t) = (tv^1, \dots, tv^n)$. Then by the geodesic equation we know

$$\Gamma_{ij}^k(\sigma_v(t)) v^i v^j = 0.$$

Multiplying t^2 on both sides of the above equation, then by the arbitrariness of v we know

$$\Gamma_{ij}^k x^i x^j = 0. \quad (3.13)$$

It follows that $g_{kl}\Gamma_{ij}^l x^i x^j = 0$. Then we know

$$\begin{aligned}
g_{kl}\Gamma_{ij}^l x^i x^j = 0 &\Rightarrow \frac{1}{2} \left(\frac{\partial g_{ik}}{\partial x^j} + \frac{\partial g_{jk}}{\partial x^i} - \frac{\partial g_{ij}}{\partial x^k} \right) x^i x^j = 0 \\
&\Rightarrow \left(\frac{\partial g_{ik}}{\partial x^j} - \frac{1}{2} \frac{\partial g_{ij}}{\partial x^k} \right) x^i x^j = 0 \\
&\Rightarrow \frac{\partial g_{ik}}{\partial x^j} x^i x^j = \frac{1}{2} \frac{\partial g_{ij}}{\partial x^k} x^i x^j = \frac{1}{2} \frac{\partial}{\partial x^k} (g_{ij} x^i x^j) - g_{ik} x^i \\
&\Rightarrow \frac{\partial}{\partial x^j} (g_{ik} x^i) x^j = \frac{1}{2} \frac{\partial}{\partial x^k} (g_{ij} x^i x^j).
\end{aligned}$$

Since $|\sigma'_v(t)|_g \equiv 1$ we know

$$g_{ij}(\sigma_v(t)) v^i v^j = \sum_{i=1}^n v^i v^i.$$

Multiplying t^2 on both sides of the above equation, then by the arbitrariness of v we know

$$g_{ij} x^i x^j = \sum_{i=1}^n x^i x^i. \quad (3.14)$$

Then we know

$$\frac{\partial}{\partial x^j} (g_{ik} x^i) x^j = \frac{1}{2} \frac{\partial}{\partial x^k} \left(\sum_{i=1}^n x^i x^i \right) = x^k \Rightarrow \frac{\partial}{\partial x^j} (g_{ik} x^i - x^k) x^j = 0$$

and hence

$$\frac{d}{dt} (g_{ik} x^i - x^k) (\sigma_v(t)) = 0 \quad \forall v.$$

Thus we get (3.12).

An alternative proof of Lemma 3.2. Consider the distance function $r(x) = d(p, x)$. Then

$$r(x) = \sqrt{\sum_{i=1}^n (x^i)^2}. \quad (3.15)$$

Consider the vector field ∂_r on U :

$$\partial_r = \sum_{i=1}^n \frac{x^i}{r} \frac{\partial}{\partial x^i} = \sum_{i=1}^n \frac{\partial r}{\partial x^i} \frac{\partial}{\partial x^i};$$

Note the following:

- (1) $\partial_r r = 1 = |\partial_r|^2$.
- (2) (Gauss Lemma) ∂_r is perpendicular to the level set of r .

Hence, $\partial_r = \nabla r$ on $U \setminus \{p\}$. It follows that

$$g^{ij} \frac{x^i}{r} \frac{\partial}{\partial x^j} = g^{ij} \frac{\partial r}{\partial x^i} \frac{\partial}{\partial x^j} = \nabla r = \partial_r = \frac{x^j}{r} \frac{\partial}{\partial x^j}. \quad (3.16)$$

Therefore, we know

$$g^{ij} x^i = x^j.$$

Then (3.12) follows. □

Problem 3.3

Let (M, g) be a Riemannian manifold of dimension $n > 2$. Define the conformal Laplacian $L : C^\infty(M) \rightarrow C^\infty(M)$ by the formula

$$Lu = -\frac{4(n-1)}{(n-2)}\Delta u + Su,$$

where Δ is the Laplace-Beltrami operator of g and S is its scalar curvature. Prove that if $\tilde{g} = e^{2f}g$ for some $f \in C^\infty(M)$, and $\tilde{L}$ denoted the conformal Laplacian with respect to $\tilde{g}$, then for every $u \in C^\infty(M)$

$$e^{\frac{n+2}{2}f}\tilde{L}u = L\left(e^{\frac{n-2}{2}f}u\right). \quad (3.17)$$

Conclude that a metric $\tilde{g}$ conformal to g has constant scalar curvature λ if and only if it can be expressed in the form $\tilde{g} = \varphi^{4/(n-2)}g$, where φ is a smooth positive solution to the Yamabe equation:

$$L\varphi = \lambda\varphi^{\frac{n+2}{n-2}}. \quad (3.18)$$

Proof. By $\tilde{g}_{ij} = e^{2f}g_{ij}$ we know $\tilde{g}^{ij} = e^{-2f}g^{ij}$, and hence

$$\tilde{\Gamma}_{ij}^k = \frac{1}{2}\tilde{g}^{kl}\left(\frac{\partial\tilde{g}_{il}}{\partial x^j} + \frac{\partial\tilde{g}_{jl}}{\partial x^i} - \frac{\partial\tilde{g}_{ij}}{\partial x^l}\right) = \Gamma_{ij}^k + \frac{\partial f}{\partial x^j}\delta_i^k + \frac{\partial f}{\partial x^i}\delta_j^k - g^{kl}\frac{\partial f}{\partial x^l}g_{ij}. \quad (3.19)$$

Note that

$$\Delta\varphi = g^{ij}\left(\frac{\partial^2\varphi}{\partial x^i\partial x^j} - \Gamma_{ij}^k\frac{\partial\varphi}{\partial x^k}\right) \quad \forall \varphi \in C^\infty(M), \quad (3.20)$$

$$\Delta(\varphi\psi) = \varphi\Delta\psi + \psi\Delta\varphi + 2\langle d\varphi, d\psi \rangle_g \quad \forall \varphi, \psi \in C^\infty(M). \quad (3.21)$$

Thus we have

$$\begin{aligned} \Delta e^{\frac{n-2}{2}f} &\stackrel{(3.20)}{=} g^{ij}\left(\frac{\partial^2 e^{\frac{n-2}{2}f}}{\partial x^i\partial x^j} - \Gamma_{ij}^k\frac{\partial e^{\frac{n-2}{2}f}}{\partial x^k}\right) \\ &= \frac{n-2}{2}g^{ij}\left(\frac{\partial}{\partial x^i}\left(\frac{\partial f}{\partial x^j}e^{\frac{n-2}{2}f}\right) - \Gamma_{ij}^ke^{\frac{n-2}{2}f}\frac{\partial f}{\partial x^k}\right) \\ &= \frac{n-2}{2}e^{\frac{n-2}{2}f}g^{ij}\left(\frac{\partial^2 f}{\partial x^i\partial x^j} + \frac{n-2}{2}\frac{\partial f}{\partial x^i}\frac{\partial f}{\partial x^j} - \Gamma_{ij}^k\frac{\partial f}{\partial x^k}\right) \\ &\stackrel{(3.20)}{=} \frac{n-2}{2}e^{\frac{n-2}{2}f}\Delta f + \frac{(n-2)^2}{4}e^{\frac{n-2}{2}f}|df|_g^2. \end{aligned}$$

Then by the following formula

$$\langle de^{\frac{n-2}{2}f}, du \rangle_g = g^{ij}\frac{\partial e^{\frac{n-2}{2}f}}{\partial x^i}\frac{\partial u}{\partial x^j} = \frac{n-2}{2}e^{\frac{n-2}{2}f}g^{ij}\frac{\partial f}{\partial x^i}\frac{\partial u}{\partial x^j} = \frac{n-2}{2}e^{\frac{n-2}{2}f}\langle df, du \rangle_g,$$

we know

$$\begin{aligned} \Delta\left(e^{\frac{n-2}{2}f}u\right) &\stackrel{(3.21)}{=} e^{\frac{n-2}{2}f}\Delta u + u\Delta e^{\frac{n-2}{2}f} + 2\langle de^{\frac{n-2}{2}f}, du \rangle_g \\ &= e^{\frac{n-2}{2}f}\left(\Delta u + \frac{n-2}{2}u\Delta f + \frac{(n-2)^2}{4}u|df|_g^2 + (n-2)\langle df, du \rangle_g\right). \end{aligned}$$

Therefore, we get

$$\begin{aligned} L\left(e^{\frac{n-2}{2}f}u\right) &= -\frac{4(n-1)}{(n-2)}\Delta\left(e^{\frac{n-2}{2}f}u\right) + Se^{\frac{n-2}{2}f}u \\ &= e^{\frac{n-2}{2}f}\left(-\frac{4(n-1)}{n-2}\Delta u - 2(n-1)u\Delta f + 2\langle df, du \rangle_g - (n-1)(n-2)u|df|_g^2 + Su\right). \end{aligned}$$

On the other hand, we have

$$\begin{aligned}
\Delta u - e^{2f} \tilde{\Delta} u &\stackrel{(3.20)}{=} g^{ij} \left(\frac{\partial^2 u}{\partial x^i \partial x^j} - \Gamma_{ij}^k \frac{\partial u}{\partial x^k} \right) - e^{2f} \tilde{g}^{ij} \left(\frac{\partial^2 u}{\partial x^i \partial x^j} - \tilde{\Gamma}_{ij}^k \frac{\partial u}{\partial x^k} \right) \\
&= -g^{ij} (\Gamma_{ij}^k - \tilde{\Gamma}_{ij}^k) \frac{\partial u}{\partial x^k} \\
&\stackrel{(3.19)}{=} g^{ij} \left(\frac{\partial f}{\partial x^j} \delta_i^k + \frac{\partial f}{\partial x^i} \delta_j^k - g^{kl} \frac{\partial f}{\partial x^l} g_{ij} \right) \frac{\partial u}{\partial x^k} \\
&= g^{jk} \frac{\partial f}{\partial x^j} \frac{\partial u}{\partial x^k} + g^{ik} \frac{\partial f}{\partial x^i} \frac{\partial u}{\partial x^k} - n g^{kl} \frac{\partial f}{\partial x^l} \frac{\partial u}{\partial x^k} = -(n-2) \langle df, du \rangle_g.
\end{aligned}$$

Then by Lemma 3.4, we know

$$\begin{aligned}
e^{\frac{n+2}{2}f} \tilde{L}u &= e^{\frac{n+2}{2}f} \left(-\frac{4(n-1)}{(n-2)} \tilde{\Delta}u + \tilde{S}u \right) \\
&= e^{\frac{n+2}{2}f} \left(-\frac{4(n-1)}{n-2} \Delta u - 2(n-1)(u\Delta f + 2\langle df, du \rangle_g) - (n-1)(n-2)u|df|_g^2 + Su \right).
\end{aligned}$$

Hence we get (3.17).

Suppose $\tilde{g} = e^{2f}g$ has constant scalar curvature λ . Set $\varphi = e^{\frac{n-2}{2}f}$. Then with $u = 1$, formula (3.17) yields that

$$L\varphi = L \left(e^{\frac{n-2}{2}f} \right) = e^{\frac{n+2}{2}f} \tilde{L}u = \tilde{S}\varphi^{\frac{n+2}{n-2}} = \lambda \varphi^{\frac{n+2}{n-2}}.$$

Then suppose $\tilde{g} = \varphi^{4/(n-2)}g$ for some positive smooth solution φ to (3.18). Setting $f = \frac{2}{n-2} \ln \varphi$, then $\tilde{g} = e^{2f}g$. With $u = 1$, formula (3.17) yields that

$$\lambda \varphi^{\frac{n+2}{n-2}} = L\varphi = L \left(e^{\frac{n-2}{2}f} \right) = e^{\frac{n+2}{2}f} \tilde{L}u = \tilde{S}\varphi^{\frac{n+2}{n-2}}.$$

Hence $\tilde{g}$ has constant scalar curvature λ . Therefore, we get the second assertion. $\square$

Lemma 3.4

Let (M, g) be a Riemannian manifold of dimension $n > 2$. Suppose $\tilde{g} = e^{2f}g$ for some $f \in C^\infty(M)$. Then

$$\tilde{S} = e^{-2f} (S - 2(n-1)\Delta f - (n-1)(n-2)|df|_g^2).$$

Proof. For the sake of convenience we write

$$(h \otimes k)_{ijkl} = h_{il}k_{jk} + h_{jk}k_{il} - h_{ik}k_{jl} - h_{jl}k_{ik}$$

By $\tilde{g}_{ij} = e^{2f}g_{ij}$ we know $\tilde{g}^{ij} = e^{-2f}g^{ij}$, and hence

$$\tilde{\Gamma}_{ij}^k = \frac{1}{2} \tilde{g}^{kl} \left(\frac{\partial \tilde{g}_{il}}{\partial x^j} + \frac{\partial \tilde{g}_{jl}}{\partial x^i} - \frac{\partial \tilde{g}_{ij}}{\partial x^l} \right) = \Gamma_{ij}^k + \frac{\partial f}{\partial x^j} \delta_i^k + \frac{\partial f}{\partial x^i} \delta_j^k - g^{ks} \frac{\partial f}{\partial x^s} g_{ij}.$$

We then have

$$\begin{aligned}
\frac{\partial \tilde{\Gamma}_{ij}^k}{\partial x^l} &= \frac{\partial \Gamma_{ij}^k}{\partial x^l} + \frac{\partial^2 f}{\partial x^l \partial x^j} \delta_i^k + \frac{\partial^2 f}{\partial x^l \partial x^i} \delta_j^k \\
&\quad + g^{kp} g^{qs} \frac{\partial g_{pq}}{\partial x^l} \frac{\partial f}{\partial x^s} g_{ij} - g^{ks} \frac{\partial^2 f}{\partial x^l \partial x^s} g_{ij} - g^{ks} \frac{\partial f}{\partial x^s} \frac{\partial g_{ij}}{\partial x^l}.
\end{aligned}$$

Then by the formula

$$\tilde{R}_{ijkl} = e^{2f} g_{lm} \left(\frac{\partial \tilde{\Gamma}_{jk}^m}{\partial x^i} - \frac{\partial \tilde{\Gamma}_{ik}^m}{\partial x^j} + \tilde{\Gamma}_{jk}^p \tilde{\Gamma}_{ip}^m - \tilde{\Gamma}_{ik}^p \tilde{\Gamma}_{jp}^m \right)$$

we know

$$\tilde{R}_{ijkl} = e^{2f} R_{ijkl} + e^{2f} \left((df \otimes df) \otimes g - (\nabla^2 f) \otimes g - \frac{1}{2} |df|_g^2 \cdot g \otimes g \right)_{ijkl}.$$

Now by the formula

$$g^{il} g^{jk} (h \otimes g)_{ijkl} = 2(n-1) \text{tr}_g h$$

we know

$$\tilde{S} = e^{-4f} g^{il} g^{jk} \tilde{R}_{ijkl} = e^{-2f} S - e^{-2f} (2(n-1) \Delta f + (n-1)(n-2) |df|_g^2).$$

Thus we get the conclusion. $\square$

4 HW4

Lemma 4.1

Let G be a Lie group. Set

$$C_g : G \rightarrow G, \quad h \mapsto ghg^{-1}, \quad (4.1)$$

and

$$Ad_g = (C_g)_{*e} : T_e G \rightarrow T_e G. \quad (4.2)$$

Then the map

$$Ad : G \rightarrow Aut(\mathfrak{g}), \quad g \mapsto Ad_g \quad (4.3)$$

is a Lie group homomorphism, and induces a Lie algebra homomorphism

$$ad = Ad_{*e} : \mathfrak{g} \rightarrow gl(\mathfrak{g}). \quad (4.4)$$

Write $ad_X = ad(X) = Ad_{*e}(X)$. Then

$$ad_X(Y) = [X, Y] = \frac{d}{dt} \Big|_{t=0} (C_{\exp(tX)})_{*e}(Y). \quad (4.5)$$

Proof. We only prove (4.5), and the rest is trivial. For any $X, Y \in \mathfrak{g}$ we have

$$ad_X(Y) = Ad_{*e}(X)(Y) = \frac{d}{dt} \Big|_{t=0} Ad_{\exp(tX)}(Y) = \frac{d}{dt} \Big|_{t=0} (C_{\exp(tX)})_{*e}(Y).$$

Extend X and Y to left-invariant vector fields $\tilde{X}$ and $\tilde{Y}$. Since

$$\frac{d}{dt} \Big|_{t=0} g \cdot \exp(tX) = (L_g)_* \frac{d}{dt} \Big|_{t=0} \exp(tX) = (L_g)_*(X_e) = \tilde{X}_g,$$

we know $\theta_t = R_{\exp(tX)}$ is the flow generated by $\tilde{X}$. Hence we compute the Lie derivative

$$\begin{aligned} [X, Y] &= (\mathcal{L}_{\tilde{X}} \tilde{Y})_e = \lim_{t \rightarrow 0} \frac{1}{t} [(\theta_{-t})_*(\tilde{Y}_{\theta_t(e)}) - Y_e] \\ &= \lim_{t \rightarrow 0} \frac{1}{t} [(R_{\exp(-tX)})_*(L_{\exp(tX)})_*(Y_e) - Y_e] \\ &= \lim_{t \rightarrow 0} \frac{1}{t} [(R_{\exp(-tX)} \circ L_{\exp(tX)})_*(Y_e) - Y_e] \\ &= \lim_{t \rightarrow 0} \frac{1}{t} [(C_{\exp(tX)})_{*e}(Y_e) - Y_e] \\ &= \frac{d}{dt} \Big|_{t=0} (C_{\exp(tX)})_{*e}(Y_e). \end{aligned}$$

Thus we get (4.5). $\square$

Problem 4.2

If h is a bi-invariant metric on a Lie group G , show that for all left-invariant vector fields X, Y, Z we have

$$h([X, Y], Z) = h(X, [Y, Z]). \quad (4.6)$$

Proof. Since X, Y and Z are left-invariant, we know

$$\begin{aligned} (L_g)_*[X, Y] &= [(L_g)_*X, (L_g)_*Y] = [X, Y], \\ (L_g)_*[Y, Z] &= [(L_g)_*Y, (L_g)_*Z] = [Y, Z]. \end{aligned}$$

Since h is also left-invariant, it suffices to verify (4.6) at point e . Since h is bi-invariant,

$$h(X_e, Z_e) = h((C_{\exp(tY)})_*X_e, (C_{\exp(tY)})_*Z_e) \quad \forall t \in \mathbb{R}.$$

Then by (4.5) we know

$$\begin{aligned} 0 &= \frac{d}{dt} \Big|_{t=0} h((C_{\exp(tY)})_*X_e, (C_{\exp(tY)})_*Z_e) \\ &= h \left(\frac{d}{dt} \Big|_{t=0} (C_{\exp(tY)})_*X_e, Z_e \right) + h \left(X_e, \frac{d}{dt} \Big|_{t=0} (C_{\exp(tY)})_*Z_e \right) \\ &= h([Y_e, X_e], Z_e) + h(X_e, [Y_e, Z_e]). \end{aligned}$$

Hence we get the conclusion.

An Alternative Proof. By definition we know

$$(\mathcal{L}_Y h)(X, Z) = Y(h(X, Z)) - h([Y, X], Z) - h(X, [Y, Z]).$$

Since X, Z and h are left-invariant, $h(X, Z)$ is constant, and hence $Y(h(X, Z)) = 0$. Since the flow of Y is given by $\theta_t = R_{\exp(tY)}$ and h is right-invariant, we know

$$\mathcal{L}_Y h = \frac{d}{dt} \Big|_{t=0} R_{\exp(tY)}^* h = 0. \quad (4.7)$$

Then the conclusion follows. $\square$

Problem 4.3

Let G be a Lie group and $\mathfrak{g}$ be its Lie algebra. Suppose $\langle \bullet, \bullet \rangle_e$ is an inner product on $\mathfrak{g}$. For any $g \in G$ and $X_g, Y_g \in T_g G$ define

$$\langle X_g, Y_g \rangle = \langle (L_{g^{-1}})_*X_g, (L_{g^{-1}})_*Y_g \rangle_e \quad (4.8)$$

- (1) $\langle \bullet, \bullet \rangle$ is a left-invariant Riemannian metric on G .
- (2) There is a bijection $\{\text{Inner products on } \mathfrak{g}\} \longleftrightarrow \{\text{Left-invariant Riemannian metrics on } G\}$.
- (3) Under the above bijection, $Ad(G)$ -invariant inner products on $\mathfrak{g}$ correspond to bi-invariant Riemannian metrics on G .

Proof. 1. Denote the Riemannian metric $\langle \bullet, \bullet \rangle$ by h . For any $g, g' \in G$ we have

$$\begin{aligned} (L_g^* h)(X_{g'}, Y_{g'}) &= \langle (L_g)_*X_{g'}, (L_g)_*Y_{g'} \rangle \\ &= \langle (L_{(gg')^{-1}})_*(L_g)_*X_{g'}, (L_{(gg')^{-1}})_*(L_g)_*Y_{g'} \rangle_e \\ &= \langle (L_{(g')^{-1}})_*X_{g'}, (L_{(g')^{-1}})_*Y_{g'} \rangle_e = h(X_{g'}, Y_{g'}). \end{aligned}$$

Hence $\langle \bullet, \bullet \rangle$ is left-invariant.

2. For each inner product $\langle \bullet, \bullet \rangle_e$ on $\mathfrak{g}$, it induces a left-invariant Riemannian metric h on G by (4.8) with $h_e = \langle \bullet, \bullet \rangle_e$. For each left-invariant Riemannian metric h on G , h_e is an inner product on $\mathfrak{g}$. Since h is left-invariant, for each $g \in G$ we know

$$h(X_g, Y_g) = (L_{g^{-1}}^* h)(X_g, Y_g) = h_e((L_{g^{-1}})_* X_g, (L_{g^{-1}})_* Y_g).$$

Therefore, each left-invariant Riemannian metric h on G is determined by h_e , and hence the above correspondence is one-to-one.

3. Suppose h is a left-invariant metric. Then $C_g^* h = h$ for all $g \in G$, and hence

$$h_e((C_g)_* X, (C_g)_* Y) = h_e(X, Y) \quad \forall g \in G, \forall X, Y \in \mathfrak{g}.$$

Note that $Ad_g = (C_g)_*$. Hence h_e is $Ad(G)$ -invariant.

On the other hand, suppose $\langle \bullet, \bullet \rangle_e$ is an $Ad(G)$ -invariant inner product on $\mathfrak{g}$. Then $\langle \bullet, \bullet \rangle_e$ induces a left-invariant Riemannian metric $\langle \bullet, \bullet \rangle$ on G by (4.8). Since $\langle \bullet, \bullet \rangle_e$ is $Ad(G)$ -invariant, for all $g, g' \in G$ and for all $X_{g'}, Y_{g'} \in T_{g'} G$ we have

$$\begin{aligned} \langle (R_{g^{-1}})_* X_{g'}, (R_{g^{-1}})_* Y_{g'} \rangle &= \langle (L_{g(g')^{-1}})_* (R_{g^{-1}})_* X_{g'}, (L_{g(g')^{-1}})_* (R_{g^{-1}})_* Y_{g'} \rangle_e \\ &= \langle (C_g)_* (L_{(g')^{-1}})_* X_{g'}, (C_g)_* (L_{(g')^{-1}})_* Y_{g'} \rangle_e \\ &= \langle (L_{(g')^{-1}})_* X_{g'}, (L_{(g')^{-1}})_* Y_{g'} \rangle_e = \langle X_{g'}, Y_{g'} \rangle. \end{aligned}$$

Hence $\langle \bullet, \bullet \rangle$ is right-invariant and hence bi-invariant. Thus we get the conclusion. $\square$

Problem 4.4

Let G be a Lie group with a bi-invariant Riemannian metric h and ∇ be the Levi-Civita connection. Then for any left-invariant vector fields X, Y on G ,

$$\nabla_X Y = \frac{1}{2}[X, Y]. \quad (4.9)$$

Proof. Let X, Y, Z be left-invariant vector fields. Since h is left-invariant, $h(X, Y)$, $h(X, Z)$, $h(Y, Z)$ are constant, and hence

$$0 = Z(h(X, Y)) = h(\nabla_Z X, Y) + h(X, \nabla_Z Y), \quad (4.10)$$

$$0 = Y(h(X, Z)) = h(\nabla_Y X, Z) + h(X, \nabla_Y Z), \quad (4.11)$$

$$0 = X(h(Y, Z)) = h(\nabla_X Y, Z) + h(Y, \nabla_X Z). \quad (4.12)$$

Considering (4.12) + (4.11) - (4.10), we get

$$2h(\nabla_X Y, Z) = h([X, Y], Z) + h(Y, [Z, X]) - h([Y, Z], X).$$

By formula (4.6) we know $2h(\nabla_X Y, Z) = h([X, Y], Z)$. From the arbitrariness of X, Y, Z we get (4.9). $\square$

Problem 4.5

Let G be a Lie group with a bi-invariant Riemannian metric.

(1) The geodesics on G are precisely the integral curves of left-invariant vector fields.

(2) The exponential map for the Lie group coincides with the exponential map of the Levi-Civita connection.

Proof. 1. Let X be a left-invariant vector field, and γ be its integral curve starting at g . Then

$$\gamma'(0) = X_g \quad \text{and} \quad \hat{\nabla}_{\frac{d}{dt}} \gamma' = (\nabla_X X) \circ \gamma = \frac{1}{2}[X, X] \circ \gamma = 0. \quad (4.13)$$

Hence γ is a geodesic. (More precisely, we have $\gamma(t) = g \cdot \exp(tX_e)$.)

Let $\gamma : I \rightarrow G$ be a geodesic with $\gamma(0) = g$ and $\gamma'(0) = v$. Then consider the left-invariant vector field X with $X_g = v$ and let $\tilde{\gamma}$ be the integral curve of X starting at g . By the preceding argument we know $\tilde{\gamma}$ is a geodesic with $\tilde{\gamma}(0) = g$ and $\tilde{\gamma}'(0) = v$. Then by the uniqueness of geodesics we know $\gamma = \tilde{\gamma}$. Hence γ is an integral curve of X .

2. Note that the exponential map of the Levi-Civita connection at e is given by $\exp_e(v) = \gamma_v(1)$ where γ_v is the geodesic with $\gamma_v(0) = e$ and $\gamma_v'(0) = v$. By the preceding argument, γ_v is just the integral curve of the left-invariant vector field X that satisfies $X_e = v$. Hence two definitions coincide. $\square$

Problem 4.6

Let M be a smooth manifold and let ∇ be any connection on TM . We can define the curvature endomorphism of ∇ by the same formula as in the Riemannian case: $\mathcal{R}(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]}Z$. Then ∇ is said to be a flat connection if $\mathcal{R}(X, Y)Z \equiv 0$. Prove that the following are equivalent:

- (1) ∇ is flat.
- (2) For every point $p \in M$, there exists a parallel local frame defined on a neighborhood of p .
- (3) For all $p, q \in M$, parallel transport along an admissible curve segment γ from p to q depends only on the path-homotopy class of γ .
- (4) Parallel transport around any sufficiently small closed curve is the identity; that is, for every $p \in M$, there exists a neighborhood U of p such that if $\gamma : [a, b] \rightarrow U$ is an admissible curve in U starting and ending at p , then $P_{ab} : T_p M \rightarrow T_p M$ is the identity map.

Proof. (4) $\Rightarrow$ (3): Suppose $F : [a, b] \times [0, 1]$ is a path-homotopy from γ_0 to γ_1 . By compactness, from (4), there exists $n \in \mathbb{N}$ such that the parallel transport around any closed curve in

$$F \left(\left[a + \frac{k(b-a)}{2^n}, a + \frac{(k+2)(b-a)}{2^n} \right] \times \left[\frac{l}{2^n}, \frac{l+2}{2^n} \right] \right)$$

is the identity, where $0 \leq k, l \leq 2^n - 2$. Then (3) follows immediately.

(3) $\Rightarrow$ (2): Fix some basis $\{v_i\}$ of $T_p M$. Following the steps 1 and 2 in Lemma 4.7, we get smooth local vector fields V_i with $V_i(p) = v_i$. Since C_ϵ is simply connected, we know each V_i is parallel from (3). Since parallel transport is a linear isomorphism, we know $\{V_i\}$ is a local parallel frame.

(2) $\Rightarrow$ (1): Let $\{V_i\}$ be a local parallel frame. Then

$$\mathcal{R}(V_i, V_j)V_k = \nabla_{V_i} \nabla_{V_j} V_k - \nabla_{V_j} \nabla_{V_i} V_k - \nabla_{[V_i, V_j]} V_k = (\nabla_{V_i} 0) - (\nabla_{V_j} 0) - 0 = 0.$$

(1) $\Rightarrow$ (4): Fix some basis $\{v_i\}$ of $T_p M$. By Lemma 4.7, we get local parallel vector fields V_i with $V_i(p) = v_i$. Since parallel transport is a linear isomorphism, we know $\{V_i\}$ is a local parallel frame. Therefore, for any curve γ in C_ϵ , we have

$$\hat{\nabla}_{\frac{d}{dt}} (\lambda^i \cdot V_i|_{\gamma(t)}) = \lambda^i \cdot \hat{\nabla}_{\frac{d}{dt}} (V_i|_{\gamma(t)}) = 0,$$

and hence the parallel transport is given by

$$P_{t_1, t_2; \gamma} (\lambda^i \cdot V_i|_{\gamma(t_1)}) = \lambda^i \cdot V_i|_{\gamma(t_2)}.$$

Thus we get the conclusion for closed curves. $\square$

Lemma 4.7

Suppose M is a smooth manifold, and ∇ is a flat connection on M . Given $p \in M$ and any vector $v \in T_p M$, there exists a parallel vector field V on a neighborhood of p such that $V_p = v$.

Proof. 1. Let $(U, \{x^i\})$ be a local chart centered at p . Shrinking the chart if necessary, we assume that the image of the coordinate map is an open cube

$$C_\epsilon = \{x : |x^i| < \epsilon, i = 1, \dots, n\}.$$

2. By the parallel transport along the x^1 -direction, from v we define

$$V(t_1, 0, \dots, 0) \quad \text{where } -\epsilon < t_1 < \epsilon. \quad (4.14)$$

By the parallel transport along the x^2 -direction, from (4.14) we define

$$V(t_1, t_2, 0, \dots, 0) \quad \text{where } -\epsilon < t_1, t_2 < \epsilon. \quad (4.15)$$

Then successively parallel transport along the x^3 through x^n -directions, we define

$$V(t_1, \dots, t_n) \quad \text{where } -\epsilon < t_1, \dots, t_n < \epsilon. \quad (4.16)$$

Since the solutions to linear ODEs smoothly depend on the initial conditions, we know V is smooth on C_ϵ .

3. By our construction, for $1 \leq i \leq n$ we know

$$\left(\nabla_{\frac{\partial}{\partial x^i}} V \right) (t_1, \dots, t_i, 0, \dots, 0) = 0 \quad \forall -\epsilon < t_1, \dots, t_i < \epsilon. \quad (4.17)$$

Since ∇ is flat, we know

$$\left(\nabla_{\frac{\partial}{\partial x^1}} \nabla_{\frac{\partial}{\partial x^2}} V \right) (t_1, t_2, 0, \dots, 0) = \left(\nabla_{\frac{\partial}{\partial x^2}} \nabla_{\frac{\partial}{\partial x^1}} V \right) (t_1, t_2, 0, \dots, 0) \quad \forall -\epsilon < t_1, t_2 < \epsilon. \quad (4.18)$$

By (4.17) we know $(\nabla_{\frac{\partial}{\partial x^2}} V)(t_1, t_2, 0, \dots, 0) = 0$. Then (4.18) yields

$$\left(\nabla_{\frac{\partial}{\partial x^2}} \nabla_{\frac{\partial}{\partial x^1}} V \right) (t_1, t_2, 0, \dots, 0) = 0 \quad \forall -\epsilon < t_1, t_2 < \epsilon.$$

By (4.17) again we know $(\nabla_{\frac{\partial}{\partial x^1}} V)(t_1, 0, \dots, 0) = 0$. Hence

$$\left(\nabla_{\frac{\partial}{\partial x^1}} V \right) (t_1, t_2, 0, \dots, 0) = 0 \quad \forall -\epsilon < t_1, t_2 < \epsilon.$$

Using the above method, by induction we know

$$\left(\nabla_{\frac{\partial}{\partial x^i}} V \right) (t_1, \dots, t_n) = 0 \quad \forall -\epsilon < t_1, \dots, t_n < \epsilon.$$

Hence V is parallel on C_ϵ . Therefore, we get the conclusion. $\square$

Problem 4.8

Show the second Bianchi identity

$$\nabla_i R_{jkpq} + \nabla_j R_{kipq} + \nabla_k R_{ijpq} = 0. \quad (4.19)$$

Proof. Note that

$$\begin{aligned} \nabla_i R_{jkpq} &= \frac{\partial R_{jkpq}}{\partial x^i} - \Gamma_{ij}^s R_{skpq} - \Gamma_{ik}^s R_{sjpq} - \Gamma_{ip}^s R_{jksq} - \Gamma_{iq}^s R_{jkps}, \\ \nabla_j R_{kipq} &= \frac{\partial R_{kipq}}{\partial x^j} - \Gamma_{jk}^s R_{sipq} - \Gamma_{ji}^s R_{kspq} - \Gamma_{jp}^s R_{kisiq} - \Gamma_{jq}^s R_{kips}, \\ \nabla_k R_{ijpq} &= \frac{\partial R_{ijpq}}{\partial x^k} - \Gamma_{ki}^s R_{sjpq} - \Gamma_{kj}^s R_{is pq} - \Gamma_{kp}^s R_{ijsq} - \Gamma_{kq}^s R_{ijps}. \end{aligned}$$

Therefore

$$\begin{aligned} &\nabla_i R_{jkpq} + \nabla_j R_{kipq} + \nabla_k R_{ijpq} \\ &= \left(\frac{\partial R_{jkpq}}{\partial x^i} - \Gamma_{ip}^s R_{jksq} - \Gamma_{iq}^s R_{jkps} \right) + \left(\frac{\partial R_{kipq}}{\partial x^j} - \Gamma_{jp}^s R_{kisiq} - \Gamma_{jq}^s R_{kips} \right) \\ &\quad + \left(\frac{\partial R_{ijpq}}{\partial x^k} - \Gamma_{kp}^s R_{ijsq} - \Gamma_{kq}^s R_{ijps} \right). \end{aligned}$$

Note that

$$\begin{aligned}
& \frac{\partial R_{jkpq}}{\partial x^i} - \Gamma_{ip}^s R_{jksq} - \Gamma_{iq}^s R_{jkps} \\
&= g \left(\nabla_{\frac{\partial}{\partial x^i}} \left(\left[\nabla_{\frac{\partial}{\partial x^j}}, \nabla_{\frac{\partial}{\partial x^k}} \right] \frac{\partial}{\partial x^p} \right), \frac{\partial}{\partial x^q} \right) + g \left(\left[\nabla_{\frac{\partial}{\partial x^j}}, \nabla_{\frac{\partial}{\partial x^k}} \right] \frac{\partial}{\partial x^p}, \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^q} \right) \\
&\quad - g \left(\left[\nabla_{\frac{\partial}{\partial x^j}}, \nabla_{\frac{\partial}{\partial x^k}} \right] \left(\Gamma_{ip}^s \frac{\partial}{\partial x^s} \right), \frac{\partial}{\partial x^q} \right) - g \left(\left[\nabla_{\frac{\partial}{\partial x^j}}, \nabla_{\frac{\partial}{\partial x^k}} \right] \frac{\partial}{\partial x^p}, \Gamma_{iq}^s \frac{\partial}{\partial x^s} \right) \\
&= g \left(\nabla_{\frac{\partial}{\partial x^i}} \nabla_{\frac{\partial}{\partial x^j}} \nabla_{\frac{\partial}{\partial x^k}} \frac{\partial}{\partial x^p} - \nabla_{\frac{\partial}{\partial x^i}} \nabla_{\frac{\partial}{\partial x^k}} \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^p}, \frac{\partial}{\partial x^q} \right) \\
&\quad - g \left(\nabla_{\frac{\partial}{\partial x^j}} \nabla_{\frac{\partial}{\partial x^k}} \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^p} - \nabla_{\frac{\partial}{\partial x^k}} \nabla_{\frac{\partial}{\partial x^j}} \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^p}, \frac{\partial}{\partial x^q} \right) \\
&= G_{ijkpq} + G_{kjipq} - G_{ikjqp} - G_{jkipq}.
\end{aligned}$$

where

$$G_{ijkpq} = g \left(\nabla_{\frac{\partial}{\partial x^i}} \nabla_{\frac{\partial}{\partial x^j}} \nabla_{\frac{\partial}{\partial x^k}} \frac{\partial}{\partial x^p}, \frac{\partial}{\partial x^q} \right).$$

It follows that

$$\begin{aligned}
& \nabla_i R_{jkpq} + \nabla_j R_{kipq} + \nabla_k R_{ijpq} \\
&= (G_{ijkpq} + G_{kjipq} - G_{ikjqp} - G_{jkipq}) + (G_{jkipq} + G_{ikjqp} - G_{kjipq} - G_{kijpq}) \\
&\quad + (G_{kijpq} + G_{jkipq} - G_{ijkpq} - G_{jikpq}) \\
&= 0.
\end{aligned}$$

We are done.

An Alternative Proof. Since $\nabla_i R_{jkpq} + \nabla_j R_{kipq} + \nabla_k R_{ijpq}$ is the component of a tensor, it suffices to verify it pointwisely. Let $\{x^i\}$ be normal coordinates centered at p . Recall that

$$R_{ijkl} = \frac{1}{2} \left(\frac{\partial^2 g_{jl}}{\partial x^i \partial x^k} + \frac{\partial^2 g_{ik}}{\partial x^j \partial x^l} - \frac{\partial^2 g_{il}}{\partial x^j \partial x^k} - \frac{\partial^2 g_{jk}}{\partial x^i \partial x^l} \right) + g_{pq} (\Gamma_{ik}^p \Gamma_{jl}^q - \Gamma_{il}^q \Gamma_{kj}^p).$$

Therefore, at p we have

$$\begin{aligned}
\nabla_i R_{jkpq}(p) &= \frac{\partial R_{jkpq}}{\partial x^i}(p) = \frac{1}{2} \left[\frac{\partial^3 g_{kp}}{\partial x^i \partial x^j \partial x^q} + \frac{\partial^3 g_{jq}}{\partial x^i \partial x^k \partial x^p} - \frac{\partial^3 g_{jp}}{\partial x^i \partial x^k \partial x^q} - \frac{\partial^3 g_{kq}}{\partial x^i \partial x^j \partial x^p} \right] (p), \\
\nabla_j R_{kipq}(p) &= \frac{\partial R_{kipq}}{\partial x^j}(p) = \frac{1}{2} \left[\frac{\partial^3 g_{ip}}{\partial x^j \partial x^k \partial x^q} + \frac{\partial^3 g_{kq}}{\partial x^j \partial x^i \partial x^p} - \frac{\partial^3 g_{kp}}{\partial x^j \partial x^i \partial x^q} - \frac{\partial^3 g_{iq}}{\partial x^j \partial x^k \partial x^p} \right] (p), \\
\nabla_k R_{ijpq}(p) &= \frac{\partial R_{ijpq}}{\partial x^k}(p) = \frac{1}{2} \left[\frac{\partial^3 g_{jp}}{\partial x^k \partial x^i \partial x^q} + \frac{\partial^3 g_{iq}}{\partial x^k \partial x^j \partial x^p} - \frac{\partial^3 g_{ip}}{\partial x^k \partial x^j \partial x^q} - \frac{\partial^3 g_{jq}}{\partial x^k \partial x^i \partial x^p} \right] (p).
\end{aligned}$$

Thus we get $\nabla_i R_{jkpq}(p) + \nabla_j R_{kipq}(p) + \nabla_k R_{ijpq}(p) = 0$. □

5 HW5

Problem 5.1

Let (M, g) be a closed orientable Riemannian manifold. Then

$$d^* \omega = -\nabla^i \omega_i = -g^{ij} \left(\frac{\partial \omega_i}{\partial x^j} - \Gamma_{ji}^k \omega_k \right) \quad (5.1)$$

for any $\omega = \omega_i dx^i \in \Omega^1(M)$.

Proof. Set $X = fg^{ij}\omega_j \frac{\partial}{\partial x^i}$. Since the expression is independent of the choice of local charts, we know $X \in \Gamma(M, TM)$ is well-defined. Note that

$$\operatorname{div} X = \nabla_i X^i = \nabla_i(fg^{ij}\omega_j) = g^{ij}(\nabla_i f) \cdot \omega_j + fg^{ij} \cdot \nabla_i \omega_j = \langle df, \omega \rangle + f \nabla^i \omega_i.$$

By the Divergence Theorem, it follows that

$$0 = \int_M \operatorname{div} X \operatorname{vol}_g = (df, \omega) + (f, \nabla^i \omega_i).$$

Hence $d^*\omega = -\nabla^i \omega_i$. Note that

$$\nabla^i \omega_i = g^{ij} \cdot \nabla_i \omega_j = g^{ij} \left(\frac{\partial \omega_j}{\partial x^i} - \Gamma_{ij}^k \omega_k \right).$$

Thus we get the conclusion. $\square$

Problem 5.2

Let $f : M \rightarrow \mathbb{R}$ be a smooth function over (M, g) . Prove the following formula by using normal coordinates at a point

$$\frac{1}{2} \Delta_g |\nabla f|^2 = |\operatorname{Hess} f|_g^2 + \operatorname{Ric}(\nabla f, \nabla f) + g(\nabla \Delta_g f, \nabla f). \quad (5.2)$$

Proof. Let $\{x^i\}$ be normal coordinates centered at p . Recall that

$$\Delta_g f = g^{ij} \left(\frac{\partial^2 f}{\partial x^i \partial x^j} - \Gamma_{ij}^k \frac{\partial f}{\partial x^k} \right).$$

Thus we know

$$\begin{aligned} g(\nabla \Delta_g f, \nabla f)(p) &= \sum_{l=1}^n \frac{\partial}{\partial x^l} \Big|_p \left(g^{ij} \frac{\partial^2 f}{\partial x^i \partial x^j} - g^{ij} \Gamma_{ij}^k \frac{\partial f}{\partial x^k} \right) \cdot \frac{\partial f}{\partial x^l}(p) \\ &= \sum_{l,i=1}^n \left(\frac{\partial^3 f}{\partial x^l \partial x^i \partial x^i} - \frac{\partial \Gamma_{ii}^k}{\partial x^l} \frac{\partial f}{\partial x^k} \right) \frac{\partial f}{\partial x^l}(p). \end{aligned}$$

Recall that

$$g_{ij} = \delta_{ij} - \frac{1}{3} \sum_{k,l} R_{iklj}(p) x^k x^l + O(|x|^3), \quad g^{ij} = \delta_{ij} + \frac{1}{3} \sum_{k,l} R_{iklj}(p) x^k x^l + O(|x|^3).$$

Hence

$$\begin{aligned} \frac{\partial^2 g^{ij}}{\partial x^t \partial x^t}(p) &= \frac{2}{3} R_{titi}(p), \\ \frac{\partial \Gamma_{ii}^k}{\partial x^l}(p) &= \frac{\partial}{\partial x^l} \Big|_p \left(\frac{1}{2} g^{ks} \left(2 \frac{\partial g_{si}}{\partial x^i} - \frac{\partial g_{ii}}{\partial x^s} \right) \right) = \frac{1}{2} \left(2 \frac{\partial^2 g_{ki}}{\partial x^l \partial x^i} - \frac{\partial^2 g_{ii}}{\partial x^l \partial x^k} \right)(p) \\ &= \frac{1}{3} (R_{kili} + R_{klji})(p) - \frac{1}{6} (R_{ilil} + R_{ilik})(p) = \frac{2}{3} R_{kili}(p). \end{aligned}$$

It follows that

$$\frac{1}{2} \Delta_g |\nabla f|^2(p) = g(\nabla \Delta_g f, \nabla f)(p) + |\operatorname{Hess} f|_g^2(p) + \sum_{i,j,l=1}^n R_{lijl} \frac{\partial f}{\partial x^i} \frac{\partial f}{\partial x^j}(p).$$

Since $\sum_{i,j,l=1}^n R_{lijl} \frac{\partial f}{\partial x^i} \frac{\partial f}{\partial x^j}(p) = \operatorname{Ric}(\nabla f, \nabla f)(p)$, we get the conclusion. $\square$

Problem 5.3

Recall that a vector field Y defined on (an open subset of) a Riemannian manifold is said to be parallel if $\nabla Y \equiv 0$.

- (1) Let $p \in \mathbb{R}^n$ and $v \in T_p \mathbb{R}^n$. Show that there is a unique parallel vector field Y on $\mathbb{R}^n$ such that $Y_p = v$.
- (2) Let $X(\varphi, \theta) = (\sin \varphi \cos \theta, \sin \varphi \sin \theta, \cos \varphi)$ be the spherical coordinate parametrization of an open subset U of the unit sphere $\mathbb{S}^2$, and let $X_\theta = X_*(\partial_\theta)$, $X_\varphi = X_*(\partial_\varphi)$ denote the coordinate vector fields associated with this parametrization. Compute $\nabla_{X_\theta}(X_\varphi)$ and $\nabla_{X_\varphi}(X_\varphi)$, and conclude that X_φ is parallel along the equator and along each meridian $\theta = \theta_0$.
- (3) Let $p = (1, 0, 0) \in \mathbb{S}^2$. Show that there is no parallel vector field W on any neighborhood of p in $\mathbb{S}^2$ such that $W_p = X_\varphi|_p$.
- (4) Use (1) and (3) to show that no neighborhood of p in $(\mathbb{S}^2, \tilde{g})$ is isometric to an open subset of $(\mathbb{R}^2, \bar{g})$.

Proof. 1. Let $\{x^i\}$ be the canonical coordinates on $\mathbb{R}^n$. For $v = v^i \frac{\partial}{\partial x^i}|_p$, the vector field $Y = v^i \frac{\partial}{\partial x^i}$ is parallel with $Y_p = v$. Given another parallel field $\tilde{Y}$ with $\tilde{Y}_p = v$, then

$$\tilde{Y}_q = Y_q = P_{0,1;\gamma}(v)$$

where $q \in \mathbb{R}^n$ and $\gamma : [0, 1] \rightarrow \mathbb{R}^n$ is a smooth curve joining p and q . Thus $\tilde{Y} = Y$, which implies the uniqueness.

2. Let g be the canonical metric on $\mathbb{R}^3$, and $\hat{g} = X^*g$ be the pullback metric on U . Then

$$\begin{aligned}\hat{g}_{\varphi\varphi} &= g(X_\varphi, X_\varphi) = (\cos \varphi \cos \theta)^2 + (\cos \varphi \sin \theta)^2 + (-\sin \varphi)^2 = 1, \\ \hat{g}_{\varphi\theta} &= \hat{g}_{\theta\varphi} = g(X_\theta, X_\varphi) = -\cos \varphi \cos \theta \sin \varphi \sin \theta + \cos \varphi \cos \theta \sin \varphi \sin \theta = 0, \\ \hat{g}_{\theta\theta} &= g(X_\theta, X_\theta) = (-\sin \varphi \sin \theta)^2 + (\sin \varphi \cos \theta)^2 = \sin^2 \varphi.\end{aligned}$$

Hence $\hat{g}^{\varphi\varphi} = 1$, $\hat{g}^{\theta\varphi} = \hat{g}^{\varphi\theta} = 0$, $\hat{g}^{\theta\theta} = \sin^{-2} \varphi$. It follows that

$$\begin{aligned}\hat{\nabla}_{\frac{\partial}{\partial \theta}} \frac{\partial}{\partial \varphi} &= \hat{\Gamma}_{\theta\varphi}^\varphi \frac{\partial}{\partial \varphi} + \hat{\Gamma}_{\theta\varphi}^\theta \frac{\partial}{\partial \theta} = \frac{1}{2} \hat{g}^{\varphi\varphi} \frac{\partial \hat{g}_{\varphi\varphi}}{\partial \theta} \frac{\partial}{\partial \varphi} + \frac{1}{2} \hat{g}^{\theta\theta} \frac{\partial \hat{g}_{\theta\theta}}{\partial \varphi} \frac{\partial}{\partial \theta} = \cot \varphi \frac{\partial}{\partial \theta}, \\ \hat{\nabla}_{\frac{\partial}{\partial \varphi}} \frac{\partial}{\partial \varphi} &= \hat{\Gamma}_{\varphi\varphi}^\varphi \frac{\partial}{\partial \varphi} + \hat{\Gamma}_{\varphi\varphi}^\theta \frac{\partial}{\partial \theta} = \frac{1}{2} \hat{g}^{\varphi\varphi} \frac{\partial \hat{g}_{\varphi\varphi}}{\partial \varphi} \frac{\partial}{\partial \varphi} - \frac{1}{2} \hat{g}^{\theta\theta} \frac{\partial \hat{g}_{\varphi\varphi}}{\partial \theta} \frac{\partial}{\partial \theta} = 0.\end{aligned}$$

Hence

$$\nabla_{X_\theta}(X_\varphi) = \cot \varphi X_\theta, \quad \nabla_{X_\varphi}(X_\varphi) = 0.$$

Along the equator we have $\nabla_{X_\theta}(X_\varphi) = 0$ since $\varphi = \pi/2$, and along any $\theta = \theta_0$ we have $\nabla_{X_\varphi}(X_\varphi) = 0$, so X_φ is parallel in both cases.

3. Suppose such W exists. Set $f = \frac{1}{2}|W|^2$. Since W is parallel, we know f is a constant, and W is a Killing vector. Thus for any vector $v \in T_p \mathbb{S}^2$ we have

$$0 = (\text{Hess } f)(v, v) = \langle \nabla_v W, \nabla_v W \rangle - R(v, W_p, W_p, v) = -R(v, W_p, W_p, v).$$

However, on the round sphere, since $W_p \neq 0$, there exists $v \in T_p \mathbb{S}^2$ with $R(v, W_p, W_p, v) \neq 0$. A contradiction.

4. WLOG assume $p = (1, 0, 0)$. Suppose $F : (U, \tilde{g}) \rightarrow (V, \bar{g})$ is an isometry, where U is a neighborhood of p in $\mathbb{S}^2$ and V is an open subset of $\mathbb{R}^2$. Let Y be the parallel vector on V with $Y_{F(p)} = dF(X_\varphi|_p)$. Then $W = (F^{-1})_* Y$ is a parallel vector field on U with $W_p = X_\varphi|_p$. A contradiction. $\square$

Problem 5.4

By analogy with the formula $\text{div } X = \text{tr}(\nabla X)$, we can define a divergence operator on tensor fields of any rank on a Riemannian manifold. If F is any smooth k -tensor field, we define the divergence of F by

$$\text{div } F = \text{tr}_g(\nabla F),$$

where the trace is taken on the last two indices of the $(k+1)$ -tensor field ∇F . If F is a smooth covariant k -tensor field and G is a smooth covariant $(k+1)$ -tensor field on a compact smooth Riemannian manifold (M, g) with boundary, then

$$\int_M \langle \nabla F, G \rangle dV_g = \int_{\partial M} \langle N^\flat \otimes F, G \rangle dV_{\hat{g}} - \int_M \langle F, \operatorname{div} G \rangle dV_g,$$

where $\hat{g}$ is the induced metric on ∂M .

Proof. Suppose $F = F_{i_1 \dots i_k} dx^{i_1} \otimes \dots \otimes dx^{i_k}$, $G = G_{j_1 \dots j_{k+1}} dx^{j_1} \otimes \dots \otimes dx^{j_{k+1}}$. Set

$$X = \{F, G\}^\# := g^{kl} g^{i_1 j_1} \dots g^{i_k j_k} F_{i_1 \dots i_k} G_{k j_1 \dots j_k} \frac{\partial}{\partial x^l}.$$

Since the expression is independent of the choice of local charts, $\{F, G\}^\# \in \Gamma(M, TM)$. Moreover,

$$\begin{aligned} \operatorname{div} X &= \nabla_l (g^{kl} g^{i_1 j_1} \dots g^{i_k j_k} F_{i_1 \dots i_k} G_{k j_1 \dots j_k}) \\ &= g^{kl} g^{i_1 j_1} \dots g^{i_k j_k} (\nabla_l F_{i_1 \dots i_k}) G_{k j_1 \dots j_k} + g^{i_1 j_1} \dots g^{i_k j_k} F_{i_1 \dots i_k} \cdot g^{kl} \nabla_l G_{k j_1 \dots j_k} \\ &= \langle \nabla F, G \rangle + \langle F, \operatorname{div} G \rangle, \end{aligned}$$

and

$$\begin{aligned} \langle X, N \rangle &= X^l N^s g_{ls} = (g^{kl} g^{i_1 j_1} \dots g^{i_k j_k} F_{i_1 \dots i_k} G_{k j_1 \dots j_k}) N^s g_{ls} \\ &= g^{kl} g^{i_1 j_1} \dots g^{i_k j_k} (N_l F_{i_1 \dots i_k}) \cdot G_{k j_1 \dots j_k} \\ &= \langle N^\flat \otimes F, G \rangle. \end{aligned}$$

Therefore, by the Divergence Theorem we know

$$\begin{aligned} \int_M \langle \nabla F, G \rangle dV_g &= \int_M (\operatorname{div} X - \langle F, \operatorname{div} G \rangle) dV_g \\ &= \int_{\partial M} \langle X, N \rangle dV_{\hat{g}} - \int_M \langle F, \operatorname{div} G \rangle dV_g \\ &= \int_{\partial M} \langle N^\flat \otimes F, G \rangle dV_{\hat{g}} - \int_M \langle F, \operatorname{div} G \rangle dV_g. \end{aligned}$$

We are done. □

6 HW6

7 HW7

Problem 6.1

Let (M, g) be a complete Riemannian manifold. Show that every Killing vector field X is complete.

Proof. Suppose γ is an integral curve of X . Then

$$\frac{d}{dt} |\gamma'(t)|^2 = 2g(\nabla_{\gamma'(t)} \gamma'(t), \gamma'(t)) = 2g(\nabla_X X, X) \circ \gamma(t) = 0.$$

Hence $|\gamma'(t)| = C$ for some constant C .

If γ is defined on a bounded interval (a, b) , then the restriction $\gamma|_{(a, b)}$ has finite length, and from completeness it can be extended to a neighborhood of $[a, b]$. This implies that γ is defined on whole $\mathbb{R}$. Therefore, X is complete. $\square$

Problem 6.2

Let (M, g) be a complete Riemannian manifold, and V be a smooth vector field with $|V|_g \leq C$ for some constant C . Show that V is a complete vector field.

Proof. Suppose $\gamma : (a, b) \rightarrow M$ is an integral curve where $\gamma(0) = p$ and (a, b) is the maximum domain with $a, b \in [-\infty, +\infty]$. Pick $a < t_1 < t_2 < b$. Then

$$d(\gamma(t_2), \gamma(t_1)) \leq \int_{t_1}^{t_2} |\gamma'(t)| dt \leq \int_{t_1}^{t_2} |V_{\gamma(t)}| dt \leq \sup_M |V| \cdot (t_2 - t_1).$$

Therefore, $\gamma(t)$ is uniformly continuous. Thus if $a > -\infty$ or $b < +\infty$, by ODE theory we can smoothly extend γ to a or b , a contradiction. Therefore, the maximum domain must be $\mathbb{R}$, which implies that V is complete. $\square$

Problem 6.3

Let (M, g) be a complete Riemannian manifold. Suppose $f \in C^\infty(M, \mathbb{R})$ satisfies

$$|\nabla f| = 1 \quad \text{and} \quad \text{Hess } f = 0. \quad (6.1)$$

We set $N = f^{-1}(0)$ and $h = g|_N$. Show that (N, h) is a totally geodesic submanifold of (M, g) .

Proof. Since $|\nabla f| = 1$, we know that df is nowhere degenerate, and thus $N = f^{-1}(0)$ is an $(n-1)$ -submanifold of M . Let B be the second fundamental form of $\iota : (N, h) \rightarrow (M, g)$. Note that

$$\langle \iota^* \nabla f, \iota_* X \rangle = \langle \nabla f, \iota_* X \rangle = df(\iota_* X) = 0 \quad \forall X \in \Gamma(N, TN).$$

It follows that

$$\begin{aligned} \langle B(X, Y), \iota^* \nabla f \rangle_g &= \langle \hat{\nabla}_X(\iota_* Y) - \iota_*(\nabla_X^h Y), \iota^* \nabla f \rangle_g = \langle \hat{\nabla}_X(\iota_* Y), \iota^* \nabla f \rangle_g \\ &= X \langle \iota_* Y, \iota^* \nabla f \rangle_g - \langle \iota_* Y, \hat{\nabla}_X(\iota^* \nabla f) \rangle_g \\ &= -\langle \iota_* Y, \nabla_X(\nabla f) \rangle_g = -(\text{Hess } f)(\iota_* X, \iota_* Y) = 0. \end{aligned}$$

Let $\iota^* TM = T^\perp N \oplus \iota_*(TN)$ be the orthogonal decomposition. Then by the dimension reason we know $T^\perp N = \text{span}\{\iota^* \nabla f\}$. Since $B \in \Gamma(N, T^* N \otimes T^* N \otimes T^\perp N)$, it follows that $B = 0$, and hence (N, h) is a totally geodesic submanifold of (M, g) . $\square$

Problem 6.4

Determine the cut locus of an arbitrary point in the n -torus $\mathbb{T}^n$ with the flat metric.

Proof. Since $\mathbb{T}^n$ is the flat torus, we know $\mathbb{T}^n$ is isometric to $\mathbb{R}^n/\mathbb{Z}^n$ where $\mathbb{R}^n/\mathbb{Z}^n$ is equipped with the canonical quotient metric. In the next we identify $\mathbb{T}^n$ with $\mathbb{R}^n/\mathbb{Z}^n$, let $\pi : \mathbb{R}^n \rightarrow \mathbb{T}^n$ be the quotient map. Fix some $p = [x] \in \mathbb{T}^n$ where $x \in \mathbb{R}^n$. Since π is a local isometry, we know

$$\pi(\exp_x(v)) = \exp_p(d\pi_x(v)) \quad \forall v \in T_x\mathbb{R}^n.$$

Let x^i be the canonical coordinates of $\mathbb{R}^n$, $\tilde{e}_i = \frac{\partial}{\partial x^i}|_x$ and $e_i = d\pi_x(\tilde{e}_i)$. Then $\pi(\exp_x(y^i \tilde{e}_i)) = \pi(x + y) = \exp_p(y^i e_i)$. We identify $T_p\mathbb{T}^n$ with $\mathbb{R}^n$ by the orthonormal basis $\{e_i\}$. Then

$$\exp_p^{-1}(p + [y]) = y + \mathbb{Z}^n.$$

Since $d(p, q) = \inf\{|\tilde{q}| : \tilde{q} \in \exp_p^{-1}(q)\}$, we know

$$\begin{aligned} d(p, p + [y]) &= |y| \quad \text{if } y \in \left[-\frac{1}{2}, \frac{1}{2}\right]^n, \\ d(p, p + [y]) &< |y| \quad \text{if } y \notin \left[-\frac{1}{2}, \frac{1}{2}\right]^n. \end{aligned}$$

Thus for $y \in \left[-\frac{1}{2}, \frac{1}{2}\right]^n$ the geodesic

$$\gamma : [0, t] \rightarrow \mathbb{T}^n, \quad t \mapsto \exp_p(ty)$$

is a minimal geodesic precisely when $t \leq 1$. Therefore, the cut locus of p is

$$\exp_p\left(\partial\left[-\frac{1}{2}, \frac{1}{2}\right]^n\right) = \left\{[x + y] : y \in \partial\left[-\frac{1}{2}, \frac{1}{2}\right]^n\right\}.$$

□

Problem 6.5

Determine the cut locus of an arbitrary point in the $\mathbb{S}^n$ with the canonical metric.

Proof. Fix some $p \in \mathbb{S}^n$. Since

$$F = \exp_p|_{B(0, \pi)} : B(0, \pi) \rightarrow \mathbb{S}^n \setminus \{-p\}$$

is a diffeomorphism, we know $\text{inj}(\mathbb{S}^n, p) = \pi$. Then by Corollary 10.2 we know

$$d(p, q) = |F^{-1}(q)| \quad \forall q \in \mathbb{S}^n \setminus \{-p\} \tag{6.2}$$

and by continuity we know $d(p, -p) = \pi$. Fix some $v \in T_p\mathbb{S}^n$ with $|v| = 1$. Set

$$\gamma : [0, t] \rightarrow \mathbb{S}^n, \quad t \mapsto \exp_p(tv).$$

If $t \leq \pi$ by the preceding discussion we know γ is a minimal geodesic. If $t > \pi$, then $l(\gamma) > \sup\{d(p, q) : q \in \mathbb{S}^n\}$, and hence γ is not a minimal geodesic. Therefore, the cut locus of p is $\exp_p(\partial B(0, \pi)) = \{-p\}$. □

Problem 6.6

Let $M = \mathbb{RP}^2$ and $p \in M$.

- (1) Find $\text{conj}(p)$. For any $q \in \text{conj}(p)$, compute $d(p, q)$.
- (2) Find $\text{cut}(p)$. For any $q \in \text{cut}(p)$, compute $d(p, q)$.

Proof. Let $\pi : \mathbb{S}^2 \rightarrow \mathbb{RP}^2$ be the quotient map. Then π is a local isometry.

(1): Since M has constant sectional curvature 1, any normal Jacobi field along a unit-speed geodesic has the form

$$J(t) = c \sin t \cdot E(t)$$

where $E(t)$ is a parallel vector field along γ with $\langle E(t), \gamma'(t) \rangle = 0, |E(t)| \equiv 1$.

Pick $x \in \pi^{-1}(p)$. Since π is a local isometry, we know

$$\pi(\exp_x(v)) = \exp_p(d\pi_x(v)) \quad \forall v \in T_x \mathbb{S}^2. \quad (6.3)$$

Since $d\pi_x$ is an isometry, we know

$$\text{conj}(p) = \exp_p(\partial B(0, \pi)) = \pi(\exp_x(\partial B(0, \pi))) = \{\pi(-x)\} = \{p\}.$$

Thus for any $q \in \text{conj}(p)$, $d(p, q) = 0$.

(2): Fix some $p \in \mathbb{RP}^2$, and pick $x \in \pi^{-1}(p)$. From (6.3) we know

$$\mathbb{RP}^2 = \exp_p(B(0, \pi/2)) \cup \exp_p(\partial B(0, \pi/2))$$

and the map

$$F = \exp_p|_{B(0, \pi/2)} : B(0, \pi/2) \rightarrow \exp_p(B(0, \pi/2))$$

is a diffeomorphism. Thus $\text{inj}(\mathbb{RP}^2, p) = \pi/2$. Then by Corollary 10.2 we know

$$d(p, q) = |F^{-1}(q)| \quad \forall q \in \exp_p(B(0, \pi/2)),$$

and by continuity we know

$$d(p, q) = \frac{\pi}{2} \quad \forall q \in \exp_p(\partial B(0, \pi/2)).$$

Fix some $v \in T_p \mathbb{RP}^2$ with $|v| = 1$. Set

$$\gamma : [0, t] \rightarrow \mathbb{RP}^2, \quad t \mapsto \exp_p(tv).$$

If $t \leq \pi/2$, by the preceding discussion we know γ is a minimal geodesic. If $t > \pi/2$, then $l(\gamma) > \sup\{d(p, q) : q \in \mathbb{RP}^2\}$, and hence γ is not a minimal geodesic. Therefore, the cut locus of p is $\exp_p(\partial B(0, \pi/2))$, and for all $q \in \text{cut}(p)$, $d(p, q) = \pi/2$. $\square$

Problem 6.7

Let (M, g) be a complete Riemannian manifold and $p \in M$. Suppose there exists some point $q \in \text{cut}(p)$ such that $d(p, q) = d(p, \text{cut}(p))$.

- (1) Show that either q is a conjugate point of p along some minimizing geodesic from p to q , or there are exactly two unit-speed minimizing geodesics from p to q , say $\gamma_1, \gamma_2 : [0, b] \rightarrow M$ such that $\gamma_1'(b) = -\gamma_2'(b)$.
- (2) If in addition that $\text{inj}(M, p) = \text{inj}(M)$, and that q is not conjugate to p along any minimizing geodesic, show that there is a closed unit-speed geodesic $\gamma : [0, 2b] \rightarrow M$ such that $\gamma(0) = \gamma(2b) = p$ and $\gamma(b) = q$ where $b = d(p, q)$.

Proof. Since $d(p, q) = d(p, \text{cut}(p))$, we know $\text{inj}(M, p) = d(p, q)$.

(1): Suppose q is not conjugate to p along any minimizing geodesic from p to q . By the classification of cut points, there exist two unit-speed minimizing geodesics $\gamma_1, \gamma_2 : [0, b] \rightarrow M$ from p to q . Suppose $\gamma'_1(b) \neq -\gamma'_2(b)$. Then we can find $v \in T_q M$ such that

$$\langle v, \gamma'_1(b) \rangle < 0, \quad \langle v, \gamma'_2(b) \rangle < 0.$$

Let $\sigma(s) = \exp_q(sv)$. Since p and q are not conjugate along γ_i , we know that $(d\exp_p)_{b\gamma'_i(0)}$ is nondegenerate. Thus for each i there is a neighborhood $U_i \subset T_p M$ of $b\gamma'_i(0)$ such that

$$F_i = \exp_p|_{U_i} : U_i \rightarrow \exp_p(U_i)$$

is a diffeomorphism for $i = 1, 2$. Since $b\gamma'_1(0) \neq b\gamma'_2(0)$, we may assume $U_1 \cap U_2 = \emptyset$. By continuity there exists $\epsilon > 0$ such that

$$\sigma(s) \in \exp_p(U_i) \quad \forall s \in (-\epsilon, \epsilon), i = 1, 2.$$

Now consider the variation

$$\alpha_i : (-\epsilon, \epsilon) \times [0, 1] \rightarrow M, \quad (s, t) \mapsto \exp_p(tF_i^{-1}(\sigma(s))).$$

Setting $\alpha_{i,s}(t) = \alpha_i(s, t)$. Then by first variation formula of length we have

$$\left. \frac{d}{ds} \right|_{s=0} l(\alpha_{i,s}) = \langle v, \gamma'_i(b) \rangle < 0 \quad \text{for } i = 1, 2.$$

Thus we can pick $s_0 \in (0, \epsilon)$ such that

$$l(\alpha_{i,s_0}) < l(\gamma_i) = b = \text{inj}(M, p) \quad \text{for } i = 1, 2.$$

By Corollary 10.2, each $\alpha_{i,s_0} : [0, 1] \rightarrow M$ is a minimal curve from p to $\sigma(s_0)$. However, Corollary 10.2 also indicates that such minimal curve is unique, but $\alpha'_{1,s_0}(0) \neq \alpha'_{2,s_0}(0)$ since $U_1 \cap U_2 = \emptyset$, a contradiction. Moreover, let γ_3 be an unit-speed minimizing geodesic from p to q which is different from γ_1 . By the preceding discussion we know $\gamma'_3(b) = -\gamma'_1(b)$. By the uniqueness of geodesics, we know $\gamma_2 = \gamma_3$, so there are exactly two unit-speed minimizing geodesics.

(2): By (1) we know that there are exactly two minimal geodesics $\gamma_1, \gamma_2 : [0, b] \rightarrow M$ joining p and q with $\gamma'_1(b) = -\gamma'_2(b)$. On the other hand, we know $p \in \text{cut}(q)$, and $d(p, q) = \text{inj}(M) \leq d(q, \text{cut}(q))$, so we can apply (1) to q , which implies $\gamma'_1(0) = -\gamma'_2(0)$. Thus γ_1 and γ_2 are connected as a closed unit-speed geodesic as desired. $\square$

Problem 6.8

Let X be a vector field on a complete Riemannian manifold and consider the map $F_t : M \rightarrow M$ given by $F_t(p) = \exp_p(tX_p)$.

- (1) For $v \in T_p M$ show that $J(t) = (dF_t)_p(v)$ is a Jacobi field along $t \mapsto \gamma(t) = \exp_p(tX_p)$ with the initial conditions $J(0) = v$ and $J'(0) = \nabla_v X$.
- (2) Select an orthonormal basis $\{e_i\}$ for $T_p M$ and let $J_i(t) = (dF_t)_p(e_i)$. Show that

$$(\det[dF_t])^2 = \det[g(J_i(t), J_j(t))]. \quad (6.5)$$

- (3) Suppose $\langle \nabla_{e_i} X, e_j \rangle = \langle e_i, \nabla_{e_j} X \rangle$ for all i, j . Show that as long as $\det[dF_t] \neq 0$ it holds

$$\frac{d^2(\det[dF_t])^{\frac{1}{n}}}{dt^2} \leq -\frac{(\det[dF_t])^{\frac{1}{n}}}{n} \text{Ric}(\gamma', \gamma'). \quad (6.6)$$

Proof. 1. Consider the map

$$\alpha : \mathbb{R} \times \mathbb{R} \rightarrow M, \quad (s, t) \mapsto \exp_{\exp_p(sv)}(tX_{\exp_p(sv)}).$$

Let $V = \left(\alpha_* \frac{\partial}{\partial s}\right) \Big|_{s=0}$ be the variation field. Then

$$\begin{aligned} V(t) &= \frac{d}{ds} \Big|_{s=0} \alpha(s, t) = \frac{d}{ds} \Big|_{s=0} \exp_{\exp_p(sv)}(tX_{\exp_p(sv)}) \\ &= \frac{d}{ds} \Big|_{s=0} F_t(\exp_p(sv)) = (dF_t)_p(v) = J(t). \end{aligned}$$

Let $\bar{\nabla}$ be the pullback connection of α and $\hat{\nabla}$ be the pullback connection of γ . Since

$$\begin{aligned} \bar{\nabla}_{\frac{\partial}{\partial t}} \bar{\nabla}_{\frac{\partial}{\partial t}} \left(\alpha_* \frac{\partial}{\partial s} \right) &= \bar{\nabla}_{\frac{\partial}{\partial s}} \bar{\nabla}_{\frac{\partial}{\partial t}} \left(\alpha_* \frac{\partial}{\partial t} \right) + R \left(\frac{\partial}{\partial t}, \frac{\partial}{\partial s} \right) \left(\alpha_* \frac{\partial}{\partial t} \right) \\ &= -R \left(\alpha_* \frac{\partial}{\partial s}, \alpha_* \frac{\partial}{\partial t} \right) \left(\alpha_* \frac{\partial}{\partial t} \right) \end{aligned}$$

we know

$$\hat{\nabla}_{\frac{d}{dt}} \hat{\nabla}_{\frac{d}{dt}} V + R(V, \gamma')\gamma' = 0.$$

Hence $J(t) = V(t)$ is a Jacobi field along γ . Since $F_0 = id$ we know

$$J(0) = V(0) = (dF_0)_p(v) = v.$$

Let $\sigma(s) = \alpha(s, 0) = \exp_p(sv)$. Then $W = \left(\alpha_* \frac{\partial}{\partial t}\right) \Big|_{t=0}$ is a vector field along σ . Note that

$$W(s) = \frac{d}{dt} \Big|_{t=0} \alpha(s, t) = \frac{d}{dt} \Big|_{t=0} \exp_{\exp_p(sv)}(tX_{\exp_p(sv)}) = X_{\exp_p(sv)} = X_{\sigma(s)}.$$

Let ∇ be the pullback connection of σ . Since $\sigma'(0) = v$, we have

$$J'(0) = V'(0) = \bar{\nabla}_{\frac{\partial}{\partial t}} \left(\alpha_* \frac{\partial}{\partial s} \right) \Big|_{s=t=0} = \bar{\nabla}_{\frac{\partial}{\partial s}} \left(\alpha_* \frac{\partial}{\partial t} \right) \Big|_{s=t=0} = \nabla_{\frac{d}{ds}} W \Big|_{s=0} = \nabla_v X.$$

2. Let $e_i(t) = P_{0,t;\gamma}(e_i)$. Then $\{e_i(t)\}$ is an orthonormal basis of $T_{\gamma(t)}M$. Write

$$J_i(t) = \sum_{j=1}^n J_{ij}(t)e_j(t) \quad \text{and} \quad [dF_t] = [J_{ij}(t)]_{1 \leq i,j \leq n}.$$

Then

$$g(J_i(t), J_j(t)) = \sum_{k=1}^n J_{ik}(t)J_{jk}(t) = [[dF_t] \cdot [dF_t]^T]_{ij}.$$

Thus

$$\det[g(J_i(t), J_j(t))] = \det([dF_t] \cdot [dF_t]^T) = (\det[dF_t])^2.$$

3. Write $A(t) = [dF_t]$. Since

$$\frac{d}{dt} \log \det A(t) = \text{tr}(A^{-1}(t)A'(t)) \quad \text{and} \quad (A^{-1})'(t) = -A^{-1}(t)A'(t)A^{-1}(t),$$

we know

$$\begin{aligned} \frac{d^2}{dt^2} \det A(t)^{\frac{1}{n}} &= \frac{1}{n} \frac{d}{dt} \left(\det A(t)^{\frac{1}{n}} \cdot \text{tr}(A^{-1}(t)A'(t)) \right) \\ &= \frac{1}{n} \det A(t)^{\frac{1}{n}} \left(\frac{1}{n} (\text{tr}(A^{-1}(t)A'(t)))^2 + \text{tr}((A^{-1})'(t)A'(t)) + \text{tr}(A^{-1}(t)A''(t)) \right) \\ &= \frac{1}{n} \det A(t)^{\frac{1}{n}} \left(\frac{1}{n} (\text{tr}(A^{-1}(t)A'(t)))^2 - \text{tr}((A^{-1}(t)A'(t))^2) + \text{tr}(A^{-1}(t)A''(t)) \right). \end{aligned}$$

Since each J_i is a Jacobi field, we know

$$J''_{ij}(t) + R(J_i, \gamma', \gamma', e_j)(t) = 0.$$

Write $[A^{-1}(t)]_{ij} = J^{ij}(t)$. Then

$$\begin{aligned} \text{tr}(A^{-1}(t)A''(t)) &= J^{ji}(t)J''_{ij}(t) = -J^{ji}(t)R(J_i, \gamma', \gamma', e_j)(t) \\ &= -\sum_{i,j,k} J^{ji}(t)J_{ik}(t)R(e_k, \gamma', \gamma', e_j)(t) = -\sum_k R(e_k, \gamma', \gamma', e_k)(t) = -\text{Ric}(\gamma', \gamma')(t). \end{aligned}$$

Setting $B(t) = A^{-1}(t)A'(t)$, then

$$\frac{d^2}{dt^2} \det A(t)^{\frac{1}{n}} + \frac{1}{n} \det A(t)^{\frac{1}{n}} \text{Ric}(\gamma', \gamma')(t) = \frac{1}{n} \det A(t)^{\frac{1}{n}} \left(\frac{1}{n} (\text{tr} B(t))^2 - \text{tr}(B(t)^2) \right).$$

Since $J_i(0) = e_i$, we know $A(0) = I_n$. Thus as long as $\det[dF_t] \neq 0$, we have $\det[dF_t] > 0$, so it is equivalent to show that

$$(\text{tr} B(t))^2 \leq n \cdot \text{tr}(B(t)^2) \quad \text{as long as} \quad \det[dF_t] \neq 0. \quad (6.7)$$

We set $C(t) = A'(t)A^T(t) - A(t)(A^T)'(t)$. Then $B(t)$ is symmetric if and only if $C(t) = 0$. Moreover, since

$$J_{ki}(t)J''_{ij}(t) = -R(J_i, \gamma', \gamma', J_k)(t) = -R(J_k, \gamma', \gamma', J_i)(t) = J_{ij}(t)J''_{kj}(t)$$

we know $A''(t)A^T(t) = A(t)(A^T)''(t)$, and hence

$$C'(t) = A''(t)A^T(t) - A(t)(A^T)''(t) = 0.$$

Note that

$$\begin{aligned} C(0)_{ij} &= J'_{ik}(0)J_{jk}(0) - J_{ik}(0)J'_{jk}(0) = \langle J'_i(0), J_j(0) \rangle - \langle J_i(0), J'_j(0) \rangle \\ &= \langle \nabla_{e_i} X, e_j \rangle - \langle e_i, \nabla_{e_j} X \rangle = 0. \end{aligned}$$

Thus $C(t) \equiv 0$, which implies that $B(t)$ is symmetric. Let $\lambda_1, \dots, \lambda_n \in \mathbb{R}$ be the eigenvalues of $B(t)$. Then $\lambda_1^2, \dots, \lambda_n^2 \in \mathbb{R}$ are the eigenvalues of $B^2(t)$. Thus

$$(\text{tr} B(t))^2 = \left(\sum_{i=1}^n \lambda_i \right)^2 \leq n \sum_{i=1}^n \lambda_i^2 = n \cdot \text{tr}(B(t)^2).$$

Hence we get (6.7), which completes the proof. $\square$

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Problem 7.1

Let $F : (M, g) \rightarrow (M, g)$ be an isometry that fixes $p \in M$. Show that $dF_p = -id$ on $T_p M$ if and only if $F^2 = id_M$ and p is an isolated fixed point.

Proof. Let $\delta = \text{inj}(M, p)$, and $B(0, \delta) = \{v \in T_p M : |v| < \delta\}$.

1. Suppose $dF_p = -id$. Then F^2 is an isometry satisfying $F^2(p) = p$ and $d(F^2)_p = id$, and hence $F^2 = id_M$ by the uniqueness of (local) isometries. Moreover, it holds that

$$F(\exp_p(v)) = \exp_{F(p)}(dF_p(v)) = \exp_p(-v) \quad \forall v \in B(0, \delta).$$

Since $\exp_p : B(0, \delta) \rightarrow B(p, \delta)$ is a diffeomorphism, we know p is an isolated fixed point.

2. Suppose $F^2 = id_M$ and p is an isolated point. Then $dF_p^2 = id$, and to prove $dF_p = -id$ it suffices to show that 1 is not an eigenvalue of dF_p . Suppose for contradiction that $dF_p(v) = v$ for some $v \neq 0$; then we know

$$F(\exp_p(tv)) = \exp_{F(p)}(dF_p(tv)) = \exp_p(tv) \quad \forall tv \in B(0, \delta),$$

which is contrary to that p is an isolated point. $\square$

Problem 7.2

Let $f : (M, g_M) \rightarrow (N, g_N)$ be a Riemannian submersion.

- (1) Show that (N, g_N) is complete if (M, g_M) is complete.
- (2) Show that f is a fibration if (M, g_M) is complete i.e., for every $x \in N$ there is a neighborhood U such that $f^{-1}(U)$ is diffeomorphic to $U \times f^{-1}(x)$.
- (3) Give a counterexample of (2) when (M, g_M) is not complete.

Proof. Let $H \subset TM$ be the subbundle such that $H_p \perp \ker(f_{*p})$. Since f is a Riemannian submersion, $df_p|_{H_p} : H_p \rightarrow T_{f(p)}N$ is a linear isometry.

(1) Fix some $x \in N$ and $v \in T_x N$. Pick some $p \in f^{-1}(x)$. Since $df_p|_{H_p} : H_p \rightarrow T_{f(p)}N$ is a linear isometry, we set

$$\bar{v}_p = (df_p|_{H_p})^{-1}(v).$$

Since (M, g_M) is complete, let $\tilde{\gamma} : \mathbb{R} \rightarrow M$ be the geodesic with $\tilde{\gamma}'(0) = \bar{v}_p$. By Proposition 10.5, we know $\gamma := f \circ \tilde{\gamma} : \mathbb{R} \rightarrow N$ is also a geodesic. Note that $\gamma(0) = x$ and $\gamma'(0) = v$. Then from the arbitrariness of x and v we know (N, g_N) is complete.

(2) Fix some $x \in N$, set $\delta = \text{inj}(N, x)$. Since f is a Riemannian submersion, the level set $f^{-1}(x)$ is an embedded submanifold of dimension $m - n$. Let $U = B(x, \delta)$ and $B(0, \delta) = \{v \in T_x N : |v| < \delta\}$. Since (M, g) is complete, we can consider the map

$$F : B(0, \delta) \times f^{-1}(x) \rightarrow f^{-1}(U), \quad (v, p) \mapsto \exp_p(\bar{v}_p) \quad \text{where } \bar{v}_p = (df_p|_{H_p})^{-1}(v).$$

From ODE theory we know F is smooth. In the next we show F is a diffeomorphism.

Step 1. First we show F is injective. Suppose $\exp_p(\bar{v}_p) = \exp_q(\bar{w}_q)$ for some $p, q \in f^{-1}(x)$ and $v, w \in B(0, \delta)$. Then by Proposition 10.5, we know

$$\exp_x(w) = f \circ \exp_q(\bar{w}_q) = f \circ \exp_p(\bar{v}_p) = \exp_x(v).$$

It follows that $v = w$. Set $\tilde{\gamma}_1 : \mathbb{R} \rightarrow M$ and $\tilde{\gamma}_2 : \mathbb{R} \rightarrow M$ by

$$\tilde{\gamma}_1(t) = \exp_p(t\bar{v}_p) \quad \text{and} \quad \tilde{\gamma}_2(t) = \exp_q(t\bar{w}_q).$$

By Proposition 10.5, both $\tilde{\gamma}_1$ and $\tilde{\gamma}_2$ are horizontal liftings of $\exp_x(tv)$. Since $\tilde{\gamma}_1(1) = \tilde{\gamma}_2(1)$, by the uniqueness of horizontal liftings (Lemma 10.3), we know $\tilde{\gamma}_1(t) \equiv \tilde{\gamma}_2(t)$ and hence we get $p = q$ for $t = 0$. Therefore, F is injective.

Step 2. Then we show F is surjective. For any $y \in f^{-1}(U)$, there exists $v \in B(0, \delta)$ with $f(y) = \exp_x(v)$. Consider the curve $\gamma(t) = \exp_x(tv)$ where $t \in \mathbb{R}$. Set $w = -\gamma'(1)$. Then by Proposition 10.5, we know

$$f \circ \exp_y(\bar{w}_y) = \exp_{f(y)}(w) = x.$$

Thus $p := \exp_y(\bar{w}_y) \in f^{-1}(x)$. Set $\tilde{\gamma}_1 : \mathbb{R} \rightarrow M$ and $\tilde{\gamma}_2 : \mathbb{R} \rightarrow M$ by

$$\tilde{\gamma}_1(t) = \exp_p(t\bar{v}_p) \quad \text{and} \quad \tilde{\gamma}_2(t) = \exp_y((1-t)\bar{w}_y).$$

By Proposition 10.5, both $\tilde{\gamma}_1$ and $\tilde{\gamma}_2$ are horizontal liftings of $\exp_x(tv)$. Since $\tilde{\gamma}_1(0) = \tilde{\gamma}_2(0)$, by the uniqueness of horizontal liftings (Lemma 10.3), we know $\tilde{\gamma}_1(t) \equiv \tilde{\gamma}_2(t)$ and hence we get $\exp_p(\bar{v}_p) = y$ for $t = 1$. Therefore, F is surjective.

Step 3. Finally we show F is a diffeomorphism. It suffices to show F^{-1} is smooth. Note that the function

$$v(y) := (\exp_x|_{B(0, \delta)})^{-1}(f(y))$$

is smooth with respect to y . Set $\gamma_v(t) = \exp_x(tv)$. Then $\gamma'_v(1)$ is smooth with respect to v , and hence is smooth with respect to y . Therefore, the function

$$p(y) := \exp_y(\bar{w}_y) \quad \text{where } w = -\gamma'_v(1)$$

is also smooth with respect to y . From point 2, we know $F^{-1}(y) = (v(y), p(y))$. Hence F^{-1} is smooth.

(3) Let $(M, g_M) = (\mathbb{R}^2 \setminus \{(0, 0)\}, g_{can})$ and $(N, g_N) = (\mathbb{R}, g_{can})$. Then the map

$$\pi : M \rightarrow N, \quad (x_1, x_2) \mapsto x_1$$

is clearly a Riemannian submersion. However, for any neighborhood U of $0 \in \mathbb{R}$, $\pi^{-1}(U) = (U \times \mathbb{R}) \setminus \{(0, 0)\}$ and $U \times \pi^{-1}(0) = U \times (\mathbb{R} \setminus \{0\})$. These two spaces cannot be diffeomorphic since $H_0(\pi^{-1}(U)) \neq H_0(U \times \pi^{-1}(0))$. $\square$

Problem 7.3

Let (M, g) be a complete, connected Riemannian manifold, and $p, q \in M$.

- (1) Show that every path-homotopy class of paths from p to q contains a geodesic segment γ .
- (2) Show that if M has non-positive sectional curvature, then γ is the unique geodesic segment in the given path homotopy class.

Proof. Let $\pi : \tilde{M} \rightarrow M$ be the universal covering map. Setting $\tilde{g} = \pi^*g$, since (M, g) is complete, one easily knows $(\tilde{M}, \tilde{g})$ is also complete. Moreover, the map $\pi : (\tilde{M}, \tilde{g}) \rightarrow (M, g)$ is a local isometry.

(1) Suppose $\alpha = [\sigma]$ is a path-homotopy class where $\sigma : [0, 1] \rightarrow M$ is a curve with $\sigma(0) = p$ and $\sigma(1) = q$. Let $\tilde{\sigma} : [0, 1] \rightarrow \tilde{M}$ be a path lifting of σ . By completeness we let $\tilde{\gamma} : [0, 1] \rightarrow \tilde{M}$ be a geodesic joining $\tilde{\sigma}(0)$ and $\tilde{\sigma}(1)$. Note that $\tilde{M}$ is simply-connected, $\tilde{\gamma}$ is path-homotopic to $\tilde{\sigma}$, and $\gamma := \pi \circ \tilde{\gamma}$ is path-homotopic to σ . Since π is a local isometry, we know γ is a geodesic. Thus we get (1).

(2) Suppose $\tau : [0, 1] \rightarrow M$ is another geodesic joining p and q that is homotopic to γ . Let $\tilde{\tau} : [0, 1] \rightarrow \tilde{M}$ be a path lifting of τ with $\tilde{\tau}(0) = \tilde{\gamma}(0)$. Since τ is path-homotopic to γ , we know $\tilde{\tau}(1) = \tilde{\gamma}(1)$. Since π is a local isometry, we know $(\tilde{M}, \tilde{g})$ also has non-positive sectional curvature, and $\tilde{\tau}$ is also a geodesic. Cartan-Hadamard Theorem then indicates that $\tilde{\tau} = \tilde{\gamma}$ since $\tilde{\tau}, \tilde{\gamma} : [0, 1] \rightarrow \tilde{M}$ are geodesics with same endpoints. Thus $\gamma = \pi \circ \tilde{\gamma} = \pi \circ \tilde{\tau} = \tau$. $\square$

Problem 7.4

Suppose (M, g) is a compact and connected Riemannian manifold. Then every nontrivial free homotopy class in M is represented by a closed geodesic that has minimum length among all admissible loops in the given free homotopy class.

Proof. Since (M, g) is compact, we know $\delta := \text{inj}(M) > 0$.

1. Let $\sigma : [a, b] \rightarrow M$ be a path with $l(\sigma) < \delta$. Then $\text{im}(\sigma) \subset B(p, \delta)$ where $p = \sigma(0)$. Since $\exp_p : B(0, \delta) \rightarrow B(p, \delta)$ is a diffeomorphism, we know σ is path-homotopic to the geodesic γ in $B(p, \delta)$ which joins $\sigma(a)$ and $\sigma(b)$. Moreover, γ is a minimal geodesic.

2. In particular, any loop σ with $l(\sigma) < \delta$ is homotopic to the constant loop.

3. Suppose $\mathcal{L}$ is a non-trivial free homotopy class. Let d be the infimum of the lengths of piecewise smooth curves belonging to $\mathcal{L}$. By point 2 we know $d \geq \delta$.

4. Let $\{\gamma_i\}$ be a sequence of piecewise smooth curves belonging to $\mathcal{L}$ such that $l_i := l(\gamma_i) \rightarrow d$. By point 1 we can suppose that γ_i is a piecewise geodesic defined on the interval $[0, l_i]$ parametrized by the arc length.

5. Let $L = \sup_{i \in \mathbb{N}} l_i$. Then for all $i \in \mathbb{N}$ and $0 \leq t_1 \leq t_2 \leq d$ we have

$$d(\gamma_i(t_1), \gamma_i(t_2)) \leq \int_{t_1}^{t_2} |\gamma_i'(t)| dt = t_2 - t_1. \quad (7.2)$$

Therefore the sequence $\{\gamma_i|_{[0, d]}\}$ is equicontinuous. Since M is compact, there exists a subsequence of $\{\gamma_i|_{[0, d]}\}$, which we denote again by $\{\gamma_i\}$, that converges uniformly to a continuous closed curve $\gamma_0 : [0, d] \rightarrow M$. Clearly, $\gamma_0 \in \mathcal{L}$.

6. By formula (7.2), for all $0 \leq t_1 \leq t_2 \leq d$ we have

$$d(\gamma_0(t_1), \gamma_0(t_2)) = \lim_{i \rightarrow \infty} d(\gamma_i(t_1), \gamma_i(t_2)) \leq t_2 - t_1.$$

Let $0 = t_0 < t_1 < \dots < t_k = d$ be a partition such that $t_i - t_{i-1} < \delta$ for all $1 \leq i \leq k$. Since

$$\gamma_0(t) \in B(\gamma_0(t_{i-1}), \delta) \quad \forall t \in [t_{i-1}, t_i],$$

by Corollary 10.2, there exists a piecewise geodesic $\gamma : [0, d] \rightarrow M$ such that each $\gamma|_{[t_{i-1}, t_i]}$ is the (unique) minimal geodesic which joins $\gamma_0(t_{i-1})$ and $\gamma_0(t_i)$. Moreover, each $\gamma|_{[t_{i-1}, t_i]}$ is path-homotopic to $\gamma_0|_{[t_{i-1}, t_i]}$, and hence γ is homotopic to γ_0 .

7. Since $\gamma \in \mathcal{L}$, we know $l(\gamma) \geq d$. On the other hand, we have

$$l(\gamma) = \sum_{i=1}^k l(\gamma|_{[t_{i-1}, t_i]}) = \sum_{i=1}^k d(\gamma_0(t_{i-1}), \gamma_0(t_i)) \leq \sum_{i=1}^k (t_i - t_{i-1}) = d.$$

Hence $l(\gamma) = d$.

8. Suppose $\gamma'(t^-) \neq \gamma'(t^+)$ for some $t \in (0, d)$. Pick some $\epsilon < \min\{t, d - t, \delta/3\}$. Then

$$d(\gamma(t - \epsilon), \gamma(t + \epsilon)) \leq l(\gamma|_{[t - \epsilon, t + \epsilon]}) \leq 2\epsilon < \delta.$$

Corollary 10.2 then indicates that $\gamma|_{[t - \epsilon, t + \epsilon]}$ is not a minimal curve, and there exists a minimal geodesic σ joining $\gamma(t - \epsilon)$ and $\gamma(t + \epsilon)$ which is path-homotopic to $\gamma|_{[t - \epsilon, t + \epsilon]}$. Then we can construct a new curve $\tilde{\gamma} \in \mathcal{L}$ with $l(\tilde{\gamma}) < l(\gamma)$, a contradiction.

9. Similarly we know $\gamma'(0) = \gamma'(d)$. Thus γ is a closed geodesic. We are done. $\square$

Problem 7.5

Let (M, g) and $(\tilde{M}, \tilde{g})$ be two Riemannian manifolds of dimension n . Fix some $p \in M$ and $\tilde{p} \in \tilde{M}$ and let $\Phi_0 : T_p M \rightarrow T_{\tilde{p}} \tilde{M}$ be some fixed linear isometry. Suppose $\delta \in (0, \min\{\text{inj}_p M, \text{inj}_{\tilde{p}} \tilde{M}\})$. If $v \in T_p M$ with $|v| < \delta$, we set $\gamma(t) = \exp_p(tv)$, $\tilde{\gamma}(t) = \exp_{\tilde{p}}(t\Phi_0(v))$ and

$$\Phi_t = P_{0, t; \tilde{\gamma}} \circ \Phi_0 \circ P_{0, t; \gamma}^{-1} : T_{\gamma(t)} M \rightarrow T_{\tilde{\gamma}(t)} \tilde{M}. \quad (7.3)$$

If $\varphi : B(p, \delta) \rightarrow B(\tilde{p}, \delta)$ is an isometry such that $\varphi(p) = \tilde{p}$ and $\varphi_{*p} = \Phi_0$, show that

$$\varphi_{*\gamma(t)} = \Phi_t.$$

Proof. To show the conclusion it suffices to show that for $t < 1$ we have

$$\varphi_{*\gamma(t)} \circ P_{0,t;\gamma} = P_{0,t;\tilde{\gamma}} \circ \varphi_{*p}. \quad (7.4)$$

Fix some $v \in T_p M$. Set $V_1(t) = \varphi_{*\gamma(t)}(P_{0,t;\gamma}(v))$ and $V_2(t) = P_{0,t;\tilde{\gamma}}(\varphi_{*p}(v))$. Then V_1 and V_2 are smooth vector fields along $\tilde{\gamma}$. Let ∇ and $\tilde{\nabla}$ be the pullback connections of γ and $\tilde{\gamma}$ respectively.

Then $\tilde{\nabla} \frac{d}{dt} V_2 = \tilde{\nabla} \frac{d}{dt} (P_{0,t;\tilde{\gamma}}(w)) = 0$ where $w = \varphi_{*p}(v)$. Since φ is a local isometry, the second fundamental form $B = 0$. Let $\{x^i\}$ and $\{y^\alpha\}$ be local charts on M and $\tilde{M}$ respectively. Set

$$\hat{e}_i = \gamma^* \frac{\partial}{\partial x^i}, \quad \tilde{e}_\alpha = \tilde{\gamma}^* \frac{\partial}{\partial y^\alpha}$$

Then for any $V(t) = f^i(t) \cdot \hat{e}_i$ we know

$$\varphi_i^\alpha = \frac{\partial \varphi^\alpha}{\partial x^i} \circ \gamma$$

and

$$\begin{aligned} \tilde{\nabla} \frac{d}{dt} (\varphi_* V) - \varphi_* (\nabla \frac{d}{dt} V) &= \tilde{\nabla} \frac{d}{dt} (f^i \varphi_i^\alpha \cdot \tilde{e}_\alpha) - \varphi_* \left((f^i)' \cdot \hat{e}_i + f^i \frac{d\gamma^j}{dt} \Gamma_{ji}^k \circ \gamma \cdot \hat{e}_k \right) \\ &= f^i \frac{d\gamma^j}{dt} \frac{\partial^2 \varphi^\alpha}{\partial x^j \partial x^i} \cdot \tilde{e}_\alpha + f^i \varphi_i^\alpha \frac{d\tilde{\gamma}^\beta}{dt} \tilde{\Gamma}_{\beta\alpha}^\gamma \circ \tilde{\gamma} \cdot \tilde{e}_\gamma - f^i \frac{d\gamma^j}{dt} \Gamma_{ji}^k \circ \gamma \cdot \varphi_k^\alpha \cdot \tilde{e}_\alpha \\ &= f^i \frac{d\gamma^j}{dt} \left(\frac{\partial^2 \varphi^\alpha}{\partial x^j \partial x^i} + \varphi_i^\alpha \varphi_j^\beta \tilde{\Gamma}_{\beta\alpha}^\gamma \circ \tilde{\gamma} - \Gamma_{ji}^k \circ \gamma \cdot \varphi_k^\alpha \right) \tilde{e}_\alpha \\ &= f^i \frac{d\gamma^j}{dt} B_{ji}^\alpha \cdot \tilde{e}_\alpha = 0. \end{aligned}$$

For $V(t) = P_{0,t;\gamma}(v)$, we then get

$$\tilde{\nabla} \frac{d}{dt} V_1 = \tilde{\nabla} \frac{d}{dt} (\varphi_* V) = \varphi_* (\nabla \frac{d}{dt} V) = 0.$$

Since $V_1(0) = V_2(0) = \varphi_{*p}(v)$, we get $V_1 \equiv V_2$ which implies (7.4). $\square$

Problem 7.6

Let (M, g) be a simply connected complete Riemannian manifold. If each exponential map is distance increasing, i.e.

$$d_g(\exp_p(v_0), \exp_p(v_1)) \geq |v_0 - v_1| \quad (7.5)$$

for all $p \in M$ and all $v_0, v_1 \in T_p M$, then (M, g) has non-positive sectional curvature.

Proof. Fix some $p \in M$ and fix some $v_0, v_1 \in T_p M$. We define $\alpha : [0, 1] \times [0, 1] \rightarrow M$ by $\alpha(s, t) = \exp_p(t((1-s)v_0 + sv_1))$. Then the variation vector field $J(t) = \alpha_* \left(\frac{\partial}{\partial s} \right) \Big|_{s=0}$ is a Jacobi field along α_0 where $\alpha_0(t) = \alpha(0, t) = \exp_p(tv_0)$. For $t \in [0, 1]$, we define $\sigma_t : [0, 1] \rightarrow M$ by

$$\sigma_t(s) = \exp_p(t((1-s)v_0 + sv_1)).$$

By distance increasing condition (7.5), for all $s \in [0, 1]$ we know

$$l(\sigma_t|_{[0,s]}) \geq d_g(\exp_p(tv_0), \exp_p(t((1-s)v_0 + sv_1))) \geq st|v_0 - v_1|.$$

In particular, we know

$$|J(t)| = |\sigma_t'(0)|_g \geq t|v_0 - v_1|, \quad \forall t \in [0, 1].$$

Note that $J(0) = 0$, and $J'(0) = \tilde{\nabla} \frac{\partial}{\partial s} \alpha_* \left(\frac{\partial}{\partial t} \right) \Big|_{(0,0)} = v_1 - v_0$. Hence

$$\begin{aligned} |J(t)|^2 &= |J'(0)|^2 t^2 - \frac{1}{3} R(J'(0), \alpha'_0(0), \alpha'_0(0), J'(0)) t^4 + o(t^4) \\ &= |v_0 - v_1|^2 t^2 - \frac{1}{3} R(v_1 - v_0, v_0, v_0, v_1 - v_0) t^4 + o(t^4) \\ &= |v_0 - v_1|^2 t^2 - \frac{1}{3} R(v_1, v_0, v_0, v_1) t^4 + o(t^4). \end{aligned}$$

Since $|J(t)| \geq t|v_0 - v_1|$, we know $R(v_1, v_0, v_0, v_1) \leq 0$. Now by the arbitrariness of p, v_0, v_1 , we know $\sec(g) \leq 0$. $\square$

Problem 7.7

Let (M, g) be a simply connected complete Riemannian manifold. If the differential of each exponential map is length increasing, i.e.

$$|(d\exp_p)_v \tilde{v}| \geq |\tilde{v}| \quad (7.6)$$

for all $p \in M$ and all $v, \tilde{v} \in T_p M$, then (M, g) has non-positive sectional curvature.

Proof. Fix some $p \in M$ and fix some $v, \tilde{v} \in T_p M$. We define $\alpha : [0, 1] \times [0, 1] \rightarrow M$ by

$$\alpha(s, t) = \exp_p(t(v + s\tilde{v})).$$

Then the variation vector field $J(t) = \alpha_* \left(\frac{\partial}{\partial s} \right) \Big|_{s=0}$ is a Jacobi field along α_0 where $\alpha_0(t) = \alpha(0, t) = \exp_p(tv)$. By our condition we know

$$|J(t)| = |(d\exp_p)_{tv}(t\tilde{v})| \geq t|\tilde{v}|, \quad \forall t \in [0, 1].$$

Note that $J(0) = 0$, $J'(0) = \bar{\nabla}_{\frac{\partial}{\partial s}} \alpha_* \left(\frac{\partial}{\partial t} \right) \Big|_{(0,0)} = \tilde{v}$. Hence

$$\begin{aligned} |J(t)|^2 &= |J'(0)|^2 t^2 - \frac{1}{3} R(J'(0), \alpha'_0(0), \alpha'_0(0), J'(0)) t^4 + o(t^4) \\ &= |\tilde{v}|^2 t^2 - \frac{1}{3} R(\tilde{v}, v, v, \tilde{v}) t^4 + o(t^4). \end{aligned}$$

Since $|J(t)| \geq t|\tilde{v}|$, we know $R(\tilde{v}, v, v, \tilde{v}) \leq 0$. Now by the arbitrariness of $p, v, \tilde{v}$, we know $\sec(g) \leq 0$. $\square$

Problem 7.8

Let (M, g) be a Cartan-Hadamard manifold. Fix $p \in M$ and two vectors $v_0, v_1 \in T_p M$. Show that for $0 < t \leq T$

$$|v_0 - v_1| \leq \frac{d(\exp_p(tv_0), \exp_p(tv_1))}{t} \leq \frac{d(\exp_p(Tv_0), \exp_p(Tv_1))}{T}. \quad (7.7)$$

Proof. Fix some $p \in M$ and two different vectors $v_0, v_1 \in T_p M$. Since (M, g) is a Cartan-Hadamard manifold, $\exp_p : T_p M \rightarrow M$ is a diffeomorphism.

1. Let $\gamma : [0, 1] \rightarrow M$ be the geodesic joining $\exp_p(v_0)$ and $\exp_p(v_1)$. Set $\tilde{\gamma} = \exp_p^{-1} \circ \gamma$. We define $\alpha : [0, 1] \times [0, 1] \rightarrow M$ by

$$\alpha(s, t) = \exp_p(t\tilde{\gamma}(s)).$$

Then the variation vector field $J_{s_0}(t) = \alpha_* \left(\frac{\partial}{\partial s} \right) \Big|_{s=s_0}$ is a Jacobi field along α_{s_0} . Since p has no conjugate point along γ , we know $|J_{s_0}(t)| > 0$ for $t > 0$. It is easy to see $\varphi_{s_0}(t) = |J_{s_0}(t)|$ is smooth, and

$$\varphi'_{s_0}(t) = |J_{s_0}|^{-1} \langle J_{s_0}, J'_{s_0} \rangle$$

Moreover, for $t > 0$ we have

$$\varphi''_{s_0}(t) = \frac{|J'_{s_0}|^2 + \langle J_{s_0}, J''_{s_0} \rangle}{|J_{s_0}|} - \frac{\langle J_{s_0}, J'_{s_0} \rangle^2}{|J_{s_0}|^3} = \frac{|J'_{s_0}|^2 |J_{s_0}|^2 - \langle J_{s_0}, J'_{s_0} \rangle^2}{|J_{s_0}|^3} - \frac{R(J_{s_0}, \alpha'_{s_0}, \alpha'_{s_0}, J_{s_0})}{|J_{s_0}|} \geq 0.$$

2. Set $f_{s_0}(t) = |J_{s_0}(t)| - t|J'_{s_0}(0)|$. Then $f''_{s_0} \geq 0$, $f'_{s_0}(0) = 0$, and $f_{s_0}(0) = 0$. Hence $f_{s_0} \geq 0$, which implies $|\gamma'(s_0)| = |J_{s_0}(1)| \geq |J'_{s_0}(0)| = |\tilde{\gamma}'(s_0)|$. Therefore,

$$d(\exp_p(v_0), \exp_p(v_1)) = \int_0^1 |\gamma'(s)| ds \geq \int_0^1 |\tilde{\gamma}'(s)| ds \geq \left| \int_0^1 \tilde{\gamma}'(s) ds \right| = |v_0 - v_1|.$$

By the arbitrariness of v_0 and v_1 we get

$$|v_0 - v_1| \leq \frac{d(\exp_p(tv_0), \exp_p(tv_1))}{t}.$$

3. Set $F_{s_0}(t) = |J_{s_0}(t)|/t$. Since $t\varphi''_{s_0}(t) \geq 0$ and $(t\varphi'_{s_0}(t) - \varphi_{s_0}(t))|_{t=0} = 0$ we know $t\varphi'_{s_0}(t) - \varphi_{s_0}(t) \geq 0$, and hence $F'_{s_0}(t) \geq 0$. Therefore, F_{s_0} is increasing, and then for $0 < t \leq 1$ we have

$$\frac{d(\exp_p(tv_0), \exp_p(tv_1))}{t} \leq \int_0^1 F_s(t)ds \leq \int_0^1 F_s(1)ds = l(\gamma) = d(\exp_p(v_0), \exp_p(v_1)).$$

By the arbitrariness of v_0 and v_1 , we get

$$\frac{d(\exp_p(tv_0), \exp_p(tv_1))}{t} \leq \frac{d(\exp_p(Tv_0), \exp_p(Tv_1))}{T}.$$

We are done. $\square$

Problem 7.9

Let (M, g) be a Cartan-Hadamard manifold. Given $p \in M$, let $f : M \rightarrow [0, +\infty)$ be the function $f(x) = \frac{1}{2}d(x, p)^2$. Show that f is strictly geodesically convex, i.e. for any (nontrivial) geodesic $\gamma : [0, 1] \rightarrow M$, the following inequality holds for all $t \in (0, 1)$:

$$f(\gamma(t)) < (1-t)f(\gamma(0)) + tf(\gamma(1)). \quad (7.8)$$

Proof. Let $q = \gamma(t)$, $v = \gamma'(t)$. Since (M, g) is a Cartan-Hadamard manifold, setting $w = \exp_q^{-1}(p)$, we know

$$\begin{aligned} f(\gamma(0)) &= \frac{1}{2}d(\exp_q(-tv), \exp_q(w))^2 \geq \frac{1}{2}|w + tv|^2, \\ f(\gamma(1)) &= \frac{1}{2}d(\exp_q((1-t)v), \exp_q(w))^2 \geq \frac{1}{2}|w - (1-t)v|^2, \\ f(\gamma(t)) &= \frac{1}{2}d(p, q)^2 = \frac{1}{2}|w|^2. \end{aligned}$$

Thus for all $t \in (0, 1)$ we have

$$\begin{aligned} (1-t)f(\gamma(0)) + tf(\gamma(1)) &\geq \frac{1}{2}[(1-t)|w + tv|^2 + t|w - (1-t)v|^2] \\ &= \frac{1}{2}|w|^2 + \frac{1}{2}t(1-t)|v|^2 > \frac{1}{2}|w|^2 = f(\gamma(t)). \end{aligned}$$

We are done. $\square$

Problem 7.10

Suppose M and N are compact smooth manifolds. Show that $M \times N$ does not admit a Riemannian metric with negative sectional curvature.

Proof. 1. Suppose both $\pi_1(M)$ and $\pi_1(N)$ are non-trivial. By Cartan's theorem, we know $\pi_1(M \times N)$ is torsion-free. Then from our hypothesis we know $\pi_1(M \times N) \cong \pi_1(M) \times \pi_1(N)$ contains a subgroup isomorphic to $\mathbb{Z} \times \mathbb{Z}$ which contradicts Preissmann's theorem.

2. Suppose one of $\pi_1(M)$ and $\pi_1(N)$ is trivial. We may assume M is simply connected. Let $\tilde{N}$ be the universal covering of N . Then Cartan-Hadamard Theorem implies that $M \times \tilde{N} \cong \mathbb{R}^{m+n}$. Since M is simply connected, we know M is orientable, and hence

$$H_{dR}^m(M) = \mathbb{R} \quad \text{where } m = \dim M.$$

Since M is compact, from Künneth formula we know

$$0 = H_{dR}^m(\mathbb{R}^{m+n}) = H_{dR}^m(M \times \tilde{N}) \cong \bigoplus_{i+j=m} H_{dR}^i(M) \otimes H_{dR}^j(\tilde{N}) \neq 0,$$

a contradiction. $\square$

Problem 7.11

- (1) Is $(\mathbb{R}^2, g_{-1} = e^{-(x^2+y^2)}(dx \otimes dx + dy \otimes dy))$ a complete Riemannian manifold?
- (2) Is $(\mathbb{R}^2, g_1 = e^{x^2+y^2}(dx \otimes dx + dy \otimes dy))$ a complete Riemannian manifold?

Proof. Generally, for $g_a = e^{a(x^2+y^2)}(dx \otimes dx + dy \otimes dy)$, we have

$$\Gamma_{ij}^k = a(\delta_{jk}x_i + \delta_{ik}x_j - \delta_{ij}x_k).$$

For each $\gamma : I \rightarrow M$ with $\gamma(t) = (x(t), y(t))$ we know

$$\ddot{\gamma}^k + \Gamma_{ij}^k \dot{\gamma}^i \dot{\gamma}^j = 0 \quad \forall k \iff \begin{cases} \ddot{x} + ax(\dot{x}^2 - \dot{y}^2) + 2ay\dot{x}\dot{y} = 0, \\ \ddot{y} + ay(\dot{y}^2 - \dot{x}^2) + 2ax\dot{x}\dot{y} = 0. \end{cases}$$

We solve in polar coordinates. For each $v = (\cos \theta, \sin \theta) \in T_0 \mathbb{R}^2 \cong \mathbb{R}^2$, the curve $\gamma(t) = (r(t) \cos \theta, r(t) \sin \theta)$ where $\ddot{r} + ar\dot{r}^2 = 0, r(0) = 0, r'(0) = 1$, is a geodesic with $\gamma(0) = 0, \gamma'(0) = (\cos \theta, \sin \theta)$. Substitute t with τ , where t is given by

$$t = \int_0^\tau e^{\frac{1}{2}ax^2} dx. \quad (7.9)$$

Then the equation $\ddot{r} + ar\dot{r}^2 = 0$ becomes

$$r''(\tau) - a\tau \cdot r'(\tau) + ar \cdot (r'(\tau))^2 = 0,$$

which has a solution $r(\tau) = \tau$. Moreover, the solution satisfies $r|_{t=0} = r|_{\tau=0} = 0$ and

$$\left. \frac{dr}{dt} \right|_{t=0} = \left. \frac{dr}{d\tau} \right|_{\tau=0} \cdot \left(\left. \frac{dt}{d\tau} \right|_{\tau=0} \right)^{-1} = 1 \cdot \left(e^{\frac{1}{2}a\tau^2} \Big|_{\tau=0} \right)^{-1} = 1.$$

Hence $r(t) = \tau(t)$ is a solution as desired, where $\tau(t)$ is given by (7.9).

1. Suppose $a > 0$. Then $\int_0^{+\infty} e^{\frac{1}{2}ax^2} dx = +\infty$ and $\int_0^{-\infty} e^{\frac{1}{2}ax^2} dx = -\infty$. Hence $r(t) = \tau(t)$ can be defined on $\mathbb{R}$. Thus $\exp_0$ can be defined on $\mathbb{R}^2$, which implies $(\mathbb{R}^2, g_a)$ is complete.

2. Suppose $a < 0$. Then $\int_0^{+\infty} e^{\frac{1}{2}ax^2} dx < +\infty$, and hence $r(t) = \tau(t)$ cannot be defined on $\mathbb{R}$. It follows that $(\mathbb{R}^2, g_a)$ is not complete.

An Alternative Proof. (1) Suppose $a = -1$. For any $p = (x, y)$, define $\gamma : [0, 1] \rightarrow \mathbb{R}^2$ by $\gamma(t) = (tx, ty)$. Then

$$\begin{aligned} d(0, p) &\leq l(\gamma) = \int_0^1 |\gamma'(t)| dt = \int_0^1 \left(\sqrt{x^2 + y^2} \cdot e^{-t^2 \cdot \frac{x^2+y^2}{2}} \right) dt \\ &= \sqrt{2} \int_0^{\sqrt{\frac{x^2+y^2}{2}}} e^{-\tau^2} d\tau \leq \sqrt{2} \int_0^{+\infty} e^{-\tau^2} d\tau = \sqrt{\frac{\pi}{2}}. \end{aligned}$$

Hence $(\mathbb{R}^2, g_{-1})$ is bounded. Since $\mathbb{R}^2$ is not compact, we know $(\mathbb{R}^2, g_{-1})$ is not complete.

(2) Suppose $a = 1$. Since $e^{x^2+y^2} \geq 1$, we know $g_1 \geq g_{can}$ and hence

$$d_{g_1}(p, q) \geq d_{g_{can}}(p, q) \quad \forall p, q \in M.$$

Therefore, any bounded set with respect to g_1 is bounded with respect to g_{can} , and hence is compact. Thus $(\mathbb{R}^2, g_1)$ is complete. $\square$

Problem 7.12

Prove the following statement: Suppose $\triangle ABC$ is a geodesic triangle in Cartan-Hadamard manifold (M, g) , and let a, b, c denote the lengths of the segments opposite the vertices A, B and C , respectively. Then the following inequalities hold:

$$(1) \quad c^2 \geq a^2 + b^2 - 2ab \cos \angle C.$$

$$(2) \quad \angle A + \angle B + \angle C \leq \pi.$$

Moreover, if the Cartan-Hadamard manifold has negative sectional curvature K , then the inequalities are strict.

Proof. 1. Suppose $A = \exp_C(v_0)$ and $B = \exp_C(v_1)$. Since (M, g) is Cartan-Hadamard manifold, we know $b = |v_0|$ and $a = |v_1|$. By Problem 7.8 we know

$$c^2 = d(A, B)^2 \geq |v_0 - v_1|^2 = |v_0|^2 + |v_1|^2 - 2|v_0||v_1| \cos \angle C = a^2 + b^2 - 2ab \cos \angle C. \quad (7.10)$$

We can choose a geodesic triangle $\Delta \tilde{A}\tilde{B}\tilde{C}$ in $(\mathbb{R}^2, g_{can})$ with $\tilde{a} = a, \tilde{b} = b$ and $\tilde{c} = c$. Then

$$\angle \tilde{A} + \angle \tilde{B} + \angle \tilde{C} = \pi$$

and from (7.10) we have

$$\angle \tilde{C} = \arccos \frac{a^2 + b^2 - c^2}{2ab} \geq \arccos(\cos \angle C) = \angle C.$$

Hence

$$\angle A + \angle B + \angle C \leq \pi.$$

2. We use the notations in Problem 7.8 with $p = C$. Since (M, g) has negative sectional curvature K , for each $s_0 \in [0, 1]$ we know

$$\varphi''_{s_0}(t) = \frac{|J'_{s_0}|^2 |J_{s_0}|^2 - \langle J_{s_0}, J'_{s_0} \rangle^2}{|J_{s_0}|^3} - \frac{R(J_{s_0}, \gamma', \gamma', J_{s_0})}{|J_{s_0}|} > 0.$$

Set $f_{s_0}(t) = |J_{s_0}(t)| - t|J'_{s_0}(0)|$. Then $f''_{s_0} > 0, f'_{s_0}(0) = 0$, and $f_{s_0}(0) = 0$. Hence $f_{s_0}(1) > 0$, which implies $|\gamma'(s_0)| = |J_{s_0}(1)| > |J'_{s_0}(0)| = |\tilde{\gamma}'(s_0)|$. Therefore,

$$d(\exp_p(v_0), \exp_p(v_1)) = \int_0^1 |\gamma'(s)| ds > \int_0^1 |\tilde{\gamma}'(s)| ds \geq \left| \int_0^1 \tilde{\gamma}'(s) ds \right| = |v_0 - v_1|.$$

Hence the inequality (1) is strict. It follows that

$$\angle \tilde{C} = \arccos \frac{a^2 + b^2 - c^2}{2ab} > \arccos(\cos \angle C) = \angle C.$$

Hence the inequality (2) is strict. □

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Problem 8.1

Let (M, g) be a connected simply connected, complete Riemannian manifold. Show the following are equivalent.

- (1) (M, g) has constant sectional curvature.
- (2) For every pair of point $p, q \in M$ and every linear isometry $\Phi_0 : T_p M \rightarrow T_q M$, there exists an isometry $\varphi : M \rightarrow M$ such that $\varphi(p) = q$ and $\varphi_{*p} = \Phi_0$.

Proof. (1) $\Rightarrow$ (2):

1. Suppose M has positive curvature. Then $M \cong \mathbb{S}^n(r) \hookrightarrow \mathbb{R}^{n+1}$. WLOG we may assume $p = (0, 0, \dots, 0, r)$. Since $O(n+1)$ acts on M transitively by isometry, we can find $A \in O(n+1)$ such that $Ap = q$.

Let $\{e_i\}$ be an orthonormal basis of $T_p M$. Then $(A^{-1})_*(\Phi_0(e_i))$ is also an orthonormal basis of $T_p M$. We can find a rotation

$$B = \begin{pmatrix} Q & 0 \\ 0 & 1 \end{pmatrix}$$

where $Q \in O(n)$, such that $Bp = p$ and $B_*(e_i) = (A^{-1})_*(\Phi_0(e_i))$. Setting $\varphi = A \circ B$, then $\varphi(p) = q$ and $\varphi_{*p} = \Phi_0$.

2. Suppose M has zero curvature. Then $M \cong \mathbb{R}^n$. WLOG we may assume $p = (0, 0, \dots, 0)$. Let $\tau_q : x \mapsto x + q$ be a translation; it is an isometry. Let $\{e_i\}$ be an orthonormal basis of $T_p M$. Then $(\tau_q^{-1})_*(\Phi_0(e_i))$ is also an orthonormal basis of $T_p M$. We can find a rotation $R \in O(n)$ such that $R_*(e_i) = (\tau_q^{-1})_*(\Phi_0(e_i))$. Setting $\varphi = \tau_q \circ R$, then $\varphi(p) = q$ and $\varphi_{*p} = \Phi_0$.

3. Suppose M has negative curvature. Then $M \cong \mathbb{H}^n(r) \hookrightarrow \mathbb{R}^{n,1}$, where $\mathbb{R}^{n,1}$ is equipped with the sub-Riemannian metric

$$dx^1 \otimes dx^1 + \dots + dx^n \otimes dx^n - dx^{n+1} \otimes dx^{n+1}.$$

WLOG we may assume $p = (0, 0, \dots, r)$. Since

$$O(n, 1) = \{A \in GL(n+1, \mathbb{R}) : A^T I_{n,1} A = I_{n,1}\}$$

acts on M transitively by isometry, we can find $A \in O(n, 1)$ such that $Ap = q$. Let $\{e_i\}$ be an orthonormal basis of $T_p M$. Then $(A^{-1})_*(\Phi_0(e_i))$ is also an orthonormal basis of $T_p M$. We can find a rotation

$$B = \begin{pmatrix} Q & 0 \\ 0 & 1 \end{pmatrix} \quad \text{where } Q \in O(n),$$

such that $Bp = p$ and $B_*(e_i) = (A^{-1})_*(\Phi_0(e_i))$. Setting $\varphi = A \circ B$, then $\varphi(p) = q$ and $\varphi_{*p} = \Phi_0$.

(2) $\Rightarrow$ (1): For any $p, q \in M$ and two-dimensional spaces $\Sigma_p \subset T_p M, \Sigma_q \subset T_q M$, let $\Phi_0 : T_p M \rightarrow T_q M$ be an isometry such that $\Phi_0(\Sigma_p) = \Sigma_q$. Then there is an isometry $\varphi : M \rightarrow M$ such that $\varphi(p) = q$, $\varphi_{*p} = \Phi_0$. Hence $\sec(\Sigma_q) = \sec(\varphi_*(\Sigma_p)) = \sec(\Sigma_p)$, so M has constant curvature. $\square$

Remark 8.2. We can also use the exponential maps to show (1) $\Rightarrow$ (2).

Problem 8.3

Let (M, g) be a complete Riemannian manifold with constant sectional curvature $K = 1$. If $\dim M = 2m$, show that (M, g) is isometric to the sphere $\mathbb{S}^{2m}$ or the real projective space $\mathbb{RP}^{2m}$.

Proof. By Myer's Theorem, we know M is compact. Let $\tilde{M}$ be the universal covering of M equipped with the pullback metric $\tilde{g}$. Since (M, g) has constant curvature 1, we then know $\tilde{M}$ is isometric to $\mathbb{S}^{2m}$.

1. If M is orientable, by Synge's theorem, M is simply connected, and hence $(M, g) = (\tilde{M}, \tilde{g})$ is isometric to $\mathbb{S}^{2m}$.

2. If M is not orientable, by Synge's theorem we know $\pi_1(M) = \mathbb{Z}_2$. Moreover, M is isometric to the Riemannian quotient of the form $\tilde{M}/\Gamma$, where $\Gamma \subset \text{Iso}(\tilde{M}, \tilde{g})$ is discrete and acts freely on $\tilde{M}$, and $\Gamma \cong \mathbb{Z}_2$. We identify $\tilde{M} = \mathbb{S}^{2m} \hookrightarrow \mathbb{R}^{2m+1}$. Suppose $\Gamma = \{id, A\}$. Since $A \in \text{Iso}(\tilde{M}, \tilde{g})$ satisfies $A^2 = id$, we can regard A as a matrix in $O(2m+1)$ with eigenvalues belonging to $\{\pm 1\}$. Since A acts freely on $\mathbb{S}^{2m}$, we then know all eigenvalues are -1 , and hence $A = -I$. It follows that (M, g) is isometric to $\mathbb{RP}^{2m}$ equipped with the canonical metric. $\square$

Problem 8.4

Let $A : \mathbb{R}^{n-1} \rightarrow \mathbb{R}^{n-1}$ be an orthogonal transformation. If $\det A = (-1)^n$, show that there exists some $v \in \mathbb{R}^{n-1} \setminus \{0\}$ such that $Av = v$.

Proof. By choosing another orthonormal basis, we may assume A has the block diagonal form

$$A = \text{diag}(R_1, \dots, R_k, -1, \dots, -1, 1, \dots, 1)$$

where

$$R_i = \begin{pmatrix} \cos \theta_i & -\sin \theta_i \\ \sin \theta_i & \cos \theta_i \end{pmatrix}.$$

We assume "−1" appears for p times and "1" appears for q times. Then $(-1)^n = \det A = (-1)^p$. Note that

$$q = n - 1 - p - 2k \equiv 1 \pmod{2}.$$

In particular $q \neq 0$, so there exists v such that $Av = v$. □

Remark 8.5. Let $A : \mathbb{R}^{n-1} \rightarrow \mathbb{R}^{n-1}$ be an orthogonal transformation. Suppose A preserves the orientation when n is even, and A changes the orientation when n is odd. Then $\det A = (-1)^n$ and hence A fixes at least one point of the unit sphere.

Problem 8.6

Let (M, g) be an n -dimensional compact oriented Riemannian manifold with positive sectional curvature. Given an isometry $f : M \rightarrow M$ such that f preserves the orientation when n is even, and f changes the orientation when n is odd. Show that f has a fixed point.

Proof. Suppose for contradiction that f has no fixed point.

1. Since M is compact we can find $p \in M$ such that $d(p, f(p)) = \inf_{q \in M} d(q, f(q)) > 0$. Let $\gamma : [0, l] \rightarrow M$ be a unit-speed minimal geodesic joining p and $f(p)$.

2. Since f is an isometry, we know $d(f(p), f(f(p))) = d(p, f(p)) = l$. Moreover, for any $q = \gamma(t)$ where $t \in (0, l)$, we have

$$d(q, f(q)) \leq d(q, f(p)) + d(f(p), f(q)) = l - t + t = l = \inf_{x \in M} d(x, f(x)).$$

Hence $d(q, f(q)) = l$. It follows that γ and $f \circ \gamma$ are smoothly jointed at $f(p)$.

3. Consider the linear isometry

$$\varphi = P_{0,l;\gamma}^{-1} \circ df_p : T_p M \rightarrow T_p M.$$

Let $V = (\gamma'(0))^\perp$ be the orthogonal complement space of $\gamma'(0)$ in $T_p M$. From point 2 we know $df_p(\gamma'(0)) = \gamma'(l)$, and hence we obtain a linear isometry

$$\psi = \varphi|_V : V \rightarrow V.$$

Since $P_{0,l;\gamma}^{-1}$ preserves the orientation, from our hypothesis we know $\det(\psi) = (-1)^n$. By linear algebra we can find $e_1 \in V$ such that $\psi(e_1) = e_1$. Hence $P_{0,l;\gamma}(e_1) = df_p(e_1)$.

4. Let $e_1(t) = P_{0,t;\gamma}(e_1)$ and consider the variation

$$\alpha : \mathbb{R} \times [0, l] \rightarrow M, \quad (s, t) \mapsto \exp_{\gamma(t)}(se_1(t)).$$

Let $V = \alpha_* \left(\frac{\partial}{\partial s} \right) \Big|_{s=0}$ be the variation field. Note that $\alpha_0 = \gamma$ and $V = e_1(t)$. By the second variation formula,

$$\frac{d^2}{ds^2} \Big|_{s=0} l(\alpha_s) = \left\langle \bar{\nabla}_s \frac{\partial}{\partial s} \alpha, \gamma' \right\rangle \Big|_{t=0}^{t=l} - \int_0^l R(V, \gamma', \gamma', V)(t) dt < 0.$$

Therefore there exists some $s \neq 0$ with

$$l(\alpha_s) < l(\alpha_0) = l(\gamma).$$

From point 3 we know $df_p(e_1) = e_1(l)$. Since f is an isometry, we then know

$$f(\alpha_s(0)) = f(\exp_p(se_1)) = \exp_{f(p)}(s \cdot df_p(e_1)) = \exp_{\gamma(l)}(se_1(l)) = \alpha_s(l).$$

This implies $l(\alpha_s) \geq d(\alpha_s(0), \alpha_s(l)) = d(\alpha_s(0), f(\alpha_s(0))) \geq \inf_{q \in M} d(q, f(q)) = l(\gamma)$, a contradiction. □

Problem 8.7

Prove that there is no compact manifold that admits both a metric of positive definite Ricci curvature and a metric of non-positive sectional curvature.

Proof. 1. Since M admits a metric of positive definite Ricci curvature and M is compact, by Myer's Theorem we know $\pi_1(M)$ is finite.

2. Since M admits a metric of non-positive sectional curvature, by Cartan's Theorem we know $\pi_1(M)$ is torsion-free, and hence $\pi_1(M)$ must be trivial. Then by Cartan-Hadamard Theorem we know M is diffeomorphic to $\mathbb{R}^n$, which is not compact, a contradiction.

An Alternative Proof. Let $\tilde{M}$ be the universal covering of M . Since M admits a metric of positive definite Ricci curvature and M is compact, $\tilde{M}$ admits a pullback metric $\tilde{g}$ with $Ric(\tilde{g}) \geq K$ for some constant $K > 0$, and hence is compact by Myer's Theorem. On the other hand, since M admits a metric of non-positive sectional curvature, $\tilde{M}$ admits a pullback metric of non-positive sectional curvature, and hence is diffeomorphic to $\mathbb{R}^n$. This implies $\tilde{M}$ is not compact, a contradiction. $\square$

Problem 8.8

In normal coordinates $\{x^1, \dots, x^n\}$ centered at $p \in M$, show that the following identities hold on a small neighborhood U of p

$$\sum_j g_{ij} x^j = x^i,$$

and

$$\begin{aligned} g_{im} &= \delta_{im} - \sum_j \frac{\partial g_{ij}}{\partial x^m} x^j, \quad \sum_j \frac{\partial g_{ij}}{\partial x^m} x^j = \sum_j \frac{\partial g_{mj}}{\partial x^i} x^j, \\ \sum_{i,j} \frac{\partial g_{ij}}{\partial x^m} x^j x^i &= \sum_{i,j} \frac{\partial g_{mj}}{\partial x^i} x^j x^i = 0, \quad \sum_{i,j} \Gamma_{ij}^k x^i x^j = 0. \end{aligned}$$

In particular, show that $\nabla_{\partial_r} \partial_r = 0$ over $U \setminus \{p\}$.

Proof. In HW3 we showed that $\sum_j g_{ij} x^j = x^i$. Therefore, we know

$$\delta_{im} - \sum_j \frac{\partial g_{ij}}{\partial x^m} x^j = \delta_{im} - \sum_j \left(\frac{\partial(g_{ij} x^j)}{\partial x^m} - g_{ij} \frac{\partial x^j}{\partial x^m} \right) = \delta_{im} - \frac{\partial x^i}{\partial x^m} + g_{im} = g_{im},$$

and hence

$$\sum_j \frac{\partial g_{ij}}{\partial x^m} x^j = g_{im} - \delta_{im} = g_{mi} - \delta_{mi} = \sum_j \frac{\partial g_{mj}}{\partial x^i} x^j.$$

It follows that

$$\sum_{i,j} \frac{\partial g_{ij}}{\partial x^m} x^j x^i = \sum_{i,j} \frac{\partial g_{mj}}{\partial x^i} x^j x^i = \sum_i (g_{mi} - \delta_{mi}) x^i = \sum_i g_{mi} x^i - x^m = 0,$$

and hence

$$\sum_{i,j} \Gamma_{ij}^k x^i x^j = \sum_{i,j,l} \frac{1}{2} g^{kl} \left(\frac{\partial g_{il}}{\partial x^j} + \frac{\partial g_{jl}}{\partial x^i} - \frac{\partial g_{ij}}{\partial x^l} \right) x^i x^j = 0.$$

In particular, we know

$$\begin{aligned} \nabla_{\partial_r} \partial_r &= \sum_{i,j} \frac{x^i}{r} \nabla_{\frac{\partial}{\partial x^i}} \left(\frac{x^j}{r} \frac{\partial}{\partial x^j} \right) \\ &= \sum_{i,j} \frac{x^i}{r} \frac{\partial}{\partial x^i} \left(\frac{x^j}{r} \right) \frac{\partial}{\partial x^j} + \sum_{i,j,k} \frac{x^i x^j}{r^2} \Gamma_{ij}^k \frac{\partial}{\partial x^k} \\ &= \frac{x^j}{r^2} \frac{\partial}{\partial x^j} - \sum_{i,j} \frac{(x^i)^2 x^j}{r^4} \frac{\partial}{\partial x^j} = 0. \end{aligned}$$

We are done. □

Problem 8.9

Let (M, g) be a Riemannian manifold and f, h be two smooth functions on M . If there is a point p such that $f(p) = h(p)$ and $f(x) \geq h(x)$ for all x near p , show that

$$\nabla f(p) = \nabla h(p), \quad \text{Hess } f|_p \geq \text{Hess } h|_p, \quad \Delta f(p) \geq \Delta h(p).$$

Proof. WLOG we may assume $h = 0$, then we need to show

$$\nabla f(p) = 0, \quad \text{Hess } f|_p \geq 0, \quad \Delta f(p) \geq 0.$$

Let $\{x^i\}$ be local coordinates near p . Then $\frac{\partial f}{\partial x^i}(p) = 0$, and $\left[\frac{\partial^2 f}{\partial x^i \partial x^j}(p)\right] \geq 0$. It follows that

$$\nabla f(p) = g^{ij}(p) \frac{\partial f}{\partial x^j}(p) \frac{\partial}{\partial x^i} \Big|_p = 0; \quad (\text{Hess } f)(p) = \frac{\partial^2 f}{\partial x^i \partial x^j}(p) dx^i \otimes dx^j \geq 0.$$

From linear algebra it follows that

$$\Delta f(p) = \text{tr}_g(\text{Hess } f)(p) \geq 0.$$

□

Problem 8.10

Let $f : [0, +\infty) \rightarrow [0, +\infty)$ and $g : [0, +\infty) \rightarrow (0, +\infty)$ be two integrable functions. If

$$\lambda(t) = \frac{f(t)}{g(t)} : [0, +\infty) \rightarrow [0, +\infty)$$

is non-increasing, show that

$$F(t) = \frac{\int_0^t f(\tau) d\tau}{\int_0^t g(\tau) d\tau} : [0, +\infty) \rightarrow [0, +\infty)$$

is non-increasing. Moreover, if there exists $0 < t_1 < t_2$ such that $F(t_1) = F(t_2)$, show that $\lambda(t) \equiv \lambda(t_1)$ for almost all $t \in [0, t_2]$.

Proof. We can assume $f(t) > 0$ for all $t \in [0, +\infty)$, otherwise we replace it by $f(t) + \epsilon g(t)$ for some $\epsilon > 0$. Given $0 < t_1 < t_2$, we need to show

$$\int_0^{t_1} f(\tau) d\tau \int_0^{t_2} g(\tau) d\tau - \int_0^{t_2} f(\tau) d\tau \int_0^{t_1} g(\tau) d\tau \geq 0.$$

Indeed,

$$\begin{aligned} & \int_0^{t_1} f(\tau) d\tau \int_0^{t_2} g(\tau) d\tau - \int_0^{t_2} f(\tau) d\tau \int_0^{t_1} g(\tau) d\tau \\ &= \int_0^{t_1} f(\tau) d\tau \int_0^{t_2} g(\tau) d\tau - \int_0^{t_1} f(\tau) d\tau \int_0^{t_1} g(\tau) d\tau - \int_{t_1}^{t_2} f(\tau) d\tau \int_0^{t_1} g(\tau) d\tau \\ &= \int_0^{t_1} f(\tau) d\tau \int_{t_1}^{t_2} g(\tau) d\tau - \int_{t_1}^{t_2} f(\tau) d\tau \int_0^{t_1} g(\tau) d\tau \\ &\geq \int_0^{t_1} \frac{f(t_1)}{g(t_1)} g(\tau) d\tau \int_{t_1}^{t_2} \frac{g(t_1)}{f(t_1)} f(\tau) d\tau - \int_{t_1}^{t_2} f(\tau) d\tau \int_0^{t_1} g(\tau) d\tau = 0. \end{aligned}$$

If $F(t_1) = F(t_2)$, then

$$\begin{aligned} \frac{f(t_1)}{g(t_1)} g(\tau) &\equiv f(\tau) \quad \text{for a.e. } \tau \in [0, t_1], \\ \frac{g(t_1)}{f(t_1)} f(\tau) &\equiv g(\tau) \quad \text{for a.e. } \tau \in [t_1, t_2], \end{aligned}$$

and hence $\lambda(t) \equiv \lambda(t_1)$ for almost all $t \in [0, t_2]$. □

Problem 8.11

Let (M, g) be a Riemannian manifold and $f : M \rightarrow \mathbb{R}$ be a Lipschitz function. Then for any $\varphi \in C_0^\infty(M, \mathbb{R})$, one has

$$-\int_M \langle \nabla \varphi, \nabla f \rangle d\text{vol}_g = \int_M \Delta \varphi \cdot f d\text{vol}_g. \quad (8.1)$$

Proof. Recall that a Lipschitz function is differentiable $\mathcal{L}^n$ -a.e. on Euclidean spaces, and this result still holds for Riemannian measure $\mu = d\text{vol}_g = \sqrt{\det g} d\mathcal{L}^n$.

On the other hand, for any Lipschitz function f and for any $\epsilon > 0$, there exists C^1 -approximation f_ϵ of f such that

$$\begin{aligned} \mu\{x \in M : f_\epsilon(x) \neq f(x)\} &< \epsilon, \\ \mu\{x \in M : \nabla f_\epsilon(x) \neq \nabla f(x)\} &< \epsilon, \\ \sup_M |\nabla f_\epsilon| &\leq C_n \text{Lip}(f). \end{aligned}$$

Then we have

$$\begin{aligned} -\int_M \langle \nabla \varphi, \nabla f \rangle d\text{vol}_g &= -\int_M \langle \nabla \varphi, \nabla f_\epsilon \rangle d\text{vol}_g + \int_{\{\nabla f_\epsilon \neq \nabla f\}} \langle \nabla \varphi, \nabla f_\epsilon - \nabla f \rangle d\text{vol}_g \\ &= \int_M \Delta \varphi \cdot f_\epsilon d\text{vol}_g + \int_{\{\nabla f_\epsilon \neq \nabla f\}} \langle \nabla \varphi, \nabla f_\epsilon - \nabla f \rangle d\text{vol}_g \\ &= \int_M \Delta \varphi \cdot f d\text{vol}_g + \int_{\{f \neq f_\epsilon\}} \Delta \varphi \cdot (f_\epsilon - f) d\text{vol}_g + \int_{\{\nabla f_\epsilon \neq \nabla f\}} \langle \nabla \varphi, \nabla f_\epsilon - \nabla f \rangle d\text{vol}_g, \end{aligned}$$

where

$$\begin{aligned} &\left| \int_{\{f \neq f_\epsilon\}} \Delta \varphi \cdot (f_\epsilon - f) d\text{vol}_g + \int_{\{\nabla f_\epsilon \neq \nabla f\}} \langle \nabla \varphi, \nabla f_\epsilon - \nabla f \rangle d\text{vol}_g \right| \\ &\leq \epsilon \|\Delta \varphi\|_{C^0} \cdot \|f_\epsilon - f\|_{C^0(\text{supp}(\varphi))} + \epsilon \|\nabla \varphi\|_{C^0} \cdot \|\nabla f_\epsilon - \nabla f\|_{C^0(\text{supp}(\varphi))}. \end{aligned}$$

Since $\sup_M |\nabla f_\epsilon| \leq C_n \text{Lip}(f)$, we deduce that

$$\|f_\epsilon - f\|_{C^0(\text{supp}(\varphi))} \leq \|f_\epsilon\|_{C^0(\text{supp}(\varphi))} + \|f\|_{C^0(M)} \leq c \text{Lip}(f) + \|f\|_{C^0(M)}.$$

Letting $\epsilon \rightarrow 0$, we obtain

$$\epsilon \|\Delta \varphi\|_{C^0} \cdot \|f_\epsilon - f\|_{C^0(\text{supp}(\varphi))} + \epsilon \|\nabla \varphi\|_{C^0} \cdot \|\nabla f_\epsilon - \nabla f\|_{C^0(\text{supp}(\varphi))} \rightarrow 0,$$

and hence we get (8.1). □

10 HW10

Definition 9.1. Let (M, g) be a Riemannian manifold and $f \in C^0(M)$. Suppose f_q is a C^2 function defined in a neighborhood U of $q \in M$.

(1) f_q is called a lower barrier function of f at q if

$$f_q(q) = f(q) \quad \text{and} \quad f_q(x) \leq f(x) \quad \forall x \in U.$$

(2) We say $\Delta f(q) \geq c$ in the barrier sense provided for all $\epsilon > 0$ there exists a lower barrier function $f_{q,\epsilon}$ of f at q such that

$$\Delta f_{q,\epsilon}(q) \geq c - \epsilon.$$

Lemma 9.2

Let (M, g) be a Riemannian manifold and $f \in C^0(M)$. If $\Delta f(x) \geq c(x)$ in the barrier sense where $c : M \rightarrow (0, +\infty)$ is a positive function, then f has no local maximum at all.

Proof. Suppose f has a local maximum p , then we can find a barrier function $f_{p,\epsilon}$ such that $f_{p,\epsilon}(p) = f(p)$, $f_{p,\epsilon}(x) \leq f(x)$ near p , and $\Delta f_{p,\epsilon}(p) > c(p)/2$. However, $f_{p,\epsilon}$ has a local maximum p as well, and this indicates $\Delta f_{p,\epsilon} \leq 0$, a contradiction. $\square$

Problem 9.3

Let (M, g) be a connected Riemannian manifold. Suppose $f \in C^0(M)$ and $\Delta f(x) \geq 0$ in the barrier sense for every $x \in M$.

- (1) If f has a local maximum at $p \in M$, then f is constant on a neighborhood of p .
- (2) f has a global maximum if and only if f is constant.

Proof. (1): Suppose the conclusion is false. It suffices to assume that $f|_{\overline{B(p,\delta)}} \leq f(p)$ and there exists $q \in \partial B(p, \delta)$ with $f(q) < f(p)$, where $0 < \delta < \text{inj}(M, p)$. Let $\{x^i\}$ be normal coordinates on $B(p, r)$ where $\delta < r < \text{inj}(M, p)$. WLOG we assume $q = \exp_p(\delta \partial_1)$. By continuity, there exist $\tau > 0$ such that $f|_{\overline{B(q,\tau)}} < f(p)$. Put

$$\varphi = x^1 - d((x^2)^2 + \cdots + (x^n)^2) \quad \text{on } B(p, r),$$

where d is chosen so large that if $x \in \partial B(p, \delta)$ and $\varphi(x) \geq 0$ then $x \in B(q, \tau)$. Since $\frac{\partial \varphi}{\partial x_1} = 1$, we know $|\nabla \varphi| > 0$. For some $a > 0$ to be determined we set

$$\psi = e^{a\varphi} - 1 \quad \text{on } B(p, r).$$

Then $\psi(p) = 0$. Moreover, if $x \in \partial B(p, \delta)$ and $\psi(x) > 0$, then $x \in B(q, \tau)$. Note that

$$\Delta \psi = (a^2 |\nabla \varphi|^2 + a \Delta \varphi) e^{a\varphi}.$$

Since φ is smooth, $|\nabla \varphi| > 0$, and $\overline{B(p, \delta)}$ is compact, we can choose a so large that

$$\Delta \psi > 0 \quad \text{on } \overline{B(p, \delta)}.$$

Since $f|_{\overline{B(q,\tau)}} < f(p)$ and $f|_{\overline{B(p,\delta)}} \leq f(p)$, from the construction of ψ , we can choose $\epsilon > 0$ sufficiently small such that

$$f_\epsilon|_{\partial B(p,\delta)} < f(p) = f_\epsilon(p) \quad \text{where } f_\epsilon = f + \epsilon \psi.$$

Hence f_ϵ has an interior maximum at some $x \in B(p, \delta)$. However,

$$\Delta f_\epsilon(x) \geq \epsilon \Delta \psi(x) > 0 \quad \forall x \in B(p, \delta)$$

in the barrier sense. Then we get a contradiction from Lemma 9.2.

(2): If $f(p)$ is a global maximum, then $\{q : f(q) = f(p)\}$ is open and closed. Since M is connected, we know that $f \equiv f(p)$ on M . $\square$

Problem 9.4

Let (M, g) be a complete Riemannian manifold and $p, q \in M$. Let $\gamma : [0, b] \rightarrow M$ be a unit-speed minimal geodesic with $\gamma(0) = p$ and $\gamma(b) = q$. For any small $\epsilon > 0$ we define

$$r_\epsilon(x) = \epsilon + d(\gamma(\epsilon), x), \quad \forall x \in M.$$

- (1) Show that $q \notin \text{cut}(\gamma(\epsilon))$ and in particular, r_ϵ is smooth at q .
- (2) Show that r_ϵ is an upper barrier function of $r(x) = d(p, x)$ at point q .

Proof. (1): Suppose for contradiction that $q \in \text{cut}(\gamma(\epsilon))$. Then either q is a conjugate point of $\gamma(\epsilon)$ or there is another minimal geodesic connecting q and $\gamma(\epsilon)$. Both cases imply that $\gamma : [0, b] \rightarrow M$ is not a minimal geodesic, a contradiction. Since $q \notin \text{cut}(\gamma(\epsilon))$ we know $d(\gamma(\epsilon), \bullet)$ is smooth at q , and hence r_ϵ is smooth at q .

(2): We have $r_\epsilon(x) = d(p, \gamma(\epsilon)) + d(\gamma(\epsilon), x) \geq d(p, x)$, and $r_\epsilon(q) = d(p, q) = r(q)$. From point (1), r_ϵ is smooth at q , so r_ϵ is an upper barrier function of r at q . $\square$

Problem 9.5

Let (M, g) be a Riemannian manifold of dimension n and $p \in M$. Suppose U is a geodesic ball around p . Suppose there exists some constant k such that

$$\text{Ric}(g) \geq (n-1)kg. \quad (9.1)$$

If there exists some point $q \in M$ with $0 < d(p, q) < \pi/\sqrt{k}$ such that

$$\Delta r(q) = (n-1) \frac{sn'_k(r(q))}{sn_k(r(q))}, \quad (9.2)$$

show that for any unit vector $v \in T_q M$ with $g(v, \partial_r|_q) = 0$,

$$R(v, \partial_r|_q, \partial_r|_q, v) = k.$$

Proof. Let S^n_k be the space form with constant curvature k . Pick $\tilde{p}, \tilde{q} \in S^n_k$ such that $d(\tilde{p}, \tilde{q}) = d(p, q)$. (From Bonnet-Myers theorem we can find such $\tilde{p}, \tilde{q}$). Let $\gamma : [0, l] \rightarrow M$ be a unit-speed minimal geodesic joining p and q , and $\tilde{\gamma} : [0, l] \rightarrow S^n_k$ be a unit-speed minimal geodesic joining $\tilde{p}$ and $\tilde{q}$.

Choose a parallel orthonormal frame $\{e_i(t)\}$ along γ such that $e_1(l) = v$ and $e_n(t) = \gamma'(t)$. Let $J_k(t) = J_k^i(t)e_i(t)$ be the Jacobi field along γ such that $J_k(0) = 0$, $J_k(l) = e_k(l)$ where $1 \leq k \leq n-1$. Let $\tilde{J}_k(t) = \tilde{J}_k^i(t)e_i(t)$ be the vector field along $\tilde{\gamma}$ given by

$$\tilde{J}_k^i(t) = \frac{sn_k(t)}{sn_k(l)} \delta_k^i.$$

Then as in the proof of Laplacian comparison theorem,

$$\begin{aligned} \Delta r(q) &= \sum_{k=1}^{n-1} I(J_k, J_k) \leq \sum_{k=1}^{n-1} I(\tilde{J}_k^i e_i, \tilde{J}_k^i e_i) \\ &= \sum_{k=1}^{n-1} \int_0^l \left(\frac{sn'_k(t)}{sn_k(l)} \right)^2 - \frac{sn_k^2(t)}{sn_k^2(l)} R(e_k(t), \gamma'(t), \gamma'(t), e_k(t)) dt \\ &= (n-1) \int_0^l \left(\frac{sn'_k(t)}{sn_k(l)} \right)^2 dt - \int_0^l \frac{sn_k^2(t)}{sn_k^2(l)} \text{Ric}(\gamma'(t), \gamma'(t)) dt \\ &\leq (n-1) \int_0^l \left(\frac{sn'_k(t)}{sn_k(l)} \right)^2 dt - \int_0^l (n-1)k \frac{sn_k^2(t)}{sn_k^2(l)} dt \\ &= \sum_{k=1}^{n-1} \int_0^l \left(\frac{sn'_k(t)}{sn_k(l)} \right)^2 - \tilde{R}(\tilde{J}_k(t), \tilde{\gamma}'(t), \tilde{\gamma}'(t), \tilde{J}_k(t)) dt \\ &= \sum_{k=1}^{n-1} \tilde{I}(\tilde{J}_k, \tilde{J}_k) = \tilde{\Delta} \tilde{r}(\tilde{q}) = (n-1) \frac{sn'_k(l)}{sn_k(l)}. \end{aligned}$$

From our hypothesis, all the inequalities must be equalities. Therefore, $J_k^i = \tilde{J}_k^i$ and

$$R(v, \partial_r|_q, \partial_r|_q, v) = R(J_1(l), \gamma'(l), \gamma'(l), J_1(l)) = -g(J_1''(l), J_1(l)) = \frac{-sn_k''(l)}{sn_k(l)} = k.$$

□

Lemma 9.6

Let $a, b : [0, +\infty) \rightarrow [0, +\infty)$ be integrable functions. Suppose $b > 0$ and a/b is nonincreasing on $[0, +\infty)$. For any $0 \leq r_1 < r_4 < +\infty$ and r_2, r_3 , we have

$$\int_{r_3}^{r_4} a(t) dt \cdot \int_{r_1}^{r_2} b(t) dt \leq \int_{r_3}^{r_4} b(t) dt \cdot \int_{r_1}^{r_2} a(t) dt, \quad (9.3)$$

and the equality holds if and only if $r_1 = r_2$ or $r_3 = r_4$, or a/b is constant on (r_1, r_4) .

Proof. We first assume $0 \leq r_1 \leq r_2 \leq r_3 \leq r_4 < +\infty$ and denote $Q(t) = a(t)/b(t)$. Then

$$\begin{aligned} \int_{r_3}^{r_4} a(t)dt \cdot \int_{r_1}^{r_2} b(t)dt &= \int_{r_3}^{r_4} b(t)Q(t)dt \cdot \int_{r_1}^{r_2} b(t)dt \\ &\leq Q(r_3) \int_{r_3}^{r_4} b(t)dt \cdot \int_{r_1}^{r_2} b(t)dt \\ &\leq \int_{r_3}^{r_4} b(t)dt \cdot \int_{r_1}^{r_2} b(t)Q(t)dt = \int_{r_3}^{r_4} b(t)dt \cdot \int_{r_1}^{r_2} a(t)dt. \end{aligned}$$

Now for $0 \leq r_1 < r_3 < r_2 < r_4 < \infty$, by the above inequality we have

$$\frac{\int_{r_2}^{r_4} a(t)dt}{\int_{r_2}^{r_4} b(t)dt} \leq \frac{\int_{r_3}^{r_2} a(t)dt}{\int_{r_3}^{r_2} b(t)dt} \leq \frac{\int_{r_1}^{r_3} a(t)dt}{\int_{r_1}^{r_3} b(t)dt}.$$

Consequently

$$\frac{\int_{r_2}^{r_4} a(t)dt + \int_{r_3}^{r_2} a(t)dt}{\int_{r_2}^{r_4} b(t)dt + \int_{r_3}^{r_2} b(t)dt} \leq \frac{\int_{r_3}^{r_2} a(t)dt}{\int_{r_3}^{r_2} b(t)dt} \leq \frac{\int_{r_1}^{r_3} a(t)dt + \int_{r_3}^{r_2} a(t)dt}{\int_{r_1}^{r_3} b(t)dt + \int_{r_3}^{r_2} b(t)dt},$$

and hence we obtain (9.3). The condition for the equality is clear. $\square$

Problem 9.7

Let (M, g) be a complete Riemannian manifold and $p \in M$. Let $B(p, \delta)$ be the metric ball centered at p with radius δ and g_k be the metric with constant sectional curvature k on $B(\tilde{p}, \delta) \setminus \{\tilde{p}\}$. If $Ric(g) \geq (n-1)kg$, then for any $0 \leq r_1 < \min(r_2, r_3) \leq \max(r_2, r_3) < r_4$, show that

$$\frac{Vol_g(B(p, r_4)) - Vol_g(B(p, r_3))}{Vol_g(B(p, r_2)) - Vol_g(B(p, r_1))} \leq \frac{Vol_{g_k}(B(\tilde{p}, r_4)) - Vol_{g_k}(B(\tilde{p}, r_3))}{Vol_{g_k}(B(\tilde{p}, r_2)) - Vol_{g_k}(B(\tilde{p}, r_1))}.$$

Proof. With respect to the normal coordinates, for $\rho \in [0, +\infty)$ we set

$$a(\rho) = \int_{S^{n-1}} I_{\Sigma(p)}(\rho, \omega) \rho^{n-1} \sqrt{\det g} \circ \Phi(\rho, \omega) d\omega, \quad \text{and} \quad b(\rho) = \int_{S^{n-1}} sn_k^{n-1}(\rho) d\omega.$$

In the proof of Bishop-Gromov volume comparison theorem, we have shown that the volume density ratio

$$\lambda(\rho, \omega) = \frac{\rho^{n-1} \sqrt{\det g} \circ \Phi(\rho, \omega)}{sn_k^{n-1}(\rho)}$$

is non-increasing in ρ . Hence a/b is non-increasing in ρ (where $b > 0$). Note that

$$\begin{aligned} \int_{r_1}^{r_2} a(\rho) d\rho &= Vol_g(B(p, r_2)) - Vol_g(B(p, r_1)), \\ \int_{r_1}^{r_2} b(\rho) d\rho &= Vol_{g_k}(B(\tilde{p}, r_2)) - Vol_{g_k}(B(\tilde{p}, r_1)). \end{aligned}$$

Then we get the conclusion from Lemma 9.6. $\square$

Problem 9.8

Let (M, g) be a complete Riemannian manifold of dimension n with $Ric(g) \geq 0$. If

$$\lim_{r \rightarrow \infty} \frac{Vol_g(B(p, r))}{r^n} \geq \frac{Vol(S^{n-1})}{n}, \tag{9.4}$$

show that (M, g) is isometric to $(\mathbb{R}^n, g_{can})$.

Proof. Note that

$$\text{Vol}(\mathbb{B}^n(r)) = r^n \text{Vol}(\mathbb{S}^{n-1})/n.$$

Then by Bishop-Gromov Volume Comparison Theorem, the function

$$\frac{\text{Vol}_g(B(p, r))}{r^n \text{Vol}(\mathbb{S}^{n-1})/n}$$

is non-increasing with respect to r , and

$$\lim_{r \rightarrow 0} \frac{\text{Vol}_g(B(p, r))}{r^n \text{Vol}(\mathbb{S}^{n-1})/n} = 1.$$

From our condition (9.4) we know

$$\lim_{r \rightarrow \infty} \frac{\text{Vol}_g(B(p, r))}{r^n \text{Vol}(\mathbb{S}^{n-1})/n} \geq 1,$$

so we must have

$$\frac{\text{Vol}_g(B(p, r))}{\text{Vol}(\mathbb{B}^n(r))} = 1, \quad \forall r \in (0, +\infty). \quad (9.5)$$

Then by the rigidity of volume comparison, we know g is a flat metric. Therefore, by Cartan-Hadamard Theorem we know the exponential map $\exp_p : (\mathbb{R}^n, g_{\text{can}}) \rightarrow (M, g)$ is a Riemannian covering map. Then from (9.5) we know $\exp_p$ must be injective. Thus (M, g) is isometric to $(\mathbb{R}^n, g_{\text{can}})$. $\square$

Theorem 9.9

Compute the Busemann functions on the upper half plane $\mathbb{H}^2 = \{(x, y) \in \mathbb{R}^2 : y > 0\}$ with canonical metric with constant sectional curvature -1.

Proof. We identify $\mathbb{H}^2$ with the halfplane $\{z = x + iy \in \mathbb{C} | y > 0\}$. Recall that the distance function in $\mathbb{H}^2$ is given by

$$d(z_1, z_2) = 2 \tanh^{-1} \frac{|z_2 - z_1|}{|z_2 - \bar{z}_1|} = \log \frac{|z_2 - \bar{z}_1| + |z_2 - z_1|}{|z_2 - \bar{z}_1| - |z_2 - z_1|}.$$

Case 1. The ray γ starts from $z_1 = x_1 + iy_1$ is a straight line $\{x_1 + it | t \geq y_1\}$. Then for an arbitrary point $z_2 = x_2 + iy_2 \in \mathbb{H}^2$ we have

$$\begin{aligned} b_\gamma(z_2) &= \lim_{t \rightarrow \infty} (d(z_2, \gamma(t)) - d(z_1, \gamma(t))) \\ &= \lim_{t \rightarrow \infty} \left(\log \frac{|z_2 - x_1 + it| + |z_2 - x_1 - it|}{|z_2 - x_1 + it| - |z_2 - x_1 - it|} - \log \frac{|i(y_1 + t)| + |i(y_1 - t)|}{|i(y_1 + t)| - |i(y_1 - t)|} \right) \\ &= \lim_{t \rightarrow \infty} \left(\log \frac{4ty_2}{(|z_2 - x_1 + it| - |z_2 - x_1 - it|)^2} - \log \frac{t}{y_1} \right) \\ &= \log \frac{4y_1y_2}{4y_2^2} = \log \frac{y_1}{y_2}. \end{aligned}$$

Case 2. The ray γ starts from $z_1 = x_1 + iy_1$ and tends to $z_2 = x_2 + 0i$. For an arbitrary point

$z_3 = x_3 + iy_3 \in \mathbb{H}^2$, we have

$$\begin{aligned}
b_\gamma(z_3) &= \lim_{t \rightarrow \infty} (d(z_3, \gamma(t)) - d(z_1, \gamma(t))) \\
&= \lim_{z \rightarrow z_2} \left(\log \frac{|z_3 - \bar{z}| + |z_3 - z|}{|z_3 - \bar{z}| - |z_3 - z|} - \log \frac{|z_1 - \bar{z}| + |z_1 - z|}{|z_1 - \bar{z}| - |z_1 - z|} \right) \\
&= \lim_{z \rightarrow z_2} \left(\log \frac{(|z_3 - \bar{z}| + |z_3 - z|)^2}{4yy_3} - \log \frac{(|z_1 - \bar{z}| + |z_1 - z|)^2}{4yy_1} \right) \\
&= \lim_{z \rightarrow z_2} \left(\log \frac{(|z_3 - \bar{z}| + |z_3 - z|)^2}{y_3} - \log \frac{(|z_1 - \bar{z}| + |z_1 - z|)^2}{y_1} \right) \\
&= \log \frac{|z_3 - z_2|^2}{y_3} - \log \frac{|z_1 - z_2|^2}{y_1} \\
&= \log \frac{(x_3 - x_2)^2 + y_3^2}{y_3} - \log \frac{(x_1 - x_2)^2 + y_1^2}{y_1}.
\end{aligned}$$

□

Theorem 9.10

Does there exist a compact Riemannian manifold (M, g) with $Ric(g) \geq 0$, $b_1(M) = 0$ and $|\pi_1(M)| = +\infty$?

Proof. Let M be the Hantzsche-Wendt manifold. Then it is a compact flat manifold with $b_1(M) = 0$ and $|\pi_1(M)| = +\infty$.

The Hantzsche-Wendt manifold can be built from two cubes that share a face. One construction proceeds as follows:

- (1) The top and bottom faces are glued to one another.
- (2) One of the remaining sides is glued to the opposite side with a 180° rotation.
- (3) One of the remaining faces on the top cube is glued to the matching face of the bottom cube, reflected across an axis parallel to the long axis of the double-cube.
- (4) Repeat step 3 for the remaining pair of faces.

□

Problem 9.11

If (M, g) is compact with $Ric \geq 0$ and has $Ric > 0$ on some tangent space $T_p M$, then $\pi_1(M)$ is finite.

Proof. Let $(\tilde{M}, \tilde{g})$ be the universal covering of (M, g) where $\tilde{g}$ is the pullback metric. Then

$$(\tilde{M}, \tilde{g}) = (N, h) \times (\mathbb{R}^k, g_{\mathbb{R}^k}),$$

where N is a compact manifold. Since $Ric > 0$ on some tangent space $T_p M$, k must be zero, and hence the universal covering is compact. This indicates that $\pi_1(M)$ is finite. □

Problem 9.12

Suppose (M^n, g) is compact and has $b_1(M) = k$. If $Ric \geq 0$ show that the universal covering splits:

$$(\tilde{M}, \tilde{g}) = (N, h) \times (\mathbb{R}^k, g_{\mathbb{R}^k}).$$

Give an example where $b_1 < n$ and $(\tilde{M}, \tilde{g}) = (\mathbb{R}^n, g_{\mathbb{R}^n})$.

Proof. 1. Since $\dim H^1(M) = b_1(M) = k$, there are k linearly independent harmonic 1-forms $\omega_1, \dots, \omega_k$. Let $X_1, \dots, X_k$ be the vector fields given by

$$\langle X_i, Y \rangle = \omega_i(Y) \quad \forall Y.$$

Since ω_i is harmonic, it is easy to see that for each i ,

$$\frac{1}{2} \Delta |X_i|^2 = |\nabla X_i|^2 + \text{Ric}(X_i, X_i) \geq 0.$$

The maximum principle then shows $|X_i|$ must be constant and $\nabla X_i = 0$. Pull the vectors $X_1, \dots, X_k$ back to $\tilde{X}_1, \dots, \tilde{X}_k$ on $\tilde{M}$. Then we get k linearly independent parallel vector fields on $(\tilde{M}, \tilde{g})$.

2. By a linear transformation, we get k linearly independent parallel vector fields $\tilde{E}_1, \dots, \tilde{E}_k$ on $(\tilde{M}, \tilde{g})$ with $\langle \tilde{E}_i, \tilde{E}_j \rangle \equiv \delta_{ij}$. Let $\tilde{\eta}_1, \dots, \tilde{\eta}_k$ be the 1-forms given by

$$\tilde{\eta}_i(Y) = \langle \tilde{E}_i, Y \rangle \quad \forall Y.$$

Since $\nabla \tilde{E}_i = 0$ we know $d\tilde{\eta}_i = 0$. Since $\tilde{M}$ is simply connected, we know $b_1(\tilde{M}) = \text{rank}(\pi_1(\tilde{M})^{ab}) = 0$, and hence $\dim H_{dR}^1(\tilde{M}) = 0$. So there exist smooth functions $\tilde{r}_1, \dots, \tilde{r}_k$ on $\tilde{M}$ such that

$$\tilde{\eta}_i = d\tilde{r}_i \quad \forall 1 \leq i \leq k.$$

It follows that $\nabla \tilde{r}_i = \tilde{E}_i$. Since $\nabla \tilde{E}_i = 0$ and $\langle \tilde{E}_i, \tilde{E}_j \rangle \equiv \delta_{ij}$, we know

$$\langle \nabla \tilde{r}_i, \nabla \tilde{r}_j \rangle \equiv \delta_{ij} \quad \text{and} \quad \text{Hess } \tilde{r}_i \equiv 0. \quad (9.6)$$

3. Since $|\nabla \tilde{r}_1| \equiv 1$ and $\text{Hess } \tilde{r}_1 = 0$, $(\tilde{M}, \tilde{g})$ splits as

$$(\tilde{M}, \tilde{g}) = (N_1, h_1) \times (\mathbb{R}, g_{\mathbb{R}})$$

where $N_1 = \tilde{r}_1^{-1}(0)$, $h_1 = \tilde{g}|_{N_1}$, and (N_1, h_1) is a totally geodesic submanifold of $(\tilde{M}, \tilde{g})$. Then, from (9.6), setting $f = \tilde{r}_2|_{N_1}$ we know

$$|\nabla f| \equiv 1 \quad \text{and} \quad \text{Hess } f \equiv 0 \quad \text{on } N_1.$$

It follows that $(\tilde{M}, \tilde{g})$ splits as

$$(\tilde{M}, \tilde{g}) = (N_2, h_2) \times (\mathbb{R}^2, g_{\mathbb{R}^2})$$

here $N_2 = \tilde{r}_1^{-1}(0) \cap \tilde{r}_2^{-1}(0)$, $h_2 = \tilde{g}|_{N_2}$, and (N_2, h_2) is a totally geodesic submanifold of (N_1, h_1) , and hence is also a totally geodesic submanifold of $(\tilde{M}, \tilde{g})$. Proceeding in this fashion, from (9.6) we then know that $(\tilde{M}, \tilde{g})$ splits as

$$(\tilde{M}, \tilde{g}) = (N, h) \times (\mathbb{R}^k, g_{\mathbb{R}^k})$$

where $N = \tilde{r}_1^{-1}(0) \cap \dots \cap \tilde{r}_k^{-1}(0)$, $h = \tilde{g}|_N$, and (N, h) is a totally geodesic submanifold of $(\tilde{M}, \tilde{g})$.

4. The Klein bottle and the Hantzsche-Wendt manifold are the desired examples. $\square$

Appendix

10.A. Injective Radius

Theorem 10.1

Let (M, g) be a Riemannian manifold, $p \in M$, and $\mathcal{E}_p$ be the domain of $\exp_p$. For each smooth curve $\tilde{\sigma} : [0, a] \rightarrow \mathcal{E}_p$ joining 0 and $\tilde{q}$ we set $\tilde{\gamma} : [0, 1] \rightarrow \mathcal{E}_p$ by $\tilde{\gamma}(t) = t\tilde{q}$. Then

$$L(\gamma) \leq L(\sigma) \quad \text{where } \sigma = \exp_p \circ \tilde{\sigma} \text{ and } \gamma = \exp_p \circ \tilde{\gamma}.$$

Proof. WLOG, we assume $\tilde{\sigma}(s) \neq 0$ for $0 < s \leq a$. Write

$$\tilde{\sigma}(s) = r(s)X(s) \quad \text{where } r(s) = |\tilde{\sigma}(s)|_{g_p}.$$

Consider the map

$$F : (0, a] \times [0, 1] \rightarrow M, \quad (s, t) \mapsto \exp_p(tX(s)).$$

Then $\sigma(s) = F(s, r(s))$ and hence

$$\sigma'(s) = \frac{\partial F}{\partial s}(s, r(s)) + r'(s) \frac{\partial F}{\partial t}(s, r(s)). \quad (10.1)$$

By Gauss lemma we know

$$g \left(\frac{\partial F}{\partial s}(s, t), \frac{\partial F}{\partial t}(s, t) \right) = 0 \quad \forall (s, t) \in (0, a] \times [0, 1]. \quad (10.2)$$

Since the exponential map keeps the norm of radial vector, we know

$$\left| \frac{\partial F}{\partial t}(s, t) \right| = 1 \quad \forall (s, t) \in (0, a] \times [0, 1]. \quad (10.3)$$

It follows from (10.1), (10.2) and (10.3) that

$$|\sigma'(s)|^2 = \left| \frac{\partial F}{\partial s}(s, r(s)) \right|^2 + (r'(s))^2 \geq (r'(s))^2.$$

Therefore,

$$L(\sigma) = \int_0^a |\sigma'(s)| ds \geq \int_0^a |r'(s)| ds \geq \int_0^a r'(s) ds = r(a).$$

Clearly $r(a) = |\tilde{q}| = L(\gamma)$. We are done. $\square$

Corollary 10.2

Let (M, g) be a Riemannian manifold, $p \in M$ and $\delta = \text{inj}(M, p)$. Then for each $q \in B(p, \delta)$, there exists a unique unit-speed minimal curve $\gamma_{p,q}$ joining p and q . More precisely, setting $F = \exp_p|_{B(0, \delta)}$, then

$$\gamma_{p,q}(t) = \exp_p(tv) \quad \text{where } 0 \leq t \leq |F^{-1}(q)|, \quad v = \frac{F^{-1}(q)}{|F^{-1}(q)|}.$$

Proof. Let $\sigma : [0, l] \rightarrow M$ be a curve joining p and q .

1. Suppose $\text{im}(\sigma) \subseteq B(p, \delta)$. Then by Theorem 10.1 we know $l(\sigma) \geq l(\gamma_{p,q})$.

2. Suppose $\text{im}(\sigma) \not\subseteq B(p, \delta)$. By continuity we can pick $t_0 \in (0, l)$ such that $\sigma(t) \in B(p, \delta)$ for all $t \in [0, t_0]$, and $|F^{-1}(\sigma(t_0))| > |F^{-1}(q)|$. Then by Theorem 10.1 we know

$$l(\sigma) \geq l(\sigma|_{[0, t_0]}) \geq l(\gamma_{p, \sigma(t_0)}) = |F^{-1}(\sigma(t_0))| > |F^{-1}(q)| = l(\gamma_{p,q}).$$

Thus $\gamma_{p,q}$ is a minimal geodesic. Since $l(\gamma_{p,q}) < \text{inj}(M, p)$, $\gamma_{p,q}$ is the unique unit-speed minimal geodesic joining p and q . $\square$

10.B. Riemannian Submersion

Lemma 10.3

Let $\pi : (M, g) \rightarrow (N, h)$ be a Riemannian submersion. Let $H \subset TM$ be the subbundle such that $H_p \perp \ker(\pi_{*p})$.

(1) For each $X \in \Gamma(N, TN)$, there exists a unique $\bar{X} \in \Gamma(M, H)$ such that $\pi_*(\bar{X}) = X$.

- (2) Let $\sigma : [a, b] \rightarrow N$ be a smooth curve, then for each $p \in \pi^{-1}(\sigma(a))$, there exists $\epsilon > 0$ and a unique smooth curve $\bar{\sigma} : [a, a + \epsilon] \rightarrow M$ such that

$$\bar{\sigma}(a) = p, \quad \pi \circ \bar{\sigma} = \sigma, \quad \bar{\sigma}'(t) \in H_{\bar{\sigma}(t)} \quad \forall t \in [a, a + \epsilon].$$

Proof. 1. Since $\pi_{*p}|_{H_p} : H_p \rightarrow T_{\pi(p)}N$ is an isometry, we can define $\bar{X}$ by

$$\bar{X}_p = (\pi_{*p}|_{H_p})^{-1}(X_{\pi(p)}).$$

In local charts $\{x^i\}$ and $\{y^j\}$ with $\pi(x^1, \dots, x^n, x^{n+1}, \dots, x^m) = (x^1, \dots, x^n)$, we know

$$\begin{aligned} X_{(y^1, \dots, y^n)} &= \sum_{i=1}^n a^i(y^1, \dots, y^n) \frac{\partial}{\partial x^i} \\ \Rightarrow \bar{X}_{(x^1, \dots, x^m)} &= \sum_{i=1}^n a^i(x^1, \dots, x^n) \frac{\partial}{\partial x^i} - \sum_{\alpha, \beta=n+1}^m g^{\alpha\beta} g_{i\beta}(x^1, \dots, x^m) \cdot a^i(x^1, \dots, x^n) \frac{\partial}{\partial x^\alpha}. \end{aligned}$$

Thus $\bar{X}$ is a smooth vector field. Therefore, we get $\bar{X}$ as desired.

2. Pick local charts $\{U, x^i\}$ and $\{V, y^j\}$ centered at p and $\sigma(a)$ respectively such that $\pi(x^1, \dots, x^m) = (x^1, \dots, x^n)$. Suppose $\sigma(t) = (\sigma^1(t), \dots, \sigma^n(t))$ in local chart. We set

$$\bar{\sigma}(t) = (\sigma^1(t), \dots, \sigma^n(t), \tau^{n+1}(t), \dots, \tau^m(t))$$

where $\tau^{n+1}(t), \dots, \tau^m(t)$ are functions to be determined. Note that

$$\bar{\sigma}'(t) = \sum_{i=1}^n (\sigma^i)'(t) \frac{\partial}{\partial x^i} \Big|_{\bar{\sigma}(t)} + \sum_{\alpha=n+1}^m (\tau^\alpha)'(t) \frac{\partial}{\partial x^\alpha} \Big|_{\bar{\sigma}(t)}$$

Hence the condition $\bar{\sigma}' \in H$ is equivalent to

$$\sum_{i=1}^n (\sigma^i)'(t) \cdot g_{i\beta}(\bar{\sigma}(t)) + \sum_{\alpha=n+1}^m (\tau^\alpha)'(t) \cdot g_{\alpha\beta}(\bar{\sigma}(t)) = 0 \quad \forall \beta = n+1, \dots, m.$$

This is a linear first-order ODE system of $\tau^{n+1}(t), \dots, \tau^m(t)$. Hence with $\tau^\alpha(a) = 0$, the solution uniquely exists. Then we get $\bar{\sigma} : [a, a + \epsilon] \rightarrow M$ as desired. $\square$

Proposition 10.4

Let $\pi : (M, g) \rightarrow (N, h)$ be a Riemannian submersion. For $X, Y \in \Gamma(N, TN)$ we have

$$\nabla_{\bar{X}}^g \bar{Y} = \overline{\nabla_X^h Y} + \frac{1}{2}[\bar{X}, \bar{Y}]^v$$

where Z^v is the orthogonal projection of Z to $\ker \pi_*$.

Proof. Suppose $T \in \Gamma(M, \ker \pi_*)$. Then Koszul formula indicates that

$$\begin{aligned} 2\langle \nabla_{\bar{X}}^g \bar{Y}, T \rangle_g &= \bar{X} \langle \bar{Y}, T \rangle_g + \bar{Y} \langle \bar{X}, T \rangle_g - T \langle \bar{X}, \bar{Y} \rangle_g \\ &\quad + \langle [\bar{X}, \bar{Y}], T \rangle_g - \langle [\bar{X}, T], \bar{Y} \rangle_g - \langle [\bar{Y}, T], \bar{X} \rangle_g \end{aligned}$$

Since $H \perp \ker \pi_*$, we know $\langle \bar{Y}, T \rangle_g = \langle \bar{X}, T \rangle_g = 0$ and hence $\bar{X} \langle \bar{Y}, T \rangle_g = \bar{Y} \langle \bar{X}, T \rangle_g = 0$. Since $\pi_* \bar{X} = X$ and $\pi_* T = 0$, we know

$$\pi_* [\bar{X}, T] = [X, 0] = 0 \Rightarrow [\bar{X}, T] \in \Gamma(M, \ker \pi_*) \Rightarrow \langle [\bar{X}, T], \bar{Y} \rangle_g = 0,$$

similarly $\langle [\bar{Y}, T], \bar{X} \rangle_g = 0$. Moreover, since $\pi_{*p}|_{H_p} : H_p \rightarrow T_{\pi(p)}N$ is an isometry, we know

$$T \langle \bar{X}, \bar{Y} \rangle_g = T \langle \langle X, Y \rangle_h \circ \pi \rangle = (\pi_* T) \langle X, Y \rangle_h = 0$$

There we use $\pi_*T = 0$ again. Hence

$$2\langle \nabla_{\bar{X}}^g \bar{Y}, T \rangle_g = \langle [\bar{X}, \bar{Y}], T \rangle_g \quad \forall T \in \Gamma(M, \ker \pi_*). \quad (10.4)$$

Suppose $\bar{Z} \in \Gamma(M, H)$ is the horizontal lifting of $Z \in \Gamma(N, TN)$. Note that

$$\bar{X}_p \langle \bar{Y}, \bar{Z} \rangle_g = \bar{X}_p(\langle Y, Z \rangle_h \circ \pi) = (\pi_* \bar{X}_p) \langle Y, Z \rangle_h = X_{\pi(p)} \langle Y, Z \rangle_h$$

and that

$$\langle [\bar{X}, \bar{Y}], \bar{Z} \rangle_g(p) = \langle \pi_* [\bar{X}, \bar{Y}], \pi_* \bar{Z} \rangle_h(\pi(p)) = \langle [X, Y], Z \rangle_h(\pi(p)).$$

Therefore, by Koszul formula, for all $Z \in \Gamma(N, TN)$ we know

$$\begin{aligned} 2\langle \nabla_{\bar{X}}^g \bar{Y}, \bar{Z} \rangle_g &= X \langle Y, Z \rangle_h \circ \pi + Y \langle X, Z \rangle_h \circ \pi - Z \langle X, Y \rangle_h \circ \pi \\ &\quad + \langle [X, Y], Z \rangle_h \circ \pi - \langle [X, Z], Y \rangle_h \circ \pi - \langle [Y, Z], X \rangle_h \circ \pi \\ &= 2\langle \nabla_X^h Y, Z \rangle_h \circ \pi = 2\langle \overline{\nabla_X^h Y}, \bar{Z} \rangle_g. \end{aligned} \quad (10.5)$$

Then the conclusion follows from (10.4) and (10.5). $\square$

Proposition 10.5

Let $\pi : (M, g) \rightarrow (N, h)$ be a Riemannian submersion. Then

$$\pi \circ \exp_p(v) = \exp_{\pi(p)}(d\pi_p(v)) \quad \forall v \in \mathcal{E}_p. \quad (10.6)$$

In particular, if $\tilde{\gamma} : [0, l] \rightarrow M$ is a geodesic, then $\pi \circ \tilde{\gamma} : [0, l] \rightarrow N$ is a geodesic. Moreover, if $\pi_*(\tilde{\gamma}'(0)) = \gamma'(0)$ and $\tilde{\gamma}'(0) \in H_{\tilde{\gamma}(0)}$, then $\tilde{\gamma}$ is the horizontal lifting of γ .

Proof. Fix some $p \in M$ and $v \in \mathcal{E}_p$. Define $\tilde{\gamma} : [0, 1] \rightarrow M$ by $\tilde{\gamma}(t) = \exp_p(tv)$, and set $\gamma = \pi \circ \tilde{\gamma}$.

1. Suppose $\gamma'(t_0) \neq 0$ with $t_0 \in (0, 1)$. Then for a sufficiently small $\epsilon > 0$, there exists a local field X defined near $\pi(\gamma(t_0))$ with $X_{\gamma(t)} = \gamma'(t)$ for $t_0 - \epsilon \leq t \leq t_0 + \epsilon$. By Lemma 10.3, let $\bar{X}$ be the horizontal lifting of X . Then $\bar{X}|_{\tilde{\gamma}(t)} = \tilde{\gamma}'(t)$. Since $\tilde{\gamma}$ is a geodesic, we know

$$(\nabla_{\bar{X}}^g \bar{X}) \circ \tilde{\gamma}(t) = 0 \quad \forall t_0 - \epsilon \leq t \leq t_0 + \epsilon.$$

From Proposition 10.4, we know

$$\overline{(\nabla_X^h X)} \circ \tilde{\gamma}(t) = (\nabla_{\bar{X}}^g \bar{X})^H \circ \tilde{\gamma}(t) = 0 \quad \forall t_0 - \epsilon \leq t \leq t_0 + \epsilon.$$

Hence $(\nabla_X^h X) \circ \gamma(t) = 0$ for $t_0 - \epsilon \leq t \leq t_0 + \epsilon$, which implies $\gamma|_{[t_0 - \epsilon, t_0 + \epsilon]}$ is a geodesic.

2. Suppose $\gamma'(t_0) \neq 0$ for some $t_0 \in (0, 1)$. Set

$$\mathcal{A} = \{\delta \in [0, 1 - t_0] : \gamma|_{[t_0, t_0 + \delta]} \text{ is a geodesic}\}.$$

By continuity we know $T_{\max} := \sup \mathcal{A} \in \mathcal{A}$. Then by point 1 we know $T_{\max} = 1 - t_0$. Similarly consider

$$\mathcal{B} = \{\delta \in [0, t_0] : \gamma|_{[t_0 - \delta, t_0]} \text{ is a geodesic}\}.$$

By continuity we know $T_{\min} := \sup \mathcal{B} \in \mathcal{B}$. Then by point 1 we know $T_{\min} = t_0$. Hence $\gamma : [0, 1] \rightarrow N$ is a geodesic with $\gamma'(0) = d\pi_p(v)$, which implies (10.6).

3. If $\gamma'(t) = 0$ for all $t \in [0, 1]$, then $\gamma(1) = \gamma(0) = \pi(p)$ which implies (10.6).

4. From (10.6), if $\tilde{\gamma} : [0, l] \rightarrow M$ is a geodesic, then $\pi \circ \tilde{\gamma} : [0, l] \rightarrow N$ is a geodesic. Now suppose in addition $\pi_*(\bar{v}) = v$ and $\bar{v} \in H_{\tilde{\gamma}(0)}$, where $\bar{v} = \tilde{\gamma}'(0)$ and $v = \gamma'(0)$. In the next we show $\tilde{\gamma}(t)$ is the horizontal lifting of $\gamma(t)$.

From (10.6) it suffices to show $\tilde{\gamma}' \in H$. Let $\{e_1, \dots, e_m\}$ be an orthonormal basis of $T_p M$ with $\ker(\pi_{*p}) = \text{span}\{e_{n+1}, \dots, e_m\}$. Fix some $t_0 > 0$ with $t_0|v| < \text{inj}(N, x)$. We define

$$\sigma_i(s) = \exp_p(t_0 \bar{v} + s e_i) \quad \text{for } n+1 \leq i \leq m.$$

Then $\sigma'_{n+1}(0), \dots, \sigma'_m(0)$ are linearly independent. From (10.6) we know $\pi \circ \sigma_i \equiv \exp_x(t_0 v)$. Hence at $q = \exp_p(t_0 \bar{v})$, we have

$$\ker(\pi_{*q}) = \text{span}\{\sigma'_{n+1}(0), \dots, \sigma'_m(0)\}.$$

Since $\langle \bar{v}, e_i \rangle = 0$ for $n+1 \leq i \leq m$, from Gauss Lemma we know

$$\langle \tilde{\gamma}'(t_0), \sigma'_i(0) \rangle = 0 \quad \text{for } n+1 \leq i \leq m.$$

Hence $\tilde{\gamma}'(t_0) \in H_{\tilde{\gamma}(t_0)}$. Set

$$\mathcal{D} = \{\delta \in [0, l] : \tilde{\gamma}'(t) \in H_{\tilde{\gamma}(t)}, \forall t \in [0, \delta]\}.$$

By continuity we know $\delta_{\max} := \sup \mathcal{D} \in \mathcal{D}$. By the preceding argument we know $\delta_{\max} = l$. Hence $\tilde{\gamma}(t)$ is the horizontal lifting of $\gamma(t)$. $\square$